STAT3405: Introduction to Bayesian Computing and Statistics
STAT4066: Bayesian Computing and Statistics

Berwin Turlach
School of Mathematics and Statistics
Berwin.Turlach@gmail.com

The University of Western Australia

Example 2: Counts of bicycles/other vehicles

Binomial data

- A survey was done of bicycle and other vehicular traffic in the neighbourhood of the campus of the University of California, Berkeley.
- Sixty city blocks were selected at random; each block was observed for an hour.

- The numbers of bicycles and other vehicles travelling along that block were recorded.
- The sampling was stratified into six types of city blocks: busy, fairly busy and residential streets, with and without bike routes, with ten blocks in each stratum.

Example 2: Counts of bicycles/other vehicles

Binomial data (ctd.)

The observed data for residential streets with a bike route

# cycles (y_i)	16	9	10	13	19	20	18	17	35	55
# other vehicles	58	90	48	57	103	57	86	112	273	64
# total (m_i)	74	99	58	70	122	77	104	129	308	119

- Can the proportion of bicyclists on residential streets with bike routes be estimated from these data?
- If so, how precise will our estimate be?
- Can we provide an interval of plausible values for the proportion of bicyclists on residential streets with bike routes?

Example 2: Counts of bicycles/other vehicles

Binomial data: the likelihood

We view the observations $y_1, y_2, \ldots, y_{10}$ as realisations of random variables $Y_1, Y_2, \ldots, Y_{10}$

From the description of the experiment, it seems plausible to start with the model $Y_i \sim \mathrm{Bin}(m_i, p)$

We assume that p is fixed for all observations, and that given p the random variables are independent of each other

Example 2: Counts of bicycles/other vehicles

Binomial data: the likelihood (ctd.)

This leads to the likelihood:

$$
\begin{aligned}
L(p|\mathbf{y}) &= P[\mathbf{Y} = \mathbf{y}|p] = P[Y_1 = y_1, Y_2 = y_2, \ldots, Y_{10} = y_{10}|p] \\
&= P[Y_1 = y_1|p] \times P[Y_2 = y_2|p] \times \cdots \times P[Y_{10} = y_{10}|p] \\
&= \prod_{i=1}^{10} P[Y_i = y_i|p] = \prod_{i=1}^{10} \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i} \\
&= p^{\sum_{i=1}^{10} y_i}(1-p)^{\sum_{i=1}^{10}(m_i - y_i)} \left(\prod_{i=1}^{10} \binom{m_i}{y_i} \right) \\
&= p^s(1-p)^{m-s} \left(\prod_{i=1}^{n} \binom{m_i}{y_i} \right)
\end{aligned}
$$

where $n = 10$ is the *sample size*, $s = \sum_{i=1}^{n} y_i$ and $m = \sum_{i=1}^{n} m_i$

Example 2: Counts of bicycles/other vehicles

Binomial data: classical analysis

```
library(stats4)
yi <- c(16, 9, 10, 13, 19, 20, 18, 17, 35, 55)
mi <- c(74, 99, 58, 70, 122, 77, 104, 129, 308, 119)
nll <- function(p) -sum(dbinom(yi, size = mi, prob = p, log = TRUE))
(res <- mle(nll, start = list(p = 0.5), lower = 0, upper = 1, method = "Brent"))
```

```
##
## Call:
## mle(minuslogl = nll, start = list(p = 0.5), method = "Brent",
##     lower = 0, upper = 1)
##
## Coefficients:
##         p
## 0.1827586
```

```
summary(res)
```

```
## Maximum likelihood estimation
##
## Call:
## mle(minuslogl = nll, start = list(p = 0.5), method = "Brent",
##     lower = 0, upper = 1)
##
## Coefficients:
##       Estimate Std. Error
## [1,] 0.1827586 0.01134683
##
## -2 log L: 117.928
```

Example 2: Counts of bicycles/other vehicles

Binomial data: classical analysis (ctd.)

Using the log likelihood:

$$l(p|\mathbf{y}) = \log(L(p|\mathbf{y})) = s\,\log(p) + (m - s)\,\log(1 - p) + c$$

where c is a constant that does not depend on p, it is (easy) to show that the maximum-likelihood estimate is

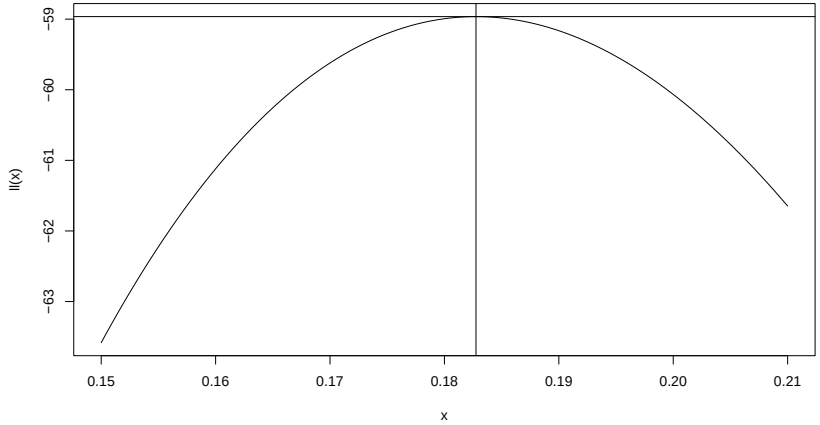
$$\hat{p} = \frac{s}{m} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}$$

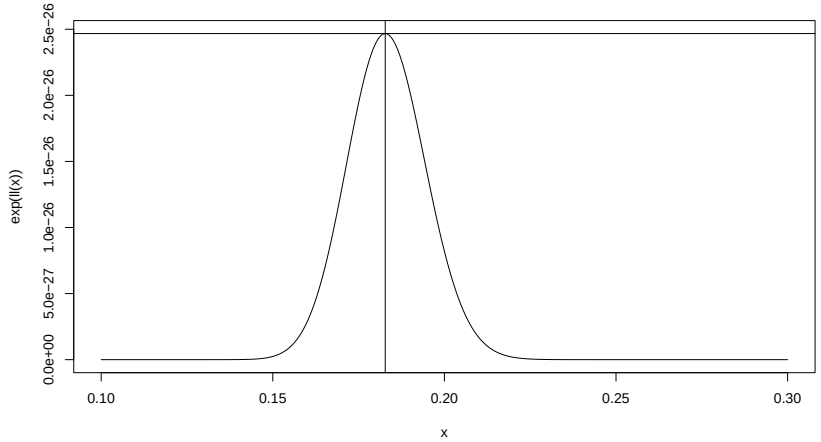
```
sum(yi)/sum(mi)
## [1] 0.1827586
```

Example 2: Counts of bicycles/other vehicles

Binomial data: classical analysis (ctd.)

```
ll <- function(p) sapply(p, function(x) -nll(x))
phat <- sum(yi)/sum(mi)
curve(ll, from = 0.15, to = 0.21)
abline(v = phat)
abline(h = ll(phat))
curve(exp(ll(x)), from = 0.1, to = 0.3, n = 401)
abline(v = phat)
abline(h = exp(ll(phat)))
```





Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior?

With a binomial sampling model for $\mathbf{Y}|p$, $s = \sum_{i=1}^{n} y_i$, $m = \sum_{i=1}^{n} m_i$ and a $\mathrm{Beta}(\alpha, \beta)$ prior for p, we have:

$$f(p|\mathbf{y}) \propto \underbrace{\prod_{i=1}^{n} \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}}_{\text{Likelihood: } L(p|\mathbf{y}) = P[\mathbf{Y}=\mathbf{y}|p]} \underbrace{\frac{1}{B(\alpha,\beta)} p^{\alpha-1}(1-p)^{\beta-1}}_{\text{prior: } f(p)}$$

$$\propto \underbrace{p^{s}(1-p)^{m-s}}_{\text{“essential” part of Likelihood}} \underbrace{p^{\alpha-1}(1-p)^{\beta-1}}_{\text{“essential” part of prior}}$$

$$\propto p^{\alpha+s-1}(1-p)^{\beta+m-s-1}$$

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

From which we can conclude that

$$f(p|\mathbf{Y} = \mathbf{y}) = \frac{1}{B(\alpha + s, \beta + m - s)} p^{\alpha+s-1}(1-p)^{\beta+m-s-1}$$

That is $p|\mathbf{Y} = \mathbf{y} \sim \mathrm{Beta}(\alpha + s, \beta + m - s)$

This shows that the family of beta distributions is the *conjugate* prior for the binomial distribution

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: Bayesian point estimation?

Traditionally, the posterior expectation of the parameter is used as a point estimate in Bayesian analysis

$$\hat{p} = \mathbb{E}[p|\mathbf{Y} = \mathbf{y}]$$

$$= \int_0^1 p\, f(p|\mathbf{Y} = \mathbf{y})\, dp$$

$$= \int_0^1 p\, \frac{1}{B(\alpha + s, \beta + m - s)} p^{\alpha+s-1}(1-p)^{\beta+m-s-1}\, dp$$

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: Bayesian point estimation? (ctd.)

Here

$$\begin{aligned}
\hat{p} = \mathbb{E}[p|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + s}{\alpha + s + \beta + m - s} = \frac{\alpha + s}{\alpha + \beta + m} \\
&= \frac{\alpha}{\alpha + \beta + m} + \frac{s}{\alpha + \beta + m} \\
&= \frac{\alpha}{\alpha + \beta} \frac{\alpha + \beta}{\alpha + \beta + m} + \frac{s}{m} \frac{m}{\alpha + \beta + m} \\
&= (1 - \lambda) \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \lambda \underbrace{\frac{s}{m}}_{\text{MLE}}
\end{aligned}$$

with $0 \leq \lambda = \frac{m}{\alpha+\beta+m} \leq 1$

That is, the Bayesian estimate $\hat{p}$ is a weighted average between the prior mean and the maximum likelihood estimate.

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: summary

Sampling model for data (likelihood):

$$Y_i|p \sim \mathrm{Bin}(m_i, p) \quad i = 1, \ldots, n, \text{ independent}$$

Prior for the parameter:

$$p \sim \mathrm{Beta}(\alpha, \beta)$$

Posterior of the parameter ($s = \sum_{i=1}^n y_i$, $m = \sum_{i=1}^n m_i$):

$$p|\mathbf{Y} = \mathbf{y} \sim \mathrm{Beta}(\alpha + s, \beta + m - s)$$

Point estimate $\hat{p}$ for p:

$$\begin{aligned}\hat{p} = \mathbb{E}[p|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + s}{\alpha + \beta + m} \\ &= \frac{\alpha + \beta}{\alpha + \beta + m} \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \frac{m}{\alpha + \beta + m} \underbrace{\frac{s}{m}}_{\text{MLE}}\end{aligned}$$

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: summary (special case of Bernoulli data)

Sampling model for data (likelihood):

$$Y_i|p \sim \mathrm{Bern}(p) \quad i = 1, \ldots, n, \text{ independent}$$

Then $S = \sum_{i=1}^{n} Y_i \sim \mathrm{Bin}(n, p)$

Prior for the parameter:

$$p \sim \mathrm{Beta}(\alpha, \beta)$$

Posterior of the parameter ($s = \sum_{i=1}^{n} y_i$):

$$p|\mathbf{Y} = \mathbf{y} \sim \mathrm{Beta}(\alpha + s, \beta + n - s)$$

Also stated as:

$$p|S = s \sim \mathrm{Beta}(\alpha + s, \beta + n - s)$$

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: summary (special case of Bernoulli data)

Point estimate $\hat{p}$ for p:

$$\begin{aligned}\hat{p} &= \mathbb{E}[p|\mathbf{Y} = \mathbf{y}] = \mathbb{E}[p|S = s] \\ &= \frac{\alpha + s}{\alpha + \beta + n} \\ &= (1 - \lambda) \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \lambda \underbrace{\bar{y}}_{\text{MLE}}\end{aligned}$$

where $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ and $0 \leq \lambda = \frac{n}{\alpha+\beta+s} \leq 1$

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

One question remains:

How to choose the *hyper-parameters* α and β?

Various proposal exists (all considered *non-informative*):

- *Haldane's prior* : $\alpha = \beta = 0$
  $$f(p) \propto p^{-1}(1-p)^{-1}$$
  This is an *improper* prior, i.e. $\int_0^1 f(p)\,dp = \infty$.
- *Jeffreys' prior* : $\alpha = \beta = \frac{1}{2}$
- *Bayes–Laplace's prior* : $\alpha = \beta = 1$
  Also called a flat prior.

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

A proposal for a non-informative prior out side of the family of beta distributions is due to Zellner:

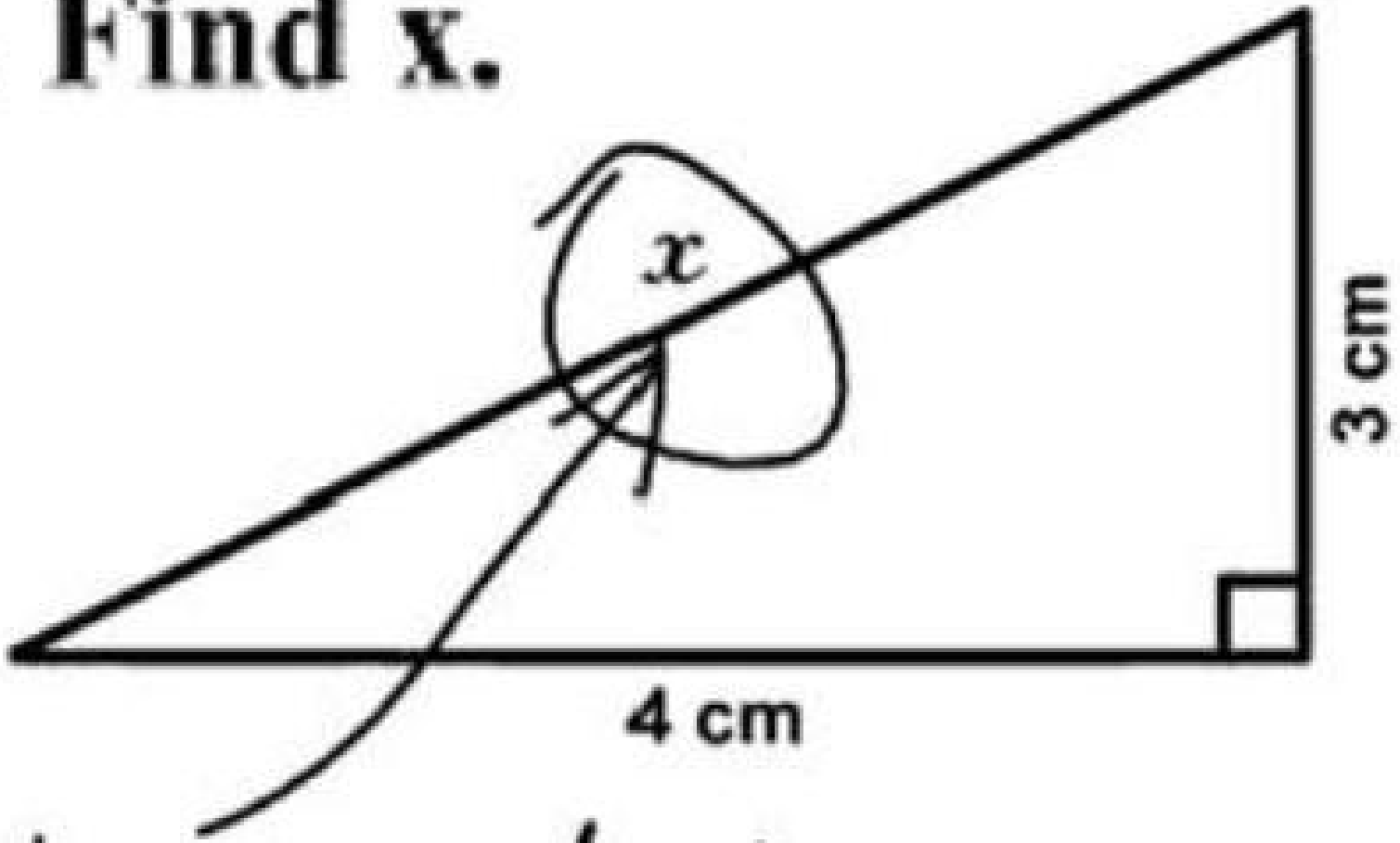
$$f(p) \propto p^p (1-p)^{1-p}$$

This would lead to the posterior ($s = \sum_{i=1}^n y_i$, $m = \sum_{i=1}^n m_i$):

$$f(p|\mathbf{Y} = \mathbf{y}) \propto p^{s+p} (1-p)^{m-s+1-p}$$

which is not a known distribution
The normalising constant has to be found by numeric integration

3. Find x.

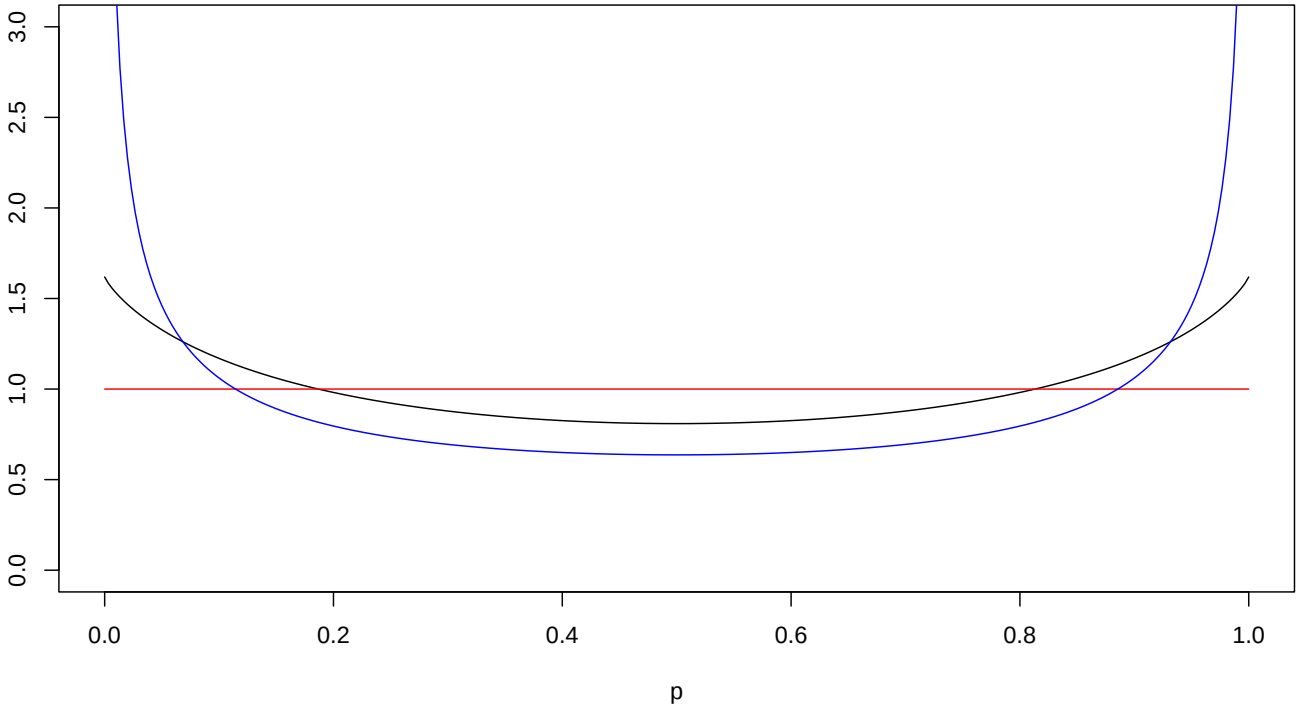


Here it is

Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

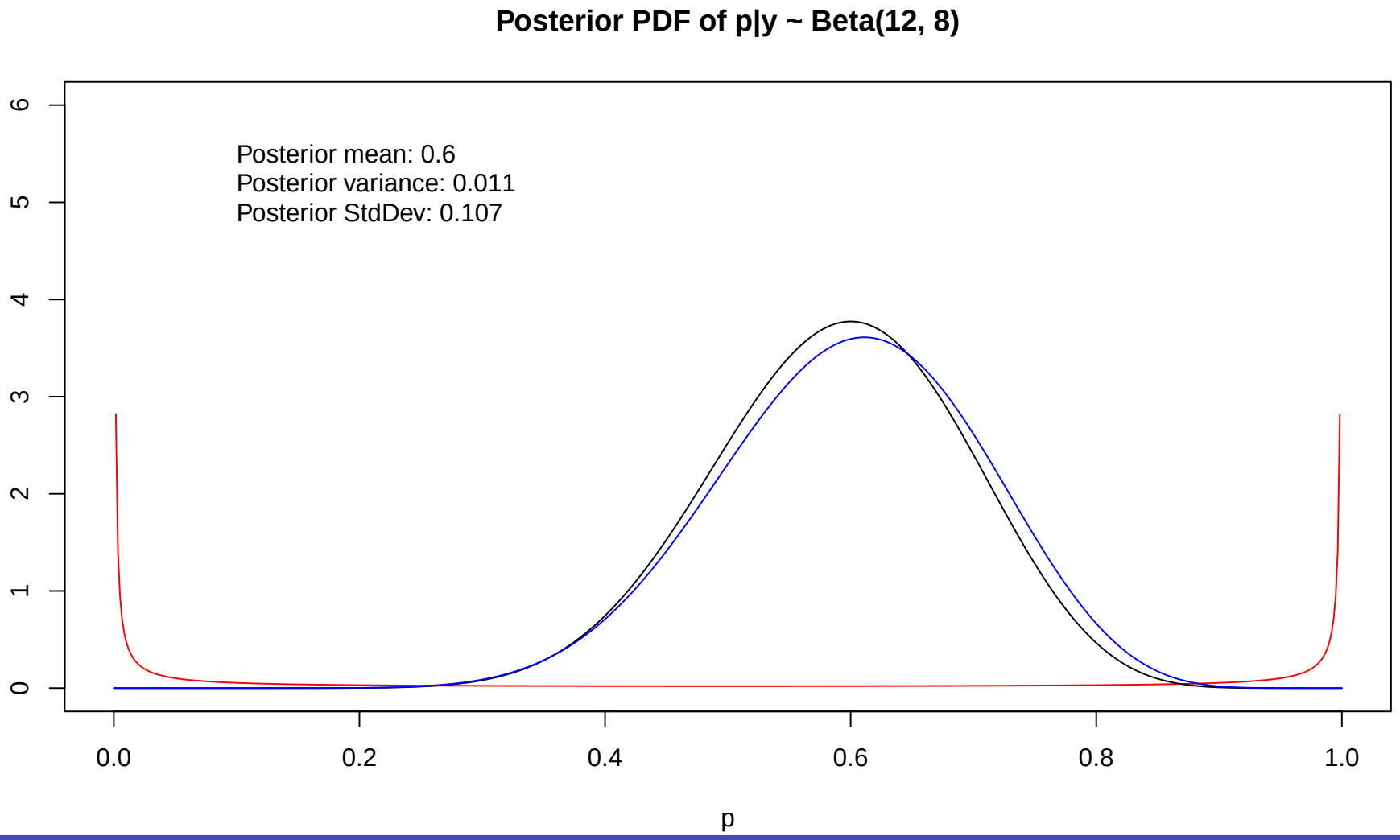
```
integrand <- function(p) p^p * (1 - p)^(1 - p)
(norm.const <- integrate(integrand, lower = 0, upper = 1)$value)

## [1] 0.6178269

curve(x^x * (1 - x)^(1 - x)/norm.const, from = 0, to = 1, n = 301, xlab = "p",
    ylab = "", ylim = c(0, 3))
curve(dbeta(x, 1, 1), from = 0, to = 1, n = 301, add = TRUE, col = "red")
curve(dbeta(x, 0.5, 0.5), from = 0, to = 1, n = 301, add = TRUE, col = "blue")
```



Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

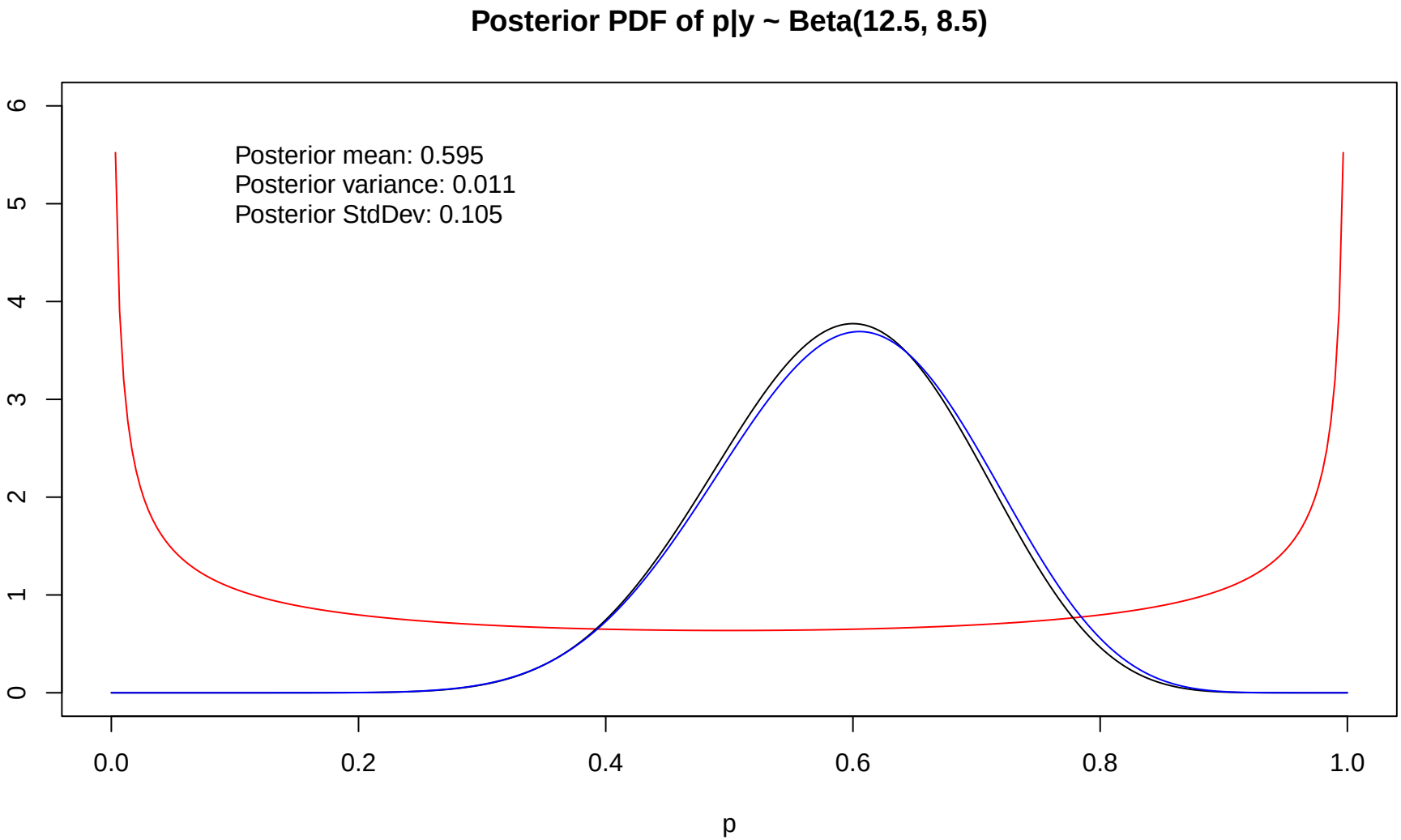
If $s = 12$ is a realisation of $S \sim \mathrm{Bin}(20, p)$ (rescaled likelihood in black) and we use Haldane's prior (red line) we obtain the posterior shown by the blue line:



Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

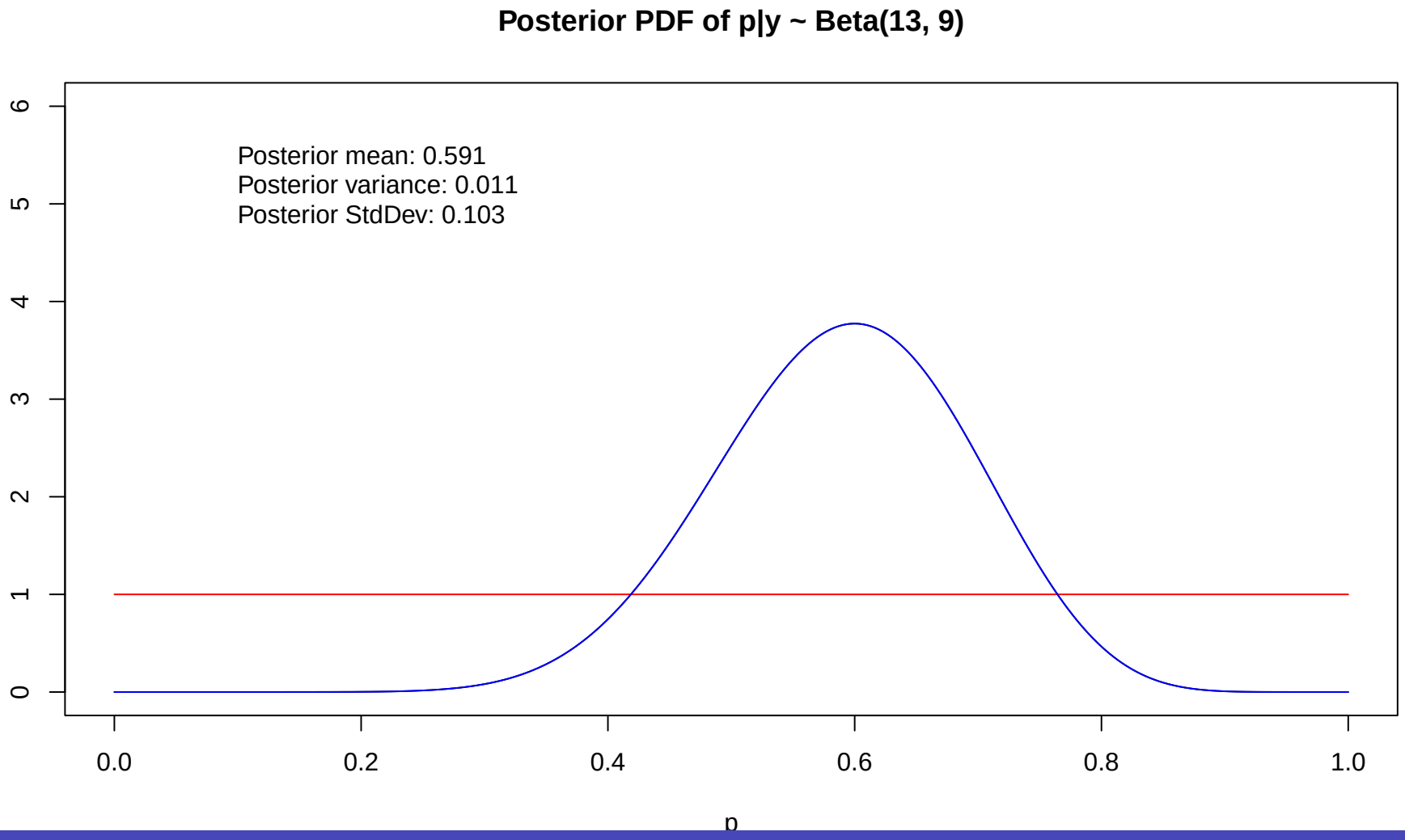
If $s = 12$ is a realisation of $S \sim \mathrm{Bin}(20, p)$ (rescaled likelihood in black) and we use Jeffreys' prior (red line) we obtain the posterior shown by the blue line:



Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

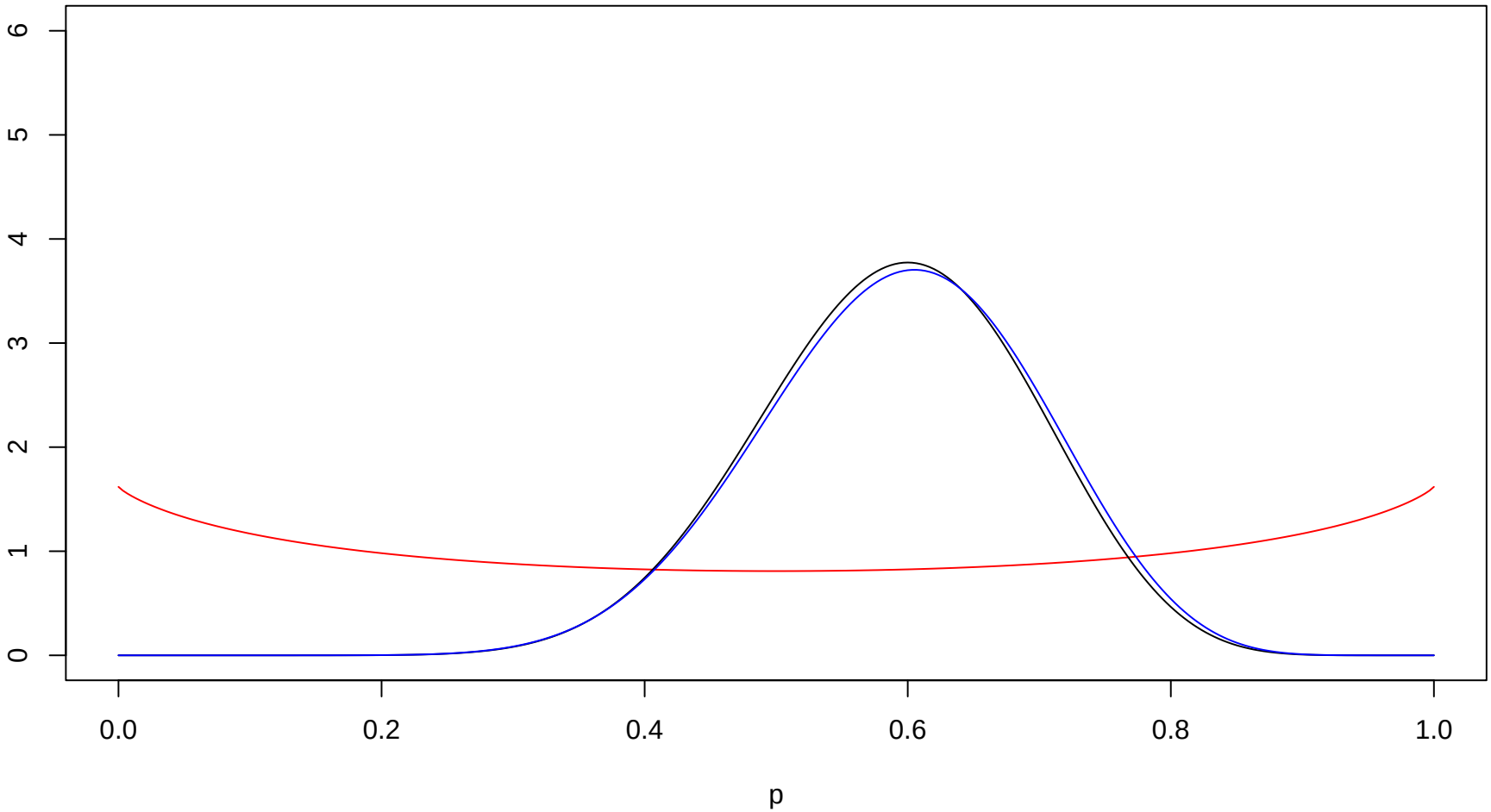
If $s = 12$ is a realisation of $S \sim \mathrm{Bin}(20, p)$ (rescaled likelihood in black) and we use Bayes–Laplace's prior (red line) we obtain the posterior shown by the blue line:



Example 2: Counts of bicycles/other vehicles

Bayesian analysis: how to choose the prior? (ctd.)

If $s = 12$ is a realisation of $S \sim \mathrm{Bin}(20, p)$ (rescaled likelihood in black) and we use Zellner's (red line) we obtain the posterior shown by the blue line:



Poisson distribution

Definition 41 (Poisson distribution)

An r.v. X has the Poisson distribution with parameter λ if the PMF of X is

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

We write this as $X \sim \mathrm{Pois}(\lambda)$.

The Poisson distribution is often used in situations where we are counting the number of successes in a particular region or interval of time, and there are a large number of trials, each with a small probability of success. The parameter λ is interpreted as the rate of occurrence of these rare events. The Poisson paradigm, if applicable, says that we can approximate the distribution of the number of events that occur by a Poisson distribution.

Example 3: Death by horse or mule kicks

Poisson data

Data from von Bortkiewicz (1898), given by Andrews and Herzberg (1985), on number of deaths by horse or mule kicks in 10 (of 14 reported) corps of the Prussian army.

Number of deaths	0	1	2	3	4
Number of corps-year	109	65	22	3	1

References:

Andrews, D.F. and Herzberg, A.M. (1985). *Data: A Collection of Problems from Many Fields for the Student and Research Worker*. Springer-Verlag, New York, NY.

von Bortkiewicz, L. (1898). *Das Gesetz der kleinen Zahlen*. Teubner, Leipzig.

Example 3: Death by horse or mule kicks

Poisson data (ctd.)

```
y.freq <- c(109, 65, 22, 3, 1)
y.obs <- rep(0:4, times = y.freq)
y.obs
```

```
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [33] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [65] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
##  [97] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [129] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [161] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [193] 2 2 2 2 3 3 3 4
```

```
table(y.obs)
```

```
## y.obs
##   0   1   2   3   4
## 109  65  22   3   1
```

```
(y.obs <- sample(y.obs))
```

```
##   [1] 1 1 1 1 0 0 1 1 0 1 0 1 2 1 0 1 1 0 0 0 1 2 0 1 1 0 0 2 1 0 1 1
##  [33] 2 1 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 0 2 0 0 1 1 1 0 0 0 2 1
##  [65] 2 0 1 0 0 1 4 1 1 1 0 0 0 1 1 0 3 0 2 1 0 1 0 0 0 1 0 0 1 0 1 1
##  [97] 1 1 1 1 0 0 0 0 0 0 0 0 2 0 1 2 0 2 0 0 1 0 0 0 0 0 1 0 1 1 2 0
## [129] 0 0 1 0 0 1 1 1 1 1 0 2 0 1 1 1 0 0 1 0 1 0 2 0 1 0 1 2 2 2 0 0
## [161] 0 0 0 0 0 3 3 0 0 0 0 2 0 1 0 0 0 1 0 2 2 0 0 1 0 0 1 0 0 2 0 0
## [193] 1 0 0 0 0 0 0 2
```

```
table(y.obs)
```

```
## y.obs
##   0   1   2   3   4
## 109  65  22   3   1
```

Example 3: Death by horse or mule kicks

Poisson data: the likelihood

We view the observations $y_1, y_2, \ldots, y_{200}$ as realisations of random variables $Y_1, Y_2, \ldots, Y_{200}$

A typical starting model for count data of (rare) events is the Poisson distribution, i.e. we model $Y_i \sim \mathrm{Pois}(\lambda)$

We assume that λ is fixed for all observations, and that given λ the random variables are independent of each other

Example 3: Death by horse or mule kicks

Poisson data: the likelihood (ctd.)

This leads to the likelihood:

$$
\begin{aligned}
L(\lambda|\mathbf{y}) &= P[\mathbf{Y} = \mathbf{y}|\lambda] = P[Y_1 = y_1, Y_2 = y_2, \ldots, Y_{200} = y_{200}|\lambda] \\
&= P[Y_1 = y_1|\lambda] \times P[Y_2 = y_2|\lambda] \times \cdots \times P[Y_{200} = y_{200}|\lambda] \\
&= \prod_{i=1}^{200} P[Y_i = y_i|\lambda] = \prod_{i=1}^{200} e^{-\lambda} \frac{\lambda^{y_i}}{y_i!} \\
&= e^{-n\lambda} \lambda^s \left(\prod_{i=1}^{n} \frac{1}{y_i!} \right)
\end{aligned}
$$

where $n = 200$ is the *sample size* and $s = \sum_{i=1}^{n} y_i$

Example 3: Death by horse or mule kicks

Poisson data: classical analysis

```
library(stats4)
nll <- function(lambda) -sum(dpois(y.obs, lambda, log = TRUE))
(res <- mle(nll, start = list(lambda = 1)))
```

```
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
## Warning in dpois(y.obs, lambda, log = TRUE): NaNs produced
```

```
##
## Call:
## mle(minuslogl = nll, start = list(lambda = 1))
##
## Coefficients:
##   lambda
## 0.610066
```

```
summary(res)
```

```
## Maximum likelihood estimation
##
## Call:
## mle(minuslogl = nll, start = list(lambda = 1))
##
## Coefficients:
##        Estimate Std. Error
## lambda 0.610066 0.05523263
##
## -2 log L: 412.2134
```

Example 3: Death by horse or mule kicks

Poisson data: classical analysis (ctd.)

Using the log likelihood:

$$l(\lambda|\mathbf{y}) = \log(L(\lambda|\mathbf{y})) = -n\lambda + s\log(\lambda) + c$$

where c is a constant that does not depend on λ, it is (easy) to show that the maximum-likelihood estimate is

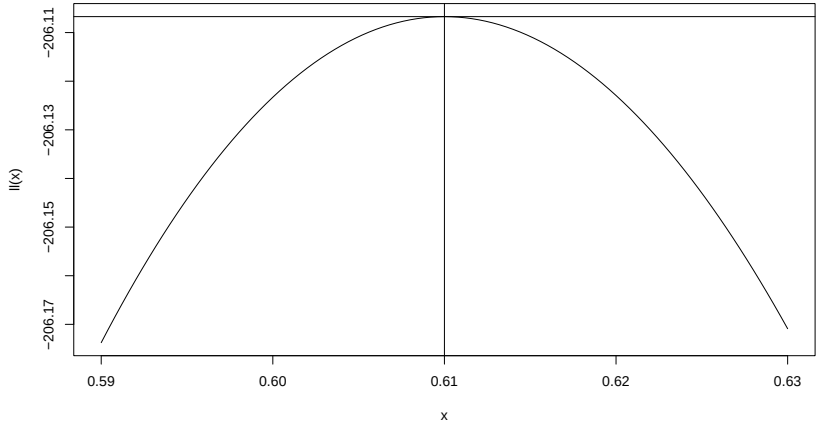
$$\hat{\lambda} = \frac{s}{n} = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{y}$$

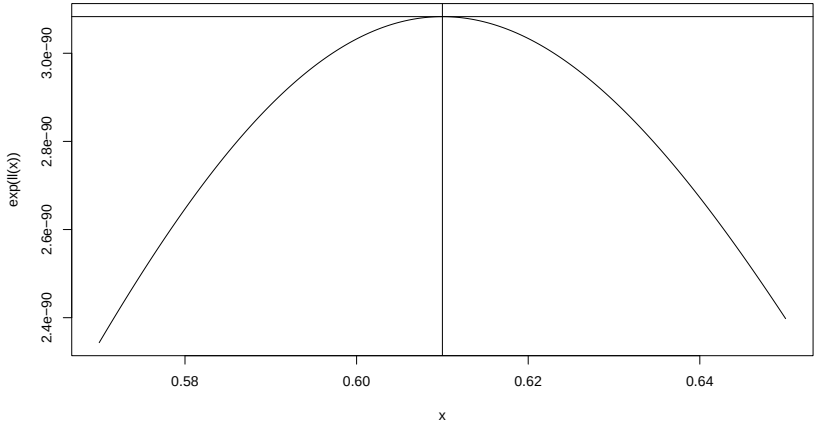
```
mean(y.obs)
## [1] 0.61
```

Example 3: Death by horse or mule kicks

Poisson data: classical analysis (ctd.)

```
ll <- function(lambda) sapply(lambda, function(x) -nll(x))
lambdahat <- mean(y.obs)
curve(ll, from = 0.59, to = 0.63)
abline(v = lambdahat)
abline(h = ll(lambdahat))
curve(exp(ll(x)), from = 0.57, to = 0.65, n = 401)
abline(v = lambdahat)
abline(h = exp(ll(lambdahat)))
```





Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior (Gamma function)

Mathematicians have studied intensively the *Gamma* function:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt, \quad x > 0$$

Which has some interesting properties and a connection to the Beta function:

$$\Gamma(x+1) = x\Gamma(x)$$

$$\Gamma(1) = 1 \qquad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma(n+1) = n! \qquad \text{if } n \text{ is an integer}$$

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior (Gamma function, ctd.)

Note for $x \in [0, \infty)$ the integrand is always non-negative

Hence

$$\frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}, \quad x > 0$$

is a non-negative function on $[0, \infty)$ that integrates to one

I.e., the PDF of a continuous random variable X with support $[0, \infty)$

Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior (Gamma function, ctd.)

For good measure, probability theorists and statisticians added a *rate* parameter β (in addition to the *shape* parameter α) to create the family of *Gamma distributions*

The continuous r.v. $X \sim \mathrm{Gamma}(\alpha, \beta)$ $(\alpha, \beta > 0)$ if its PDF is:

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x \in [0, \infty)$$

Note, $\mathbb{E}[X] = \frac{\alpha}{\beta}$ and $\mathrm{Var}[X] = \frac{\alpha}{\beta^2}$

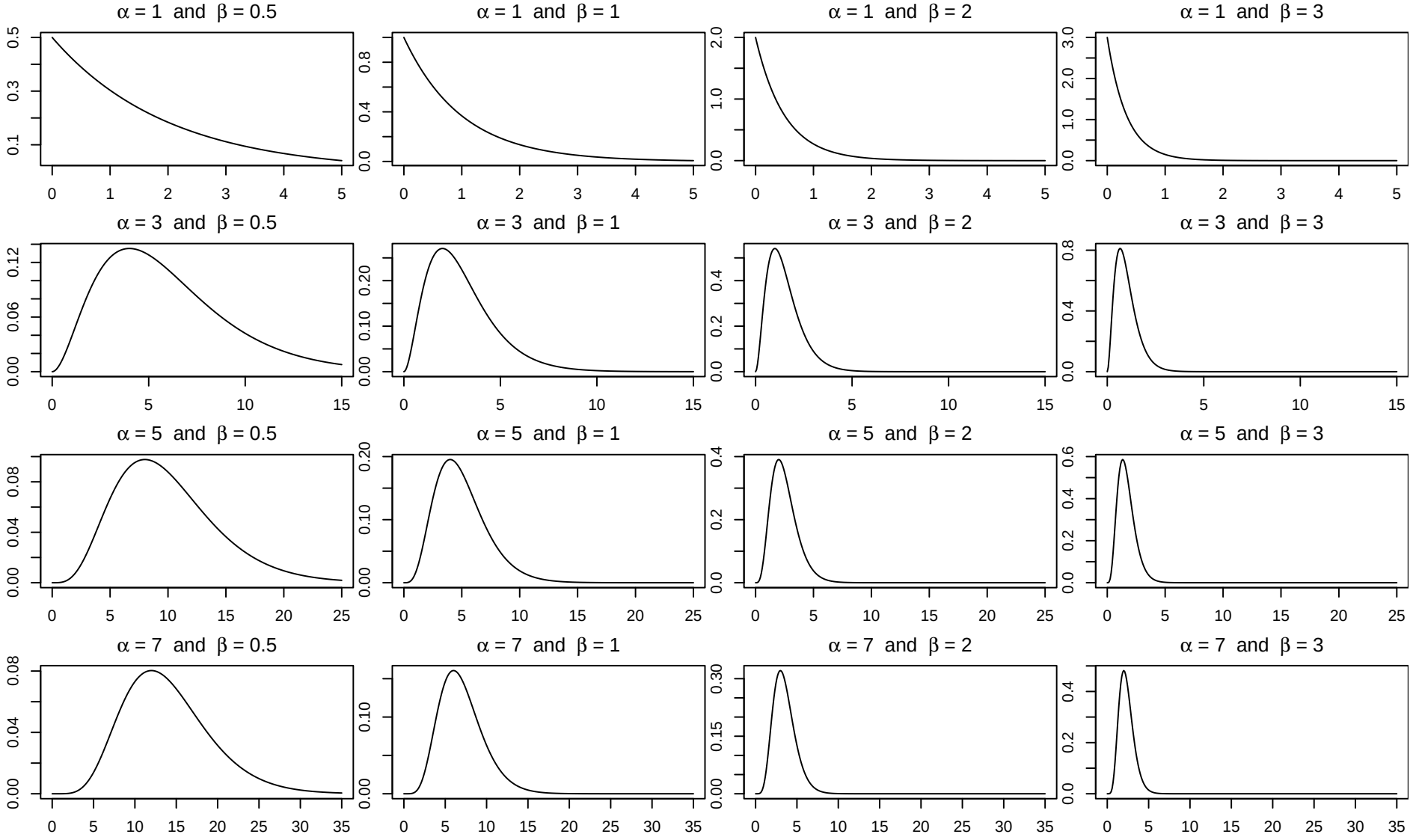
Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior (Gamma function ctd.)

```
alpha.vals <- c(1, 3, 5, 7)
beta.vals <- c(0.5, 1, 2, 3)
par(mfrow = c(4,4), mar = c(2.1, 2.1, 2.1, 0))
for(a in alpha.vals){
  for(b in beta.vals){
    curve(dgamma(x, a, b), from = 0, to = 5*a,
          n = 301, xlab = "x", ylab = "",
          main = bquote(paste(alpha, " = ", .(a), "  and  ", beta, " = ", .(b))))
  }
}
```

Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior (Gamma function ctd.)



Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior? (ctd.)

With a Poisson sampling model for $\mathbf{Y}|\lambda$, $s = \sum_{i=1}^{n} y_i$, and a $\mathrm{Gamma}(\alpha, \beta)$ prior for λ, we have:

$$f(\lambda|\mathbf{y}) \propto \underbrace{\prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{y_i}}{y_i!}}_{\text{Likelihood: } L(\lambda|\mathbf{y})=P[\mathbf{Y}=\mathbf{y}|\lambda]} \underbrace{\frac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}}_{\text{prior: } f(\lambda)}$$

$$\propto \underbrace{e^{-n\lambda}\lambda^{s}}_{\text{“essential” part of Likelihood}} \underbrace{\lambda^{\alpha-1}e^{-\beta\lambda}}_{\text{“essential” part of prior}}$$

$$\propto \lambda^{\alpha+s-1}e^{-(\beta+n)\lambda}$$

Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior? (ctd.)

From which we can conclude that

$$f(\lambda|\mathbf{Y} = \mathbf{y}) = \frac{(\beta + n)^{\alpha+s}}{\Gamma(\alpha + s)} \lambda^{\alpha+s-1} e^{-(\beta+n)\lambda}$$

That is $\lambda|\mathbf{Y} = \mathbf{y} \sim \mathrm{Gamma}(\alpha + s, \beta + n)$

This shows that the family of gamma distributions is the *conjugate* prior for the Poisson distribution

Example 3: Death by horse or mule kicks

Bayesian analysis: Bayesian point estimation?

Traditionally, the posterior expectation of the parameter is used as a point estimate in Bayesian analysis

$$\hat{\lambda} = \mathbb{E}[\lambda | \mathbf{Y} = \mathbf{y}]$$

$$= \int_0^\infty \lambda \, f(\lambda | \mathbf{Y} = \mathbf{y}) \, d\lambda$$

$$= \int_0^\infty \lambda \, \frac{(\beta + n)^{\alpha+s}}{\Gamma(\alpha + s)} \lambda^{\alpha+s-1} e^{-(\beta+n)\lambda} \, d\lambda$$

Example 3: Death by horse or mule kicks

Bayesian analysis: Bayesian point estimation? (ctd.)

Here

$$\begin{aligned}\hat{\lambda} = \mathbb{E}[\lambda|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + s}{\beta + n} = \frac{\alpha}{\beta + n} + \frac{s}{\beta + n} \\ &= \frac{\beta}{\beta + n}\frac{\alpha}{\beta} + \frac{n}{\beta + n}\frac{s}{n} \\ &= (1 - \kappa) \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}} + \kappa \underbrace{\bar{y}}_{\text{MLE}}\end{aligned}$$

with $0 \leq \kappa = \frac{n}{\beta+n} \leq 1$

That is, the Bayesian estimate $\hat{\lambda}$ is a weighted average between the prior mean and the maximum likelihood estimate.

Example 3: Death by horse or mule kicks

Bayesian analysis: summary

Sampling model for data (likelihood):

$$Y_i|\lambda \sim \mathrm{Pois}(\lambda) \quad i = 1, \ldots, n, \text{ independent}$$

Prior for the parameter:

$$\lambda \sim \mathrm{Gamma}(\alpha, \beta)$$

Posterior of the parameter ($s = \sum_{i=1}^{n} y_i$):

$$\lambda|\mathbf{Y} = \mathbf{y} \sim \mathrm{Gamma}(\alpha + s, \beta + n)$$

Point estimate $\hat{\lambda}$ for λ:

$$\begin{aligned}\hat{\lambda} = \mathbb{E}[\lambda|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + s}{\beta + n} \\ &= \frac{\beta}{\beta + n} \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}} + \frac{n}{\beta + n} \underbrace{\bar{y}}_{\text{MLE}}\end{aligned}$$

Example 3: Death by horse or mule kicks

Bayesian analysis: how to choose the prior? (ctd.)

One question remains:

How to choose the *hyper-parameters* α and β?

Various proposal exists (all considered *non-informative*):

- *Uniform/flat prior* : $\alpha = 1, \beta = 0$
$$f(\lambda) \propto 1, \quad \text{if } \lambda > 0$$
This is an *improper* prior, i.e. $\int_0^\infty f(\lambda)\, d\lambda = \infty$.
- *Jeffreys' prior* : $\alpha = \frac{1}{2}, \beta = 0$
$$f(\lambda) \propto \lambda^{-1/2}, \quad \text{if } \lambda > 0$$
This is an *improper* prior, i.e. $\int_0^\infty f(\lambda)\, d\lambda = \infty$.

Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior

Prior: $\lambda \sim$ "$\text{Gamma}(1, 0)$"

Data: $n = 200$ observations from $\text{Pois}(\lambda)$

$s = \sum_{i=1}^{n} y_i = 122$

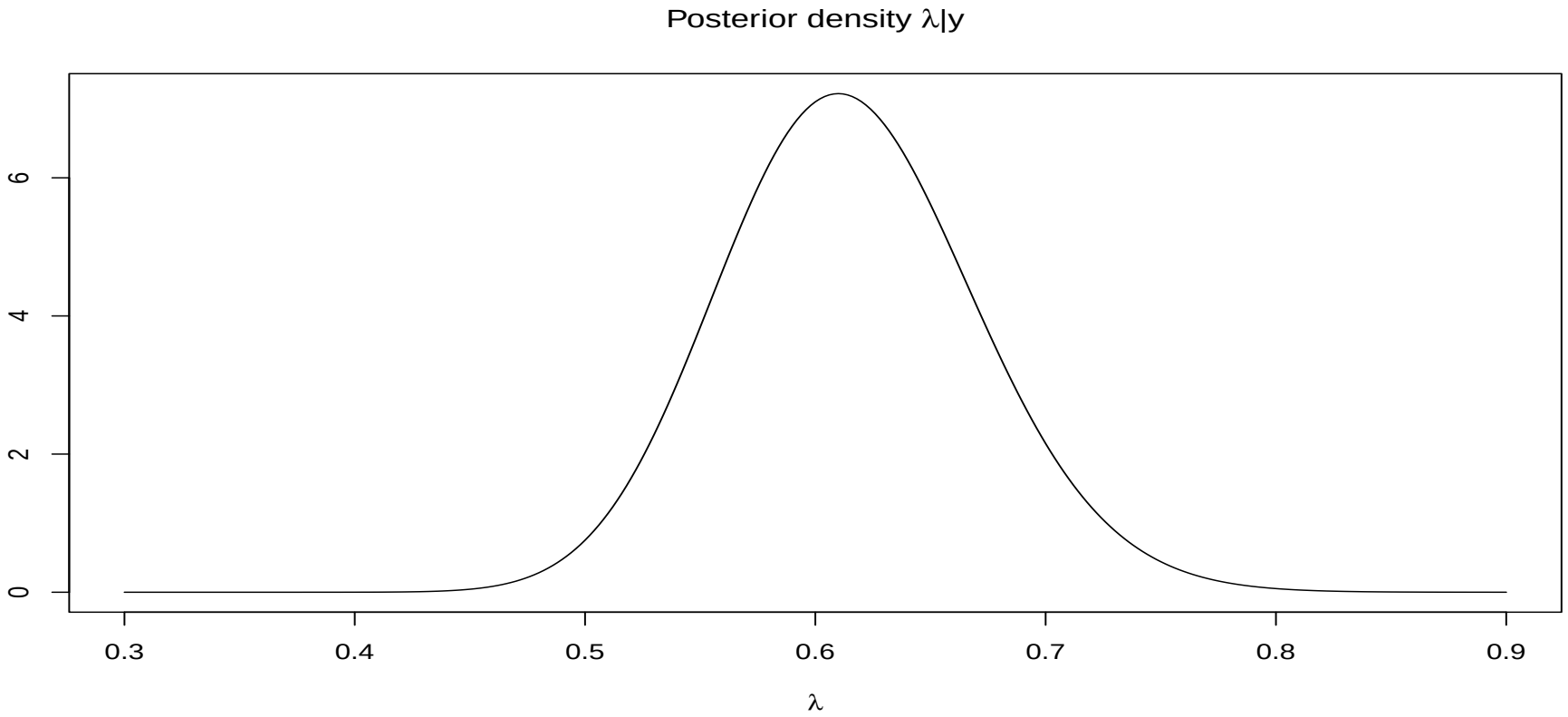
Posterior: $\lambda | \mathbf{Y} = \mathbf{y} \sim \text{Gamma}(123, 200)$

```
alpha <- 1
beta <- 0
n <- length(y.obs)
s <- sum(y.obs)
alpha.post <- alpha + s
beta.post <- beta + n
(post.mean <- alpha.post/beta.post)  ## posterior mean

## [1] 0.615

(post.var <- alpha.post/beta.post^2)  ## posterior variance

## [1] 0.003075

(post.sd <- sqrt(post.var))  ## posterior StdDev

## [1] 0.05545268
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

```
curve(dgamma(x, shape = alpha.post, rate = beta.post), from = 0.3, to = 0.9,
    n = 301, xlab = expression(lambda), ylab = "", main = expression("Posterior density " *
        lambda * "|y"))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

Writing the BUGS code for this model is fairly simple:

```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dpois(lambda)
  }

  ## prior
  u ~ dflat()
  lambda <- abs(u)
}
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

```
library(BRugs)

## Welcome to BRugs connected to OpenBUGS version 3.2.3

modelCheck("Poisson1.bug")

## model is syntactically correct

bugsData(list(y = y.obs, N = length(y.obs)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(u = 1), list(u = 2), list(u = 0.5), list(u = 3))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 3: Death by horse or mule kicks

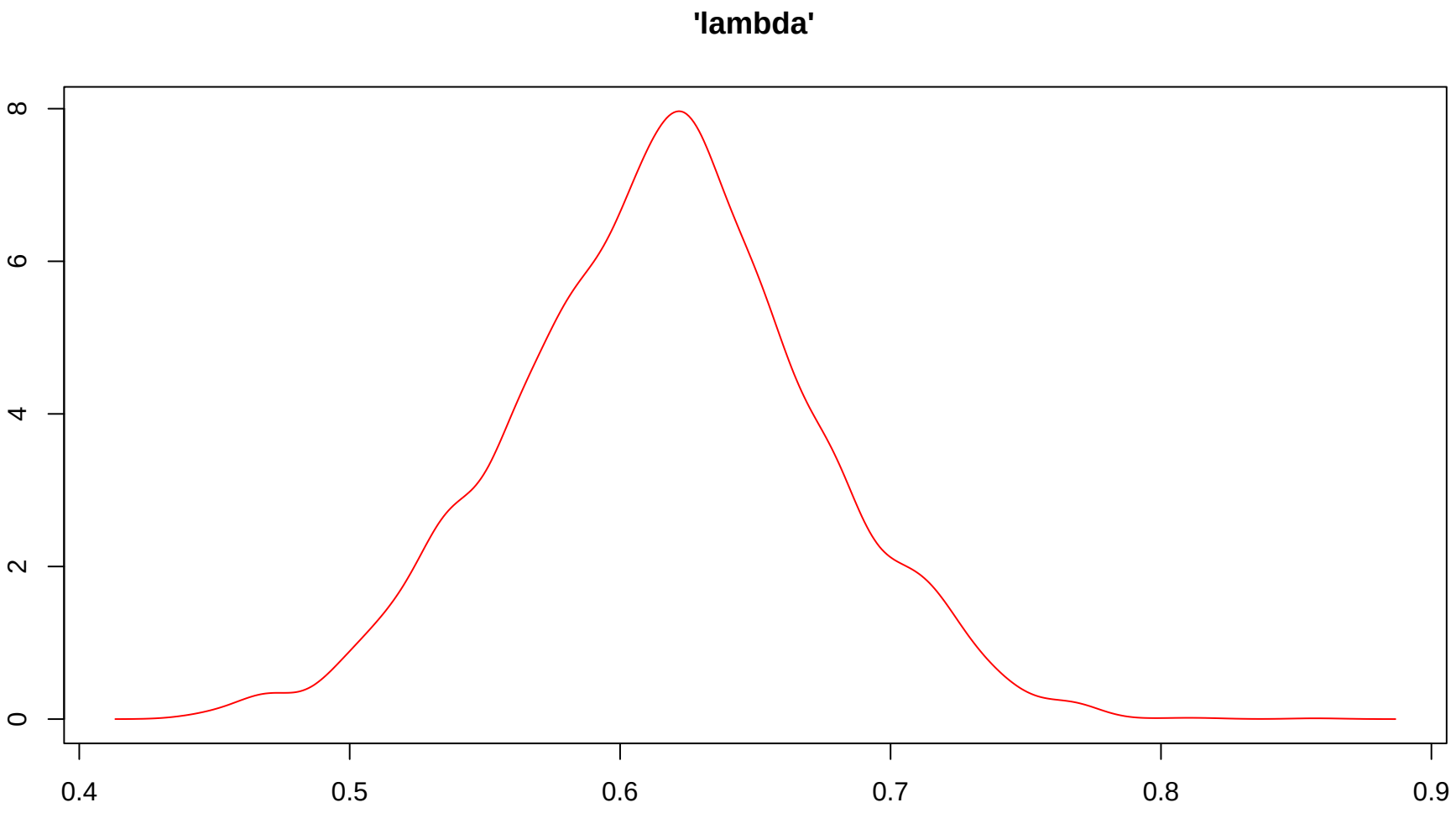
Bayesian analysis: using a flat prior (ctd.)

```
samplesSet("lambda")
## monitor set for variable 'lambda'
modelUpdate(3000)
## 3000 updates took 0 s
samplesStats("*", beg = 1001)
##          mean      sd MC_error val2.5pc median val97.5pc start sample
## lambda 0.6159 0.05552 0.001982   0.5068 0.6173    0.7248  1001   8000
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

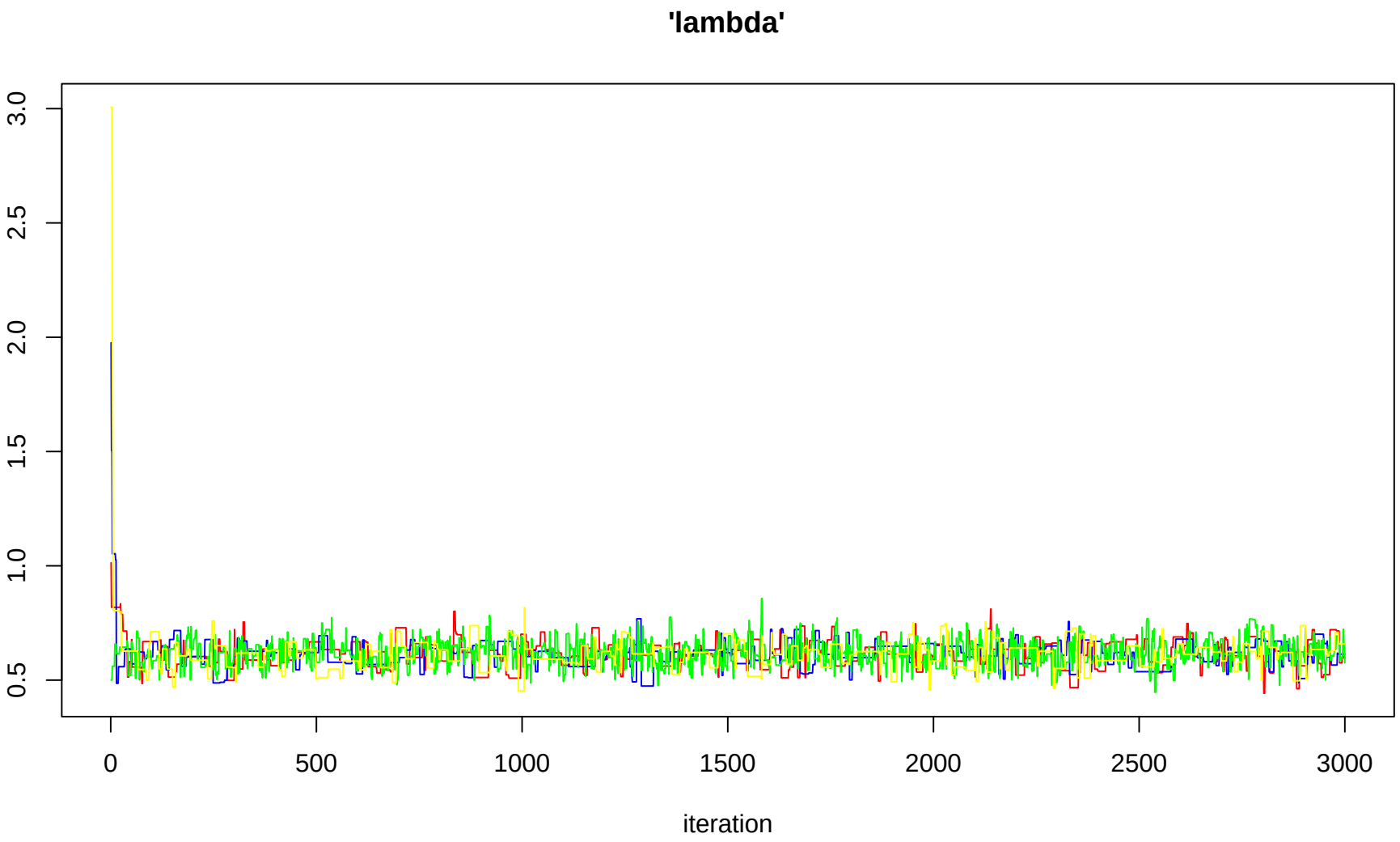
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

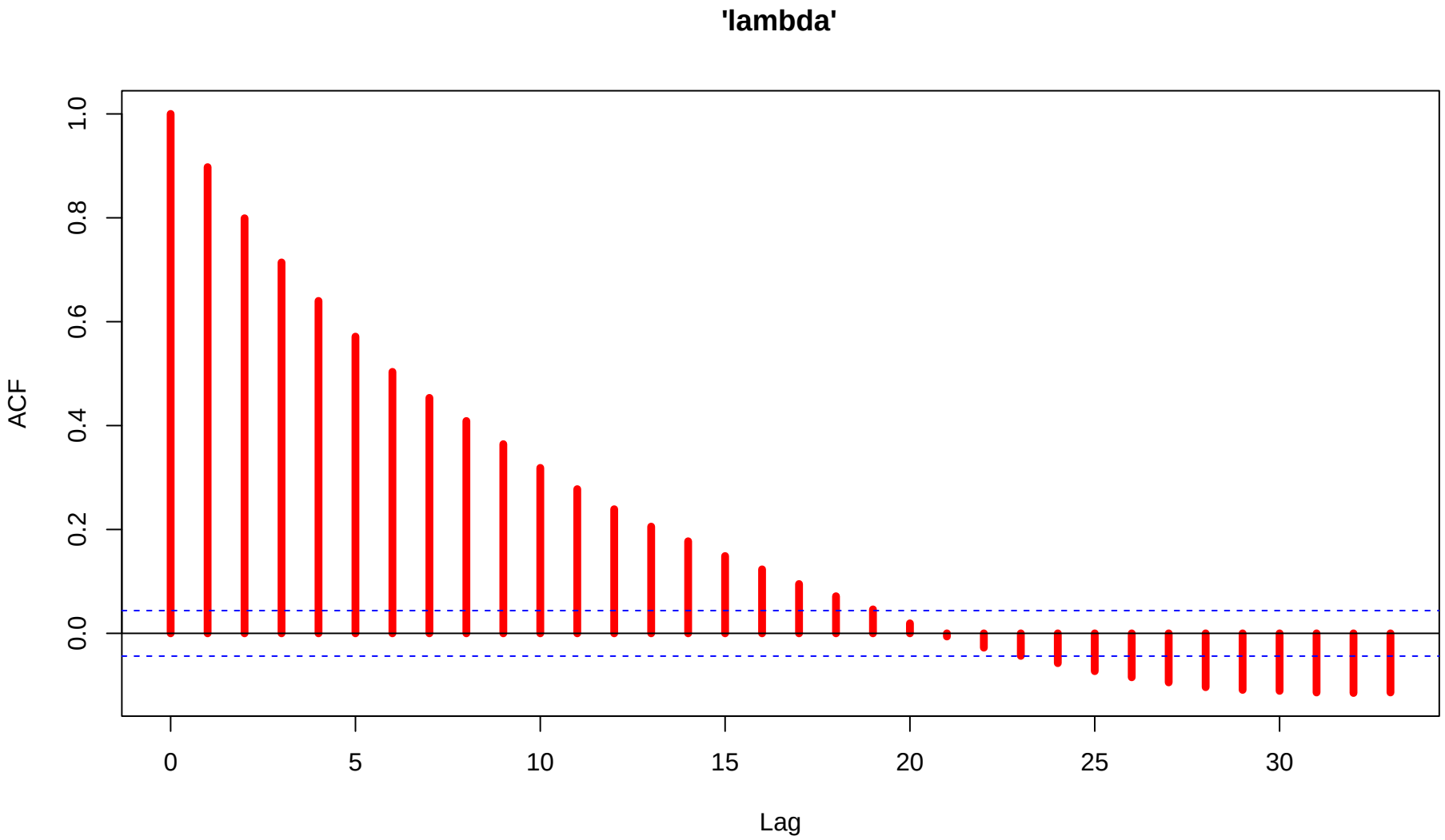
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

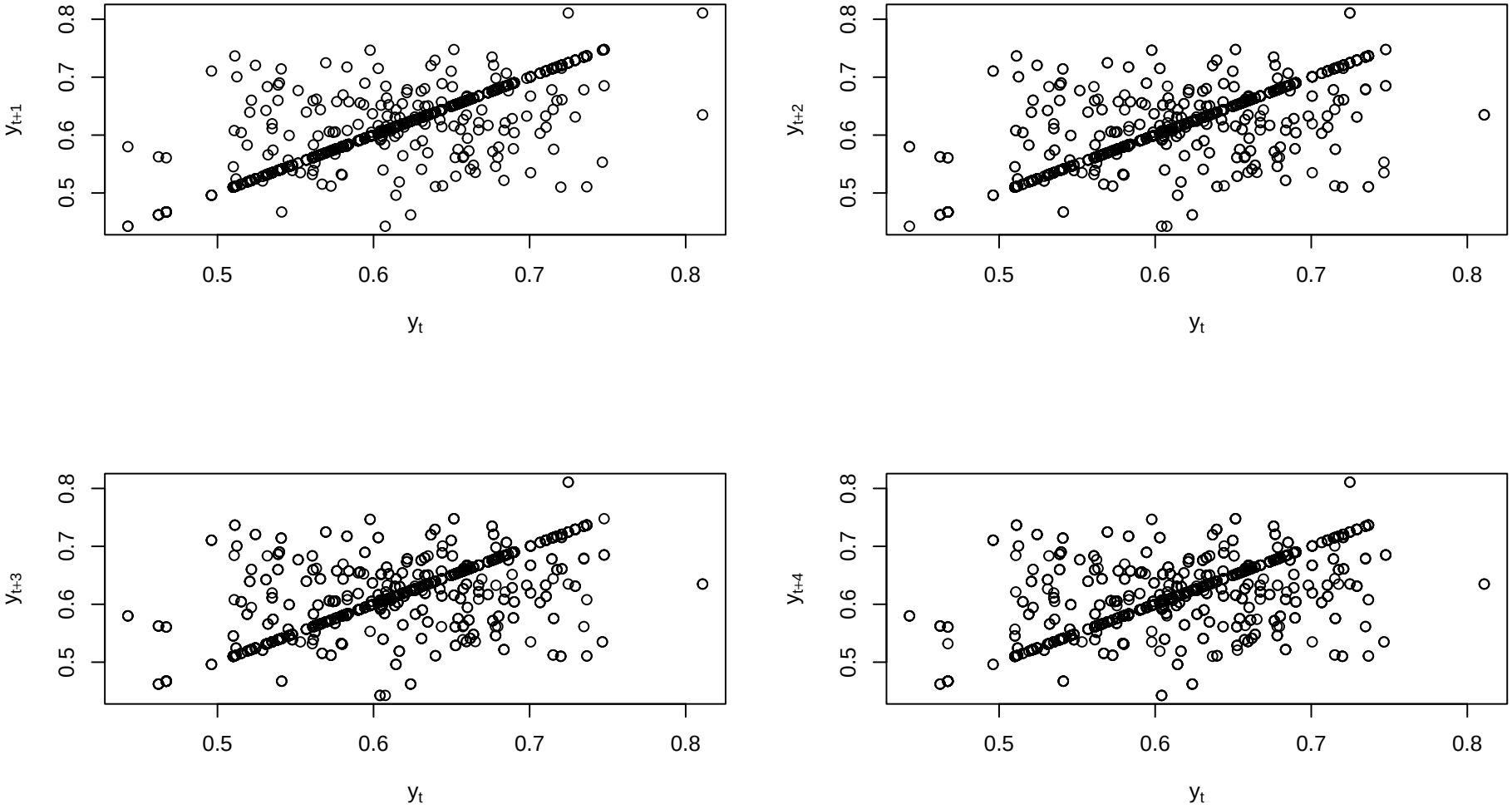
Bayesian analysis: using a flat prior (ctd.)

```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

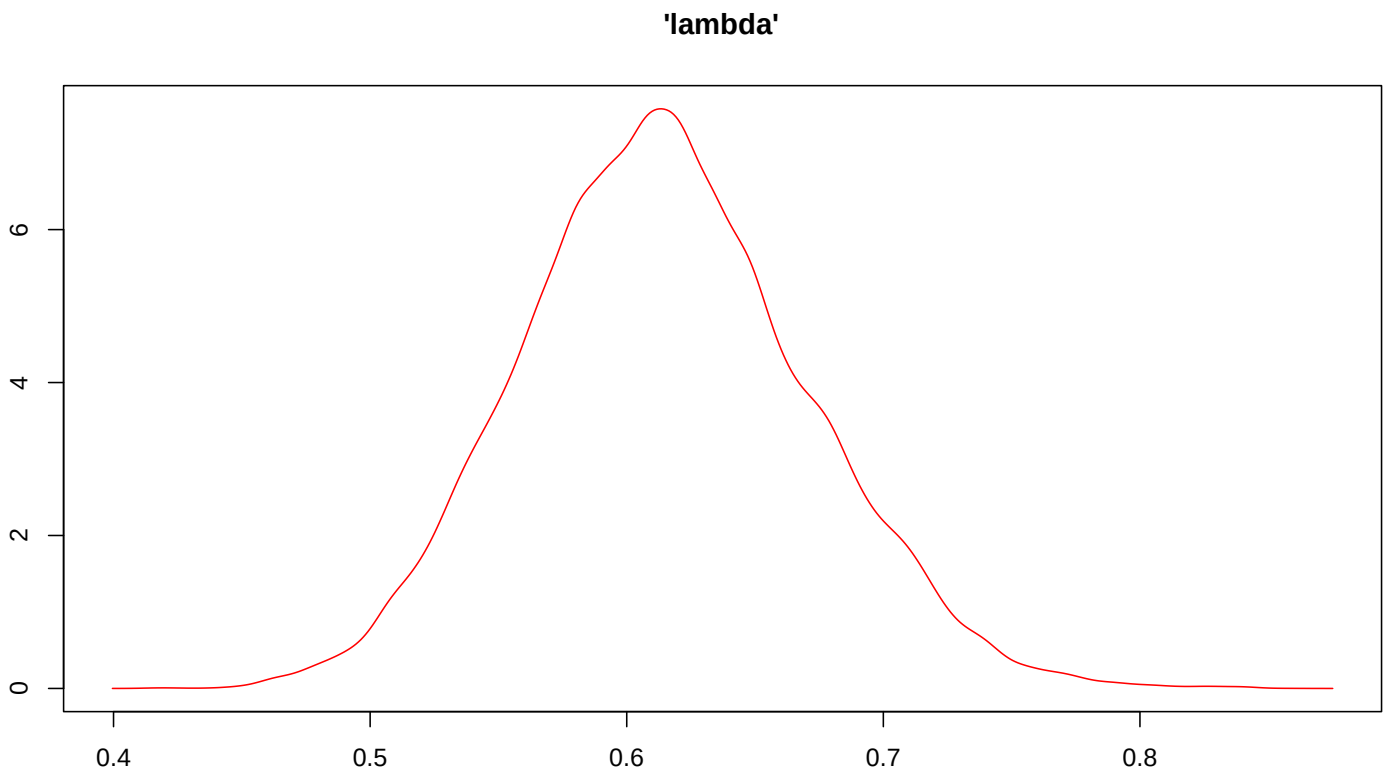
Bayesian analysis: using a flat prior (ctd.)



Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

```
modelUpdate(28000)

## 28000 updates took 3 s

samplesStats("*", beg = 1001)

##          mean      sd  MC_error val2.5pc median val97.5pc start
## lambda 0.6143 0.05549 0.0005825   0.5098 0.6124    0.7264  1001
##        sample
## lambda 120000

samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using a flat prior (ctd.)

Analysing the data on death by horse or mule kicks using a flat prior for λ leads to an estimate of $\hat{\lambda} = 0.61$ deaths per year (per corps).

This estimate is based on running four chains, each for 31 000 iterations and discarding the first 1 000 iterations as burn-in. Trace plots indicated that the chains had converged and were mixing well.

Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior

Prior: $\lambda \sim$ "$\text{Gamma}\left(\frac{1}{2}, 0\right)$"

Data: $n = 200$ observations from $\text{Pois}(\lambda)$

$s = \sum_{i=1}^{n} y_i = 122$

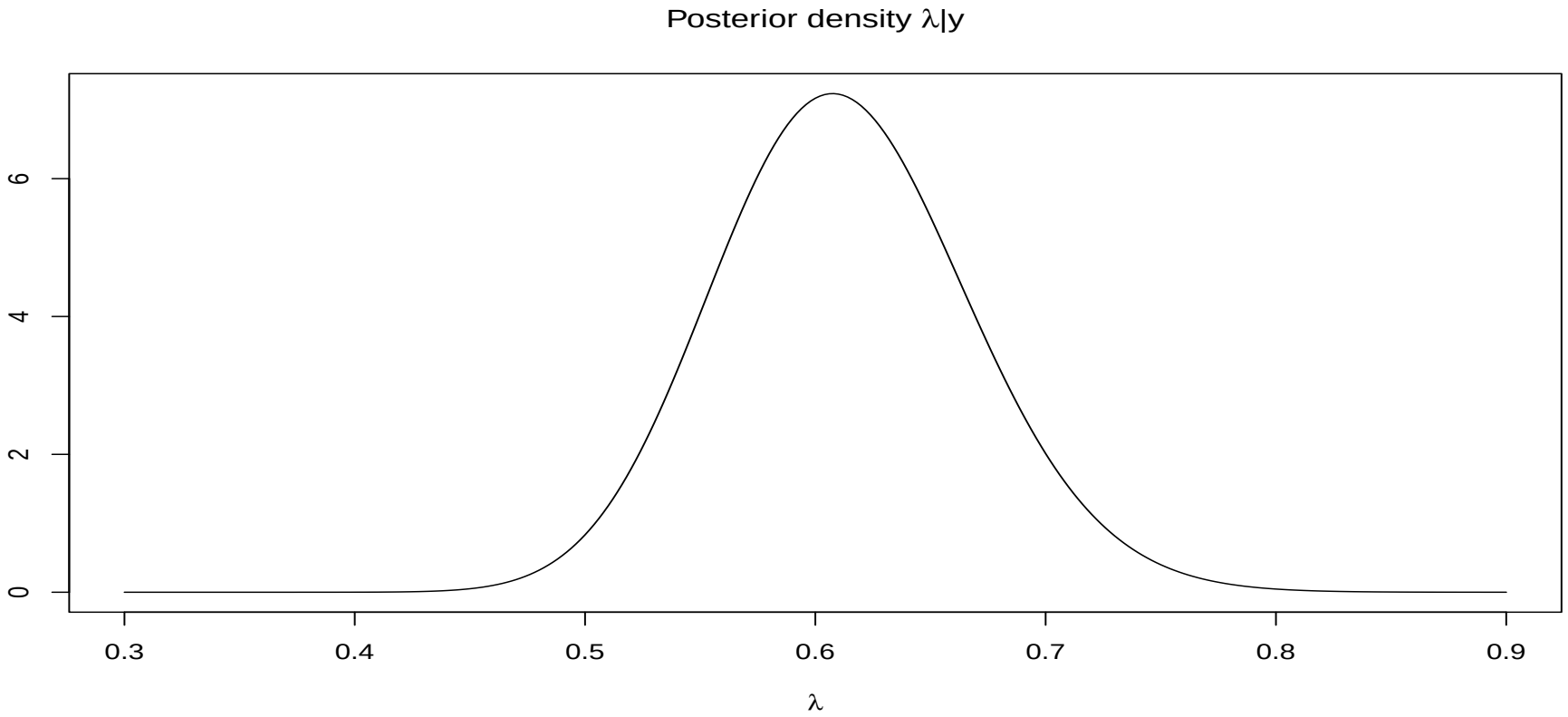
Posterior: $\lambda | \mathbf{Y} = \mathbf{y} \sim \text{Gamma}(122.5, 200)$

```
alpha <- 0.5
beta <- 0
n <- length(y.obs)
s <- sum(y.obs)
alpha.post <- alpha + s
beta.post <- beta + n
(post.mean <- alpha.post/beta.post)  ## posterior mean

## [1] 0.6125

(post.var <- alpha.post/beta.post^2)  ## posterior variance

## [1] 0.0030625

(post.sd <- sqrt(post.var))  ## posterior StdDev

## [1] 0.05533986
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

```
curve(dgamma(x, shape = alpha.post, rate = beta.post), from = 0.3, to = 0.9,
    n = 301, xlab = expression(lambda), ylab = "", main = expression("Posterior density " *
        lambda * "|y"))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

To use Jeffreys' prior, we have to use the following trick.

Let U be a non-negative continuous random variable with "uniform" distribution on $(-\infty, \infty)$, i.e.

$$f_U(u) \propto 1, \quad \text{for all } u \in \mathbb{R}$$

Then the distribution of

$$\lambda = u^2$$

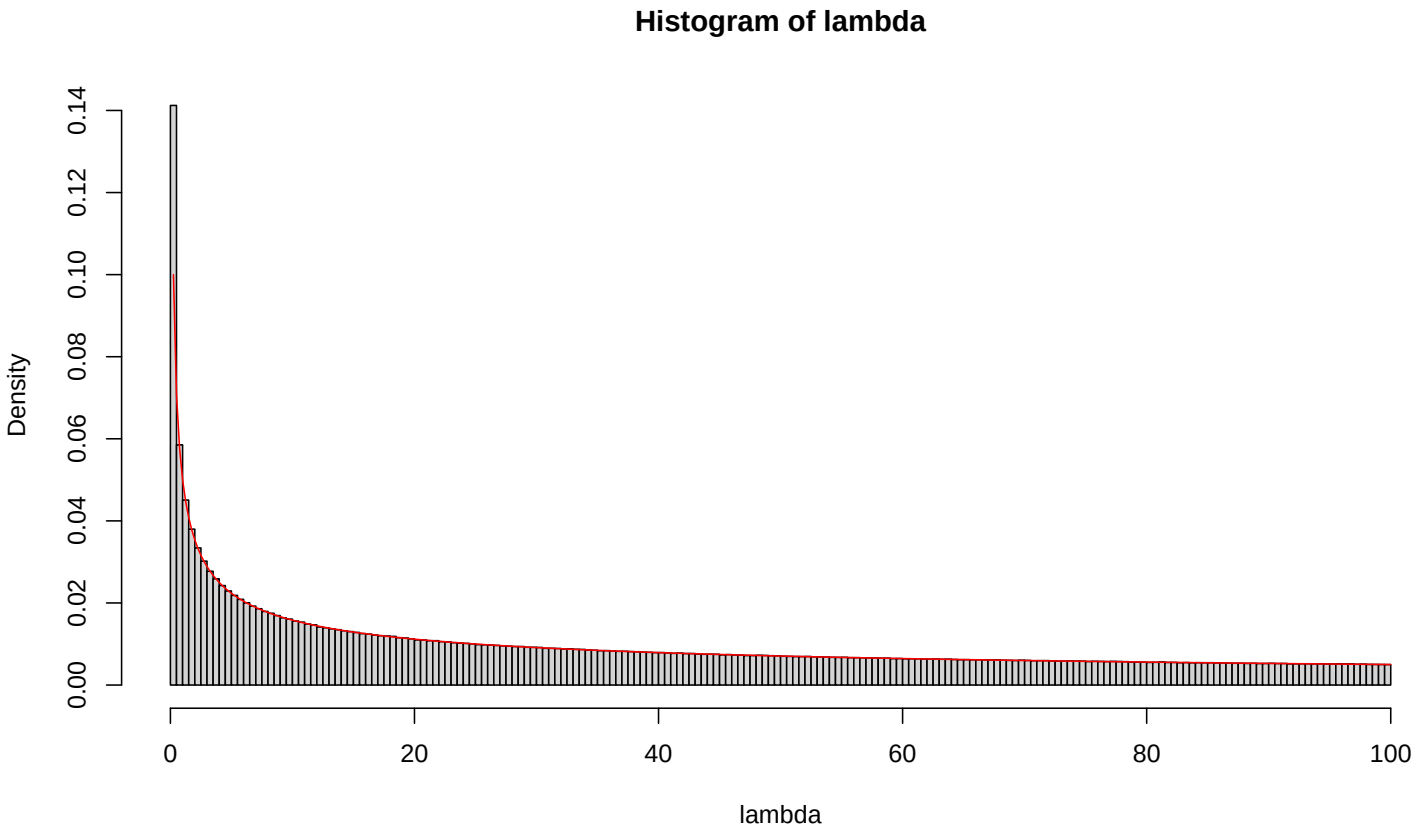
is

$$f(\lambda) \propto \lambda^{-1/2}$$

Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

```
u <- runif(1e+07, min = -10, max = 10)
lambda <- u^2
hist(lambda, probability = TRUE, breaks = 200)
curve(0.05 * 1/sqrt(x), from = 0, to = 100, n = 401, add = TRUE, col = "red")
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

Writing the BUGS code for this model is fairly simple:

```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dpois(lambda)
  }

  ## prior
  u ~ dflat()
  lambda <- pow(u, 2)
}
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

```
library(BRugs)
modelCheck("Poisson2.bug")

## model is syntactically correct

bugsData(list(y = y.obs, N = length(y.obs)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(u = 1), list(u = 2), list(u = 0.5), list(u = 1.5))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 3: Death by horse or mule kicks

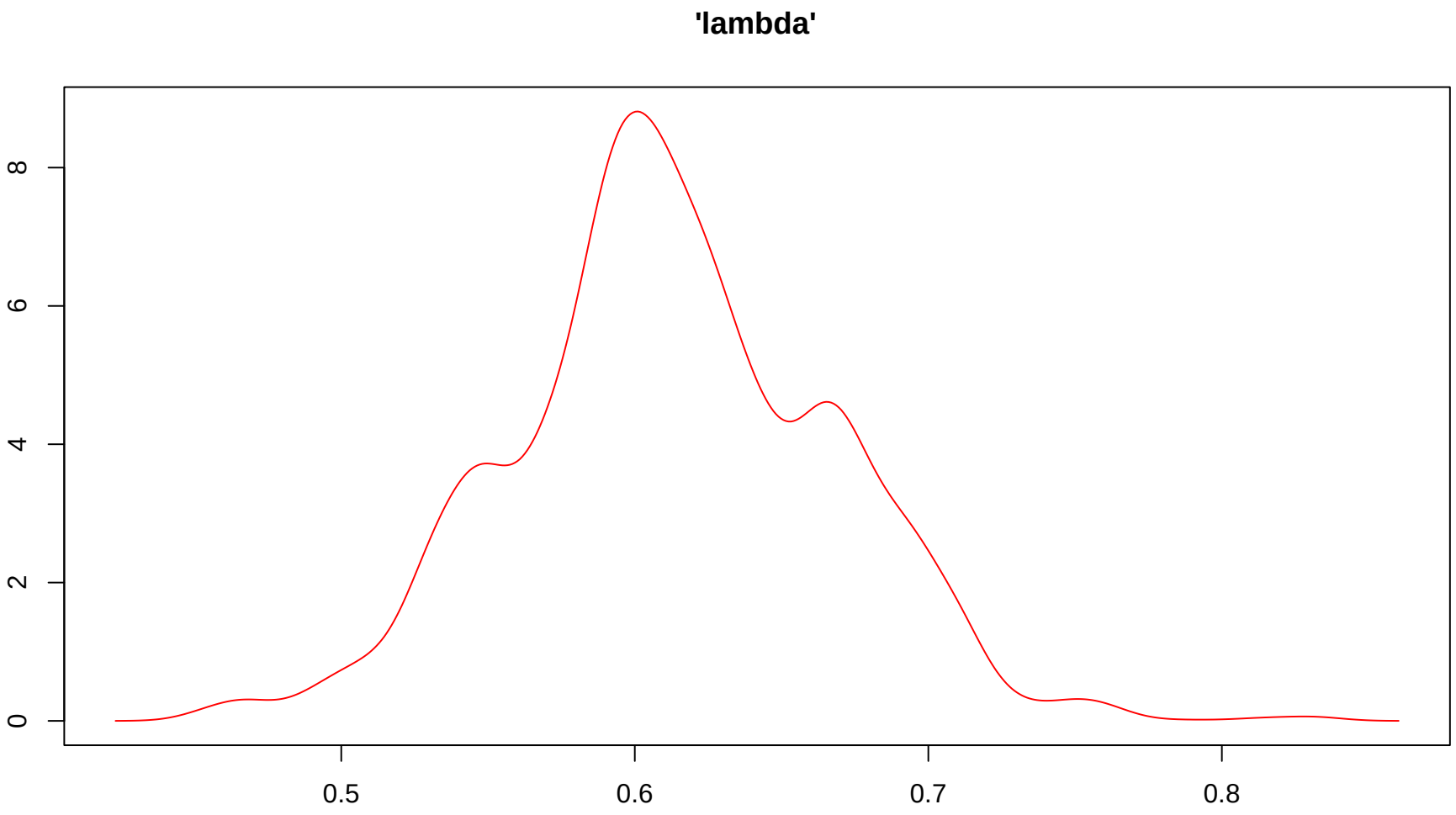
Bayesian analysis: using Jeffreys' prior (ctd.)

```
samplesSet("lambda")
## monitor set for variable 'lambda'
modelUpdate(3000)
## 3000 updates took 0 s
samplesStats("*", beg = 1001)
##          mean      sd MC_error val2.5pc median val97.5pc start sample
## lambda 0.6127 0.05404 0.002074   0.5089 0.6086    0.7114  1001   8000
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

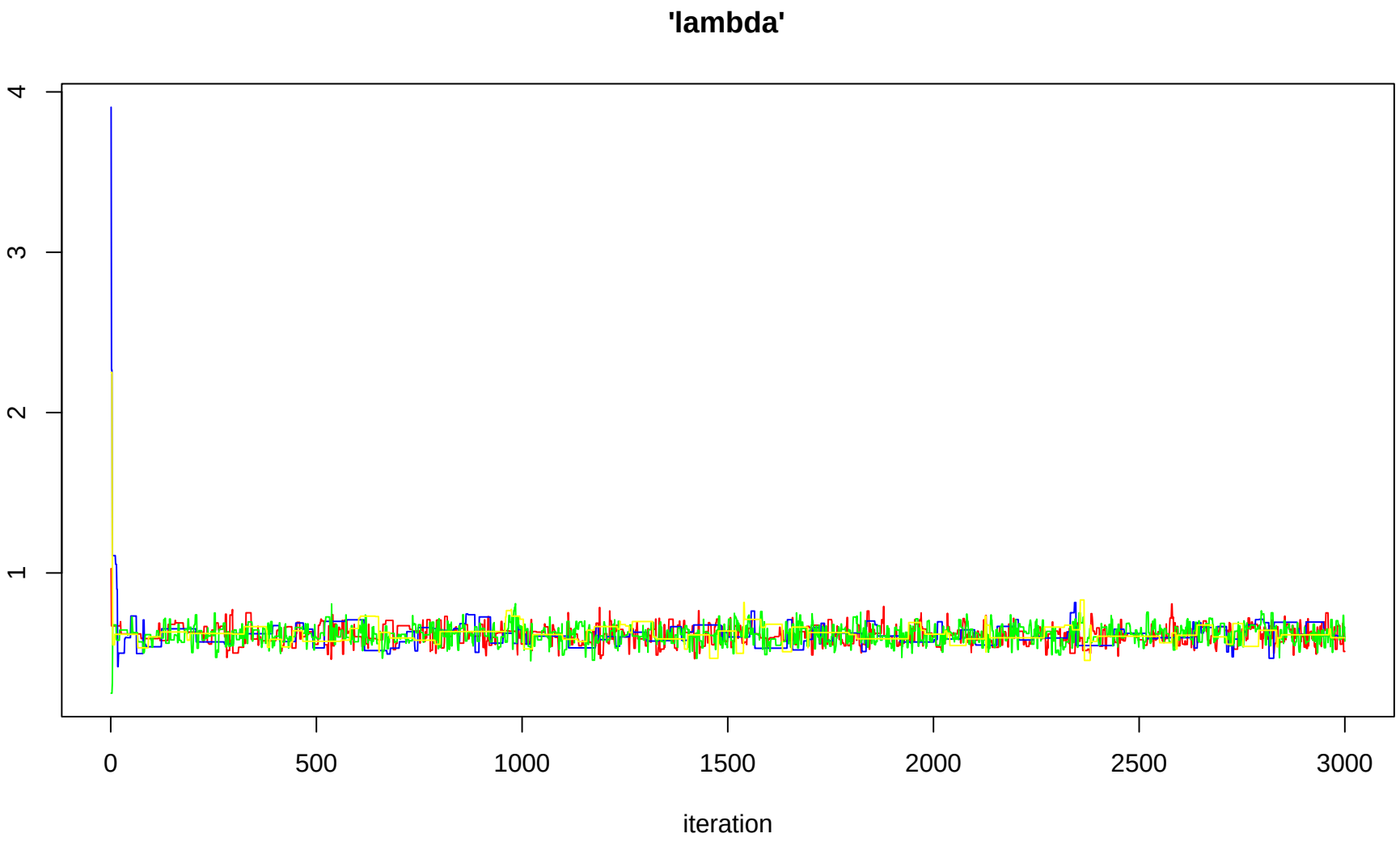
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

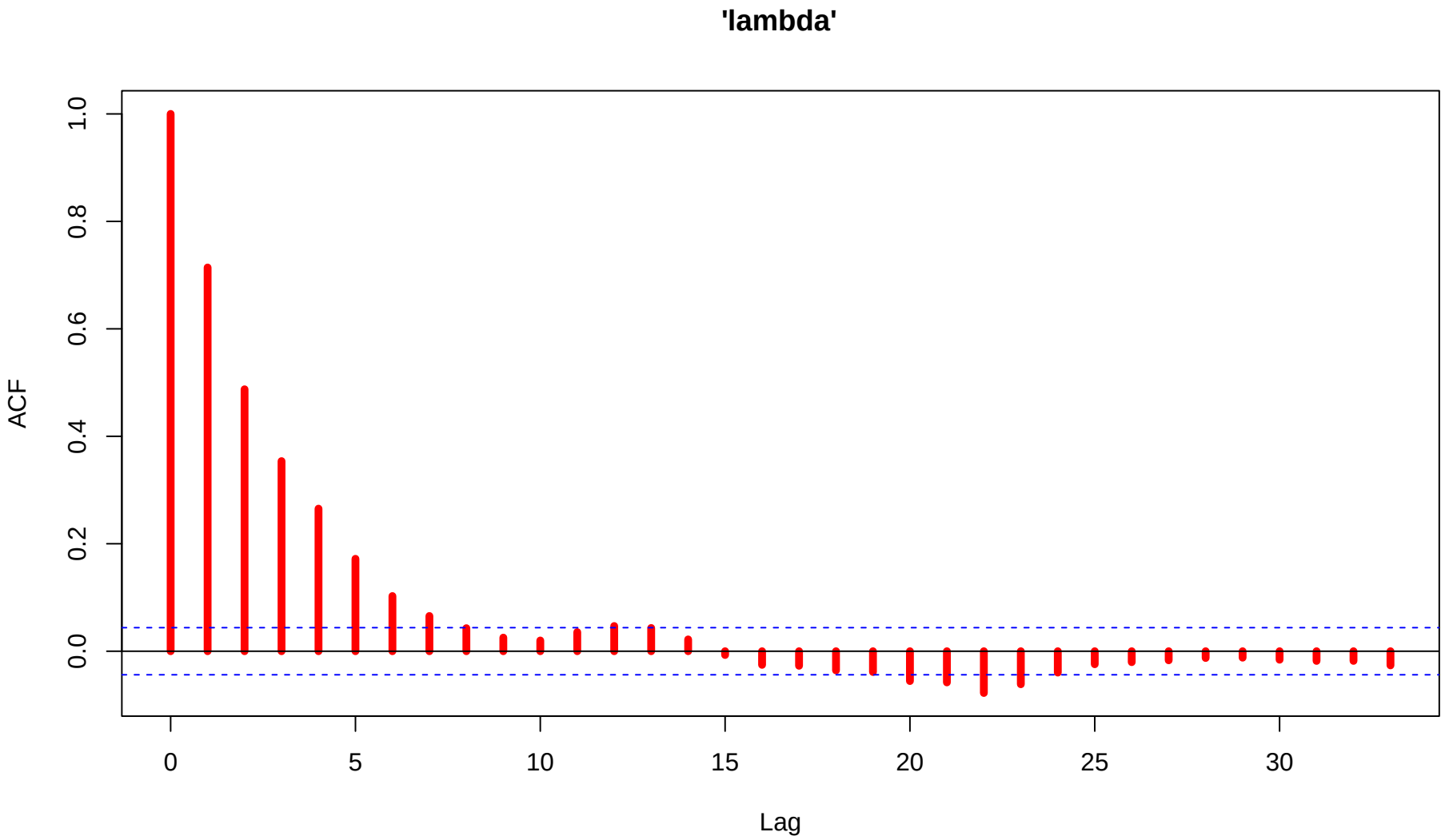
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

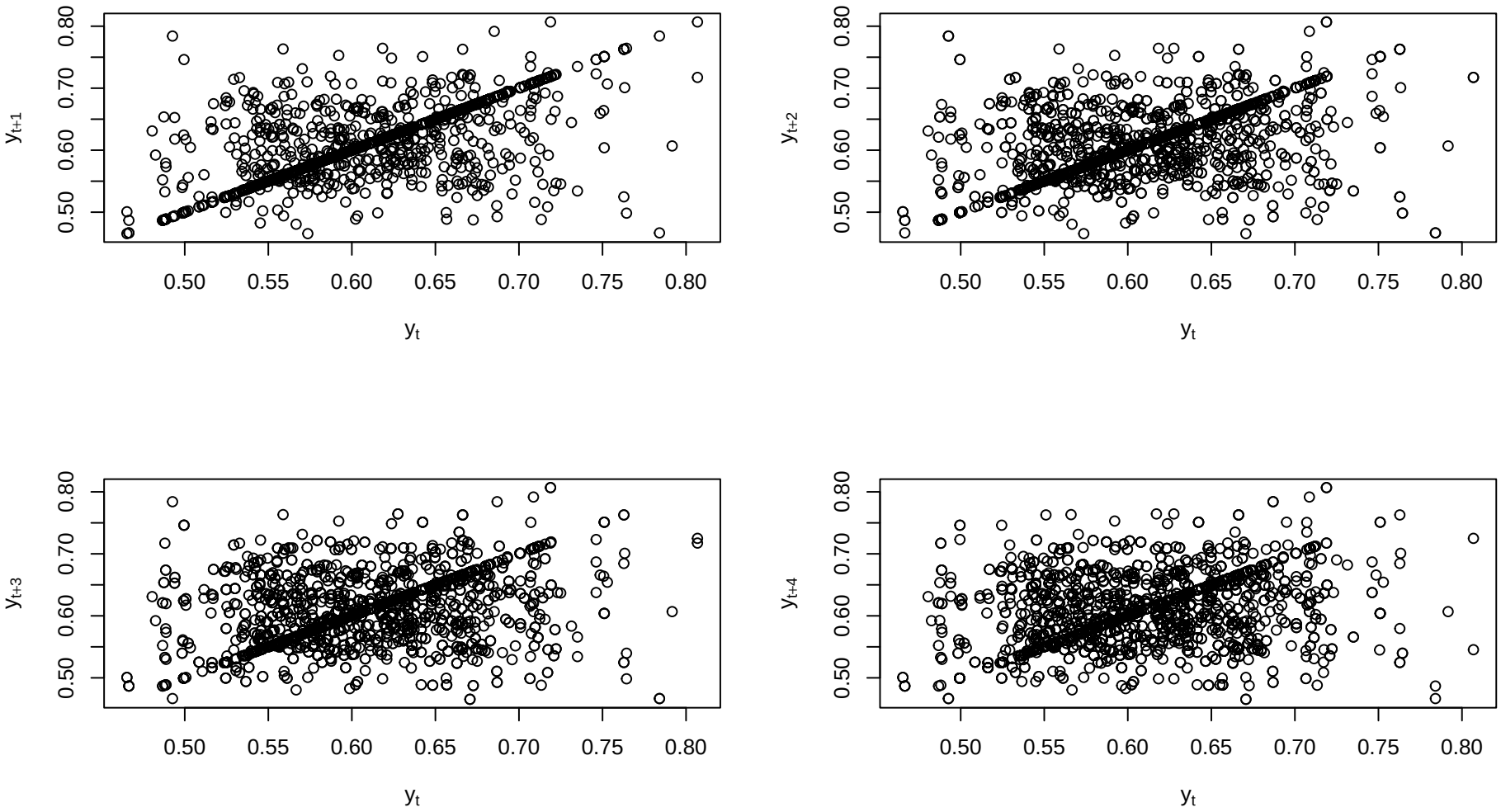
Bayesian analysis: using Jeffreys' prior (ctd.)

```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

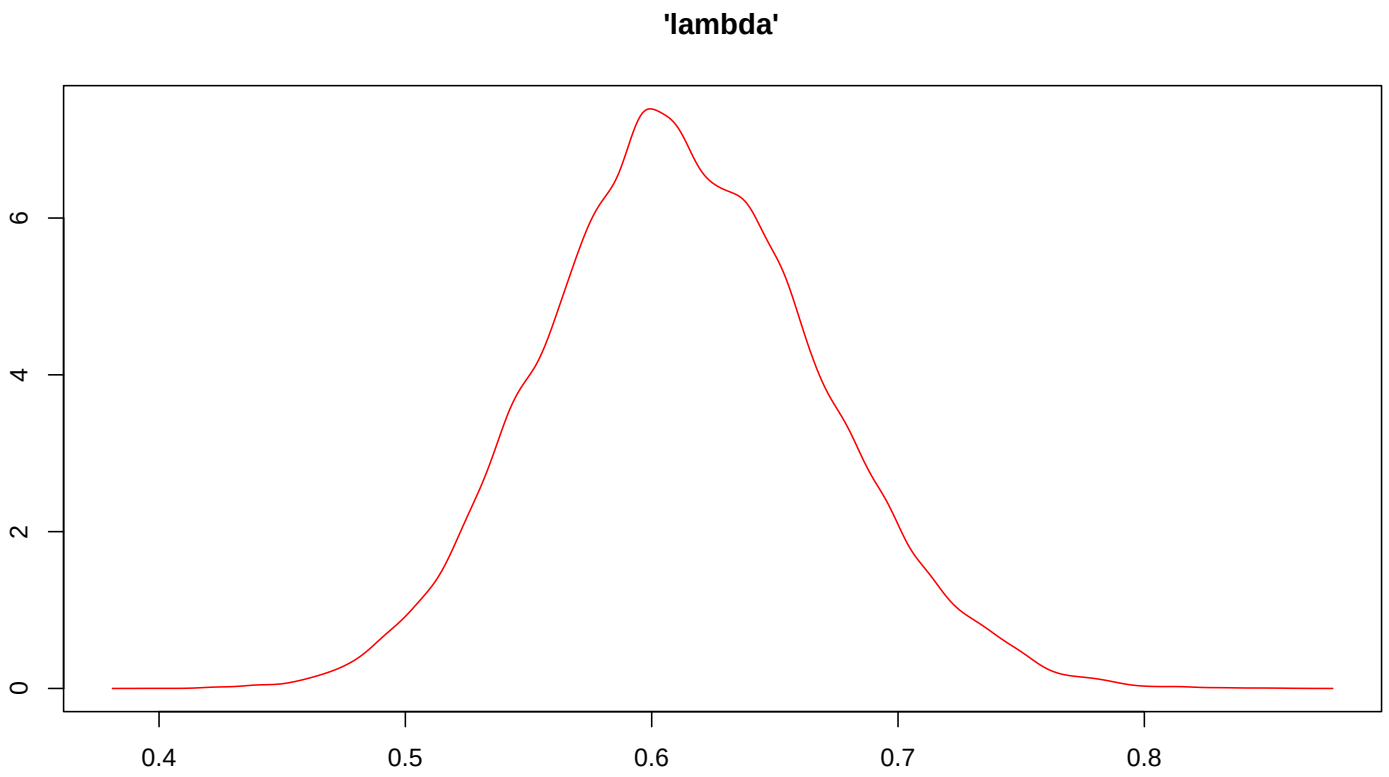
Bayesian analysis: using Jeffreys' prior (ctd.)



Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

```
modelUpdate(28000)

## 28000 updates took 3 s

samplesStats("*", beg = 1001)

##          mean      sd  MC_error val2.5pc median val97.5pc start
## lambda 0.6124 0.05605 0.0006574   0.5066 0.6103    0.7273  1001
##        sample
## lambda 120000

samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using Jeffreys' prior (ctd.)

Analysing the data on death by horse or mule kicks using Jeffreys' prior for λ leads to an estimate of $\hat{\lambda} = 0.61$ deaths per year (per corps).

This estimate is based on running four chains, each for 31 000 iterations and discarding the first 1 000 iterations as burn-in. Trace plots indicated that the chains had converged and were mixing well.

Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior

Writing the BUGS code for this model is fairly simple:

```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dpois(lambda)
  }

  ## prior
  lambda ~ dgamma(0.5, 0.00001)
}
```

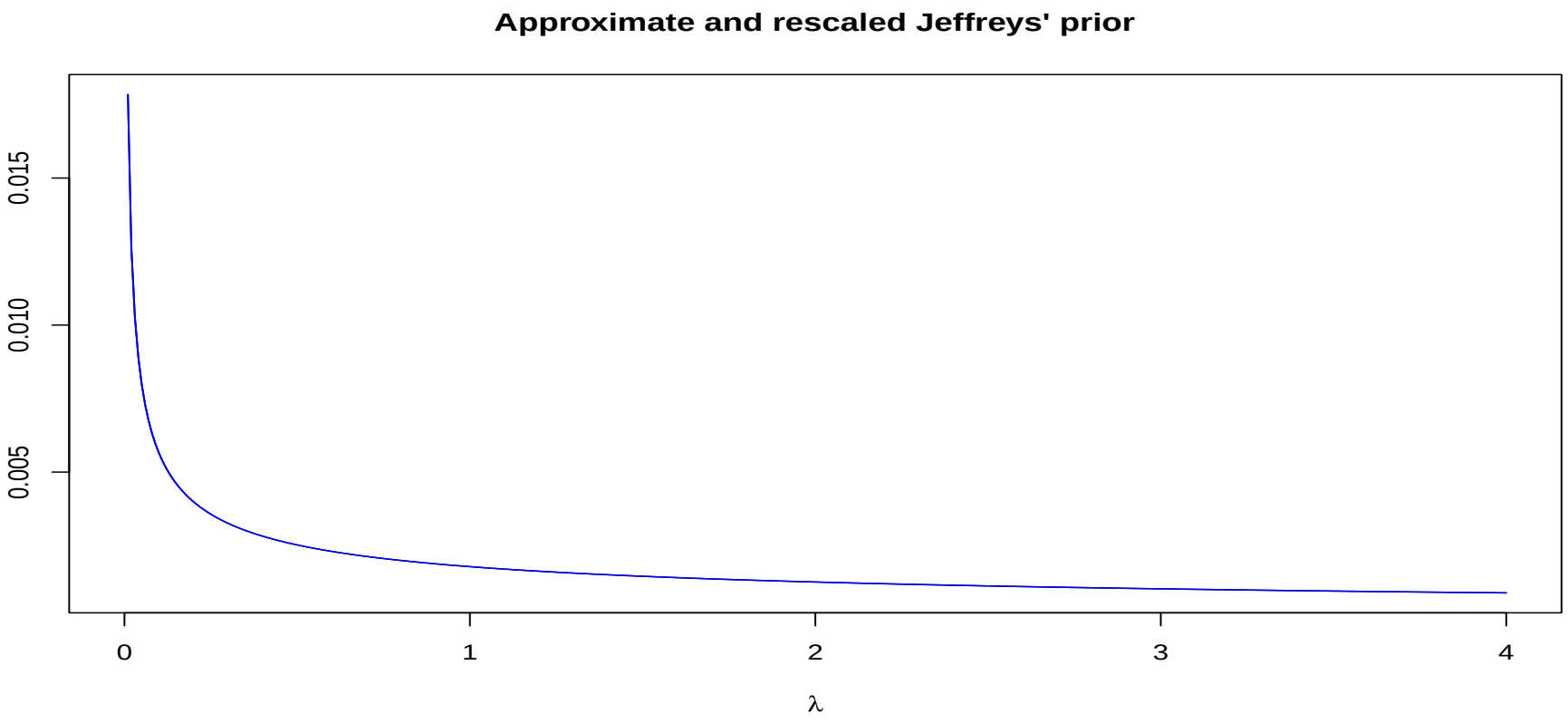
Lunn, D., Jackson, C., Best, N., Thomas, A. and Spiegelhalter, D. (2013). *The BUGS Book: A Practical Introduction to Bayesian Analysis*, CRC/Chapman & Hall, Chapter 5.2.6, p. 85ff

Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

Visually this approximate prior (black line) is indistinguishable from the (rescaled) Jeffreys' prior (blue line):

```
curve(dgamma(x, 0.5, 1e-05), from = 0, to = 4, n = 401, xlab = expression(lambda),
    ylab = "", main = "Approximate and rescaled Jeffreys' prior")  ## approximate prior
curve(1/sqrt(x) * 1e-05^0.5/gamma(0.5), from = 0, to = 4, n = 401, add = TRUE,
    col = "blue")
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
library(BRugs)
modelCheck("Poisson3.bug")

## model is syntactically correct

bugsData(list(y = y.obs, N = length(y.obs)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(lambda = 1), list(lambda = 2), list(lambda = 3), list(lambda = 4))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 3: Death by horse or mule kicks

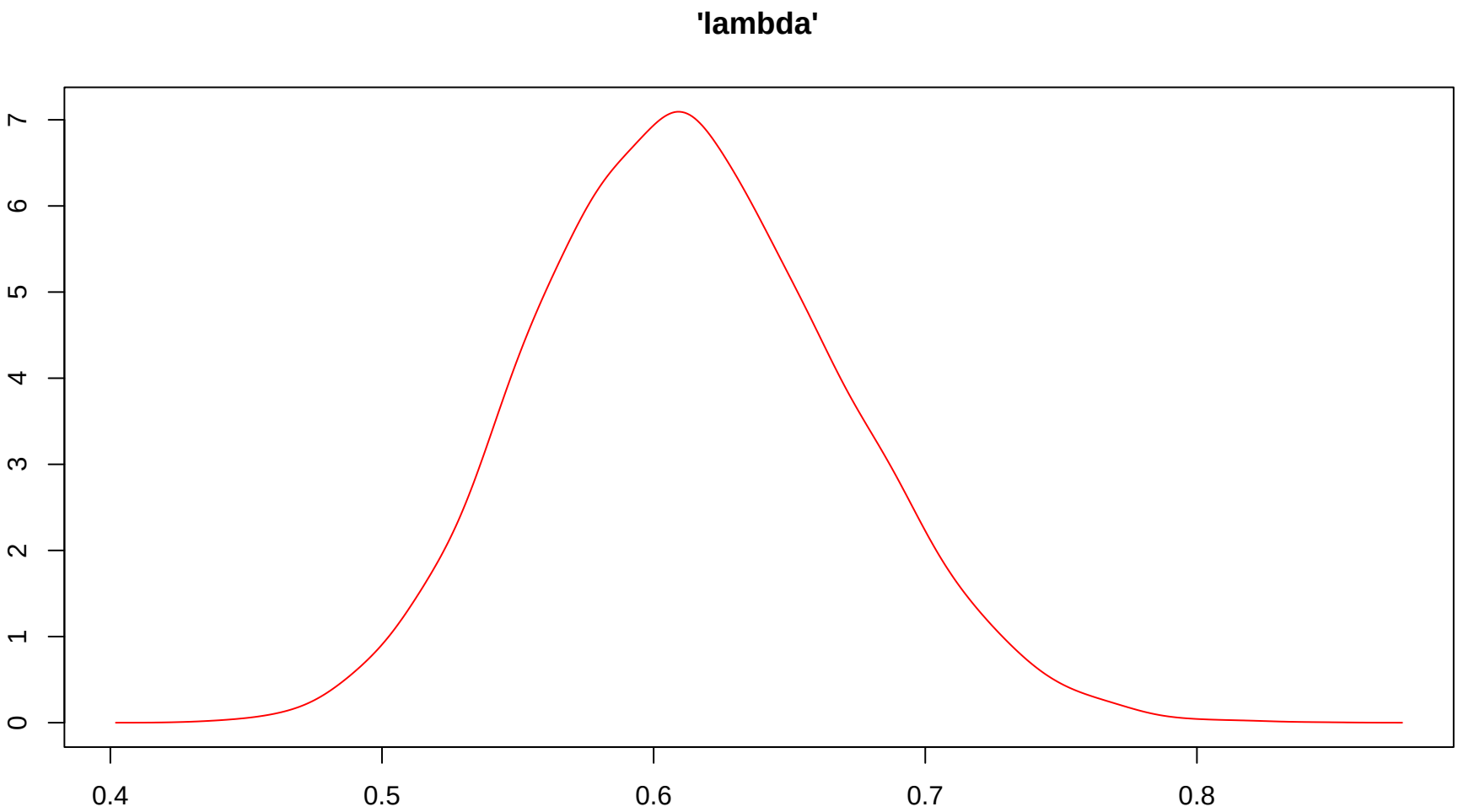
Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
samplesSet("lambda")
## monitor set for variable 'lambda'
modelUpdate(3000)
## 3000 updates took 0 s
samplesStats("*", beg = 1001)
##         mean      sd  MC_error val2.5pc median val97.5pc start sample
## lambda 0.613 0.05604 0.0005537   0.5101 0.6108    0.7276  1001   8000
```

Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

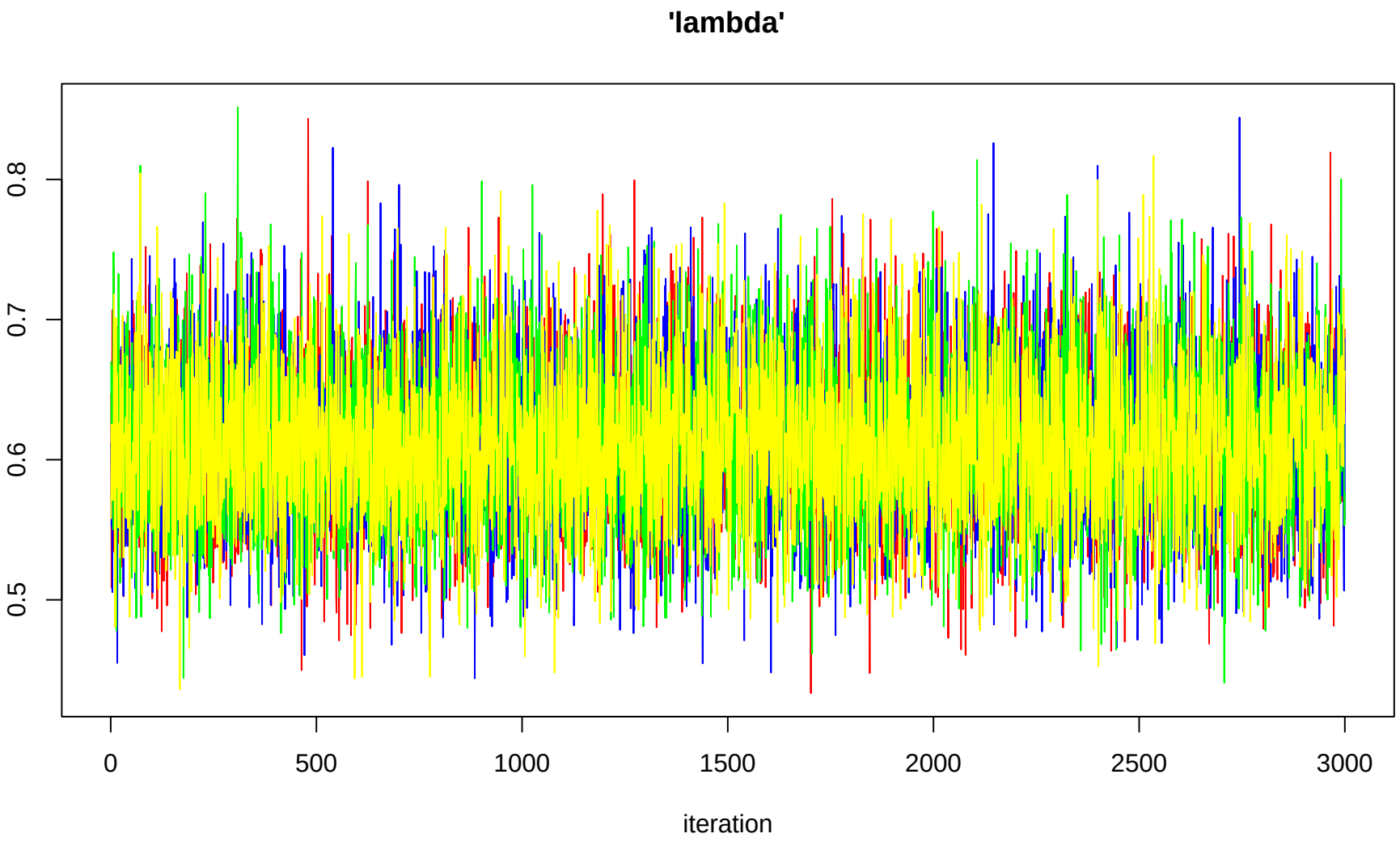
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

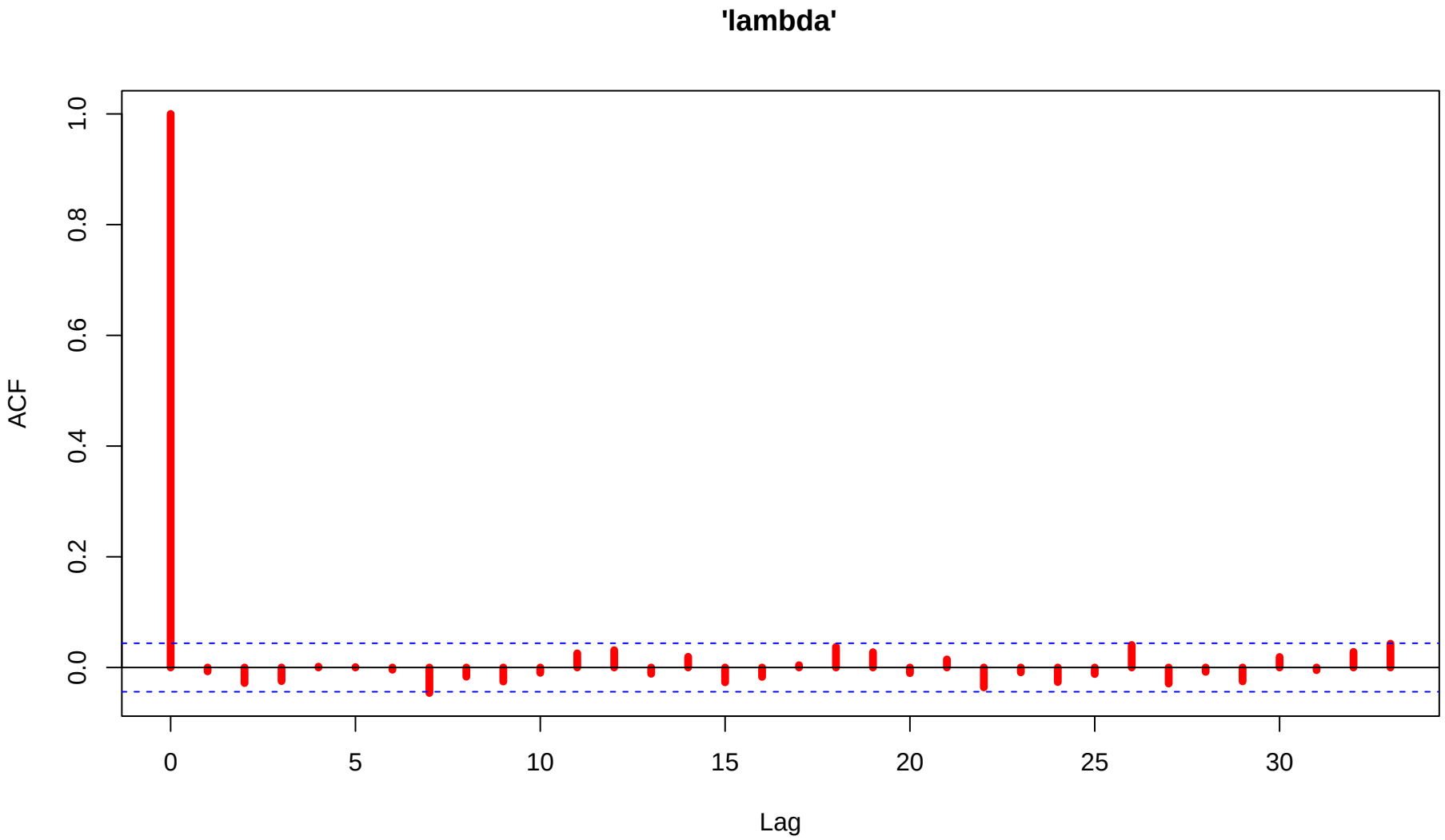
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

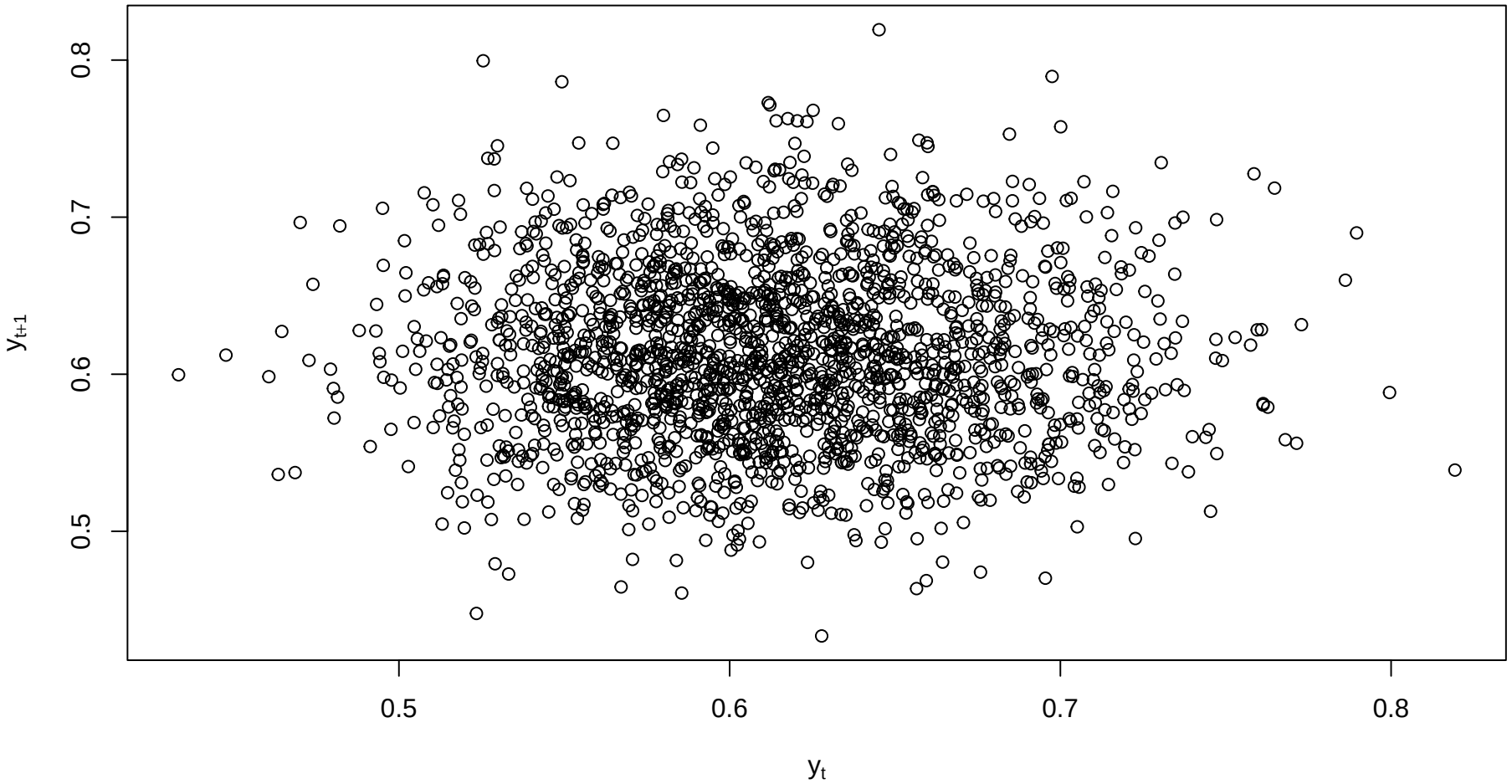
Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

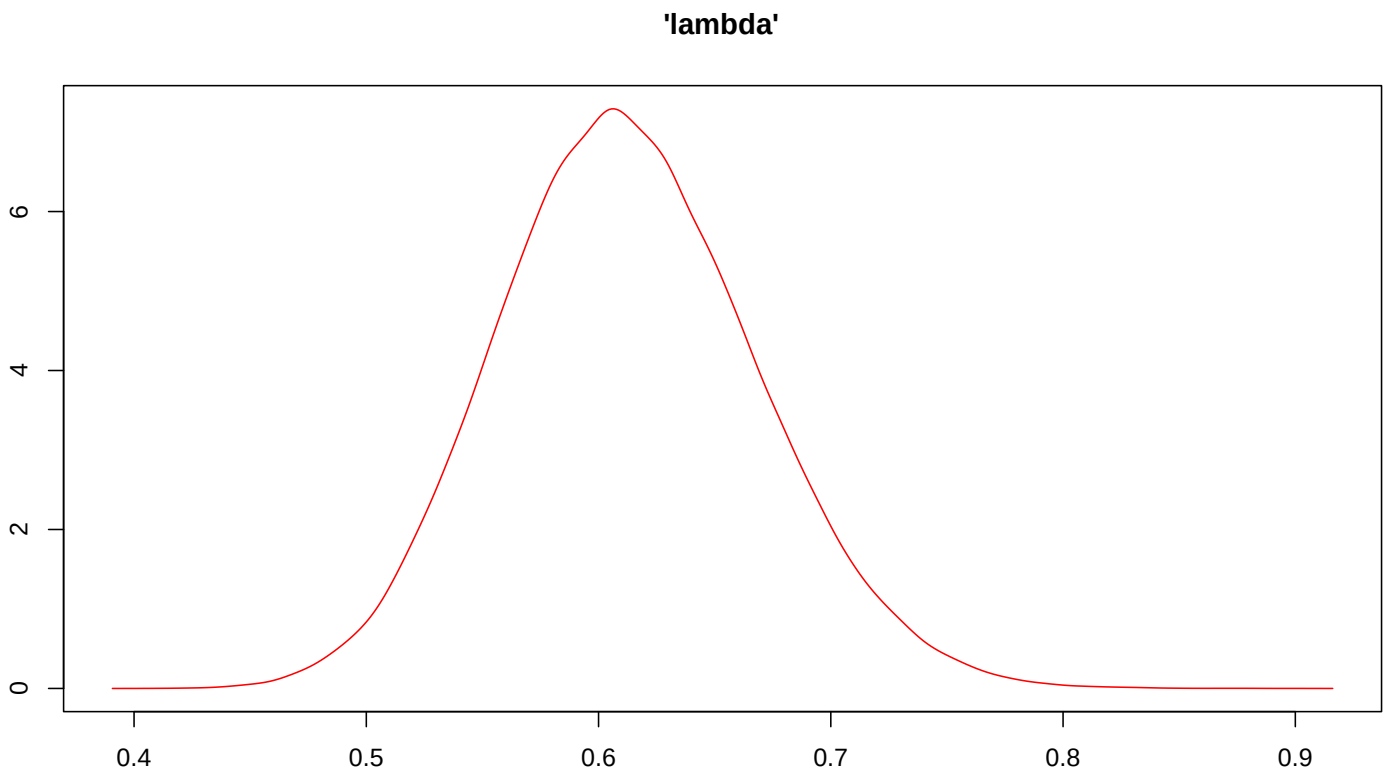
Bayesian analysis: using an approximate Jeffreys' prior (ctd.)



Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
modelUpdate(28000)

## 28000 updates took 0 s

samplesStats("*", beg = 1001)

##          mean      sd  MC_error val2.5pc median val97.5pc start
## lambda 0.6124 0.05544 0.0001569   0.5091 0.6105     0.726  1001
##        sample
## lambda 120000

samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 3: Death by horse or mule kicks

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

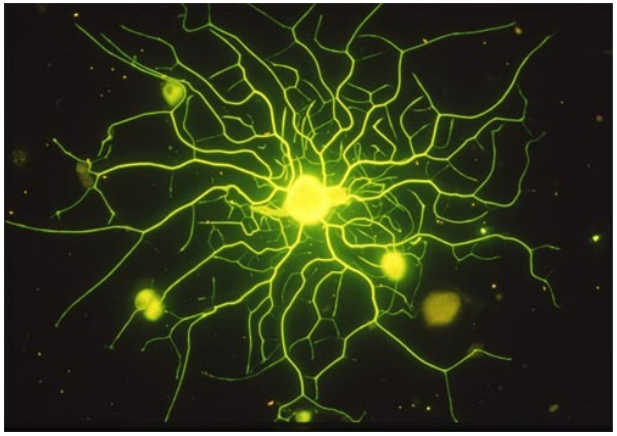
Analysing the data on death by horse or mule kicks using a gamma prior with parameters $\alpha = 0.5$ and $\beta = 10^{-5}$, which can be viewed as an approximation to Jeffreys' prior (over a wide range of values of the parameter), for λ leads to an estimate of $\hat{\lambda} = 0.612$ deaths per year (per corps).

This estimate is based on running four chains, each for 31 000 iterations and discarding the first 1 000 iterations as burn-in. Trace plots indicated that the chains had converged and were mixing well.

Example 4: Nerve Pulses

Exponential data

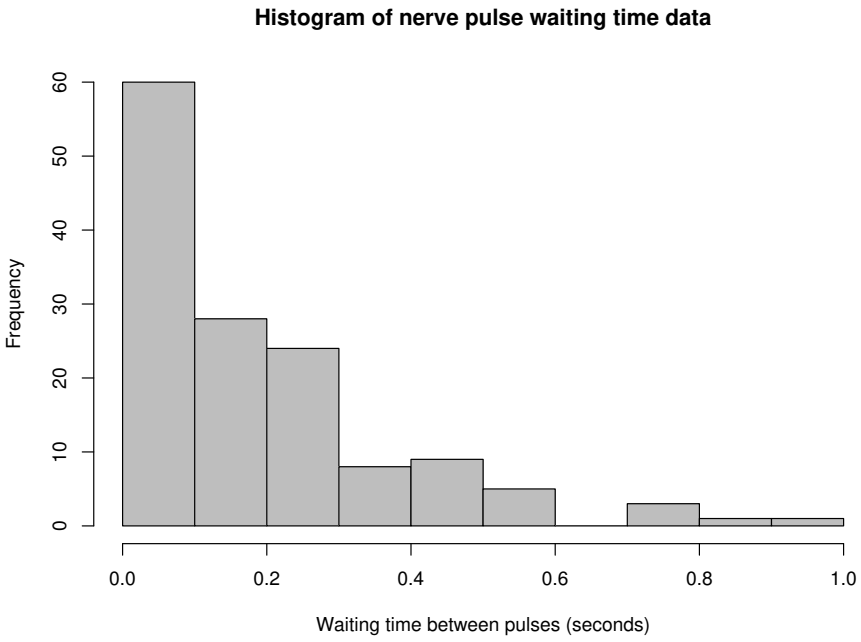
- Sir Bernard Katz was a Nobel prize winning biophysicist, noted for his work on nerve biochemistry.
- In his work with Dr P. Fatt, Professor Katz looked at pulses along nerve fibres in frogs.
- This case study concerns waiting time (in seconds) recorded between 140 consecutive pulses.



Example 4: Nerve Pulses

Exponential data (ctd.)

- What is an appropriate statistical model for these data?
- Can we estimate the true arrival rate of the pulses, and quantify the precision of our estimate?



Data source: Fatt, P., & Katz, B. (1952). Spontaneous subthreshold activity at motor nerve endings. *Journal of Physiology*, **117**, 109–128.

Exponential distribution

Definition 42

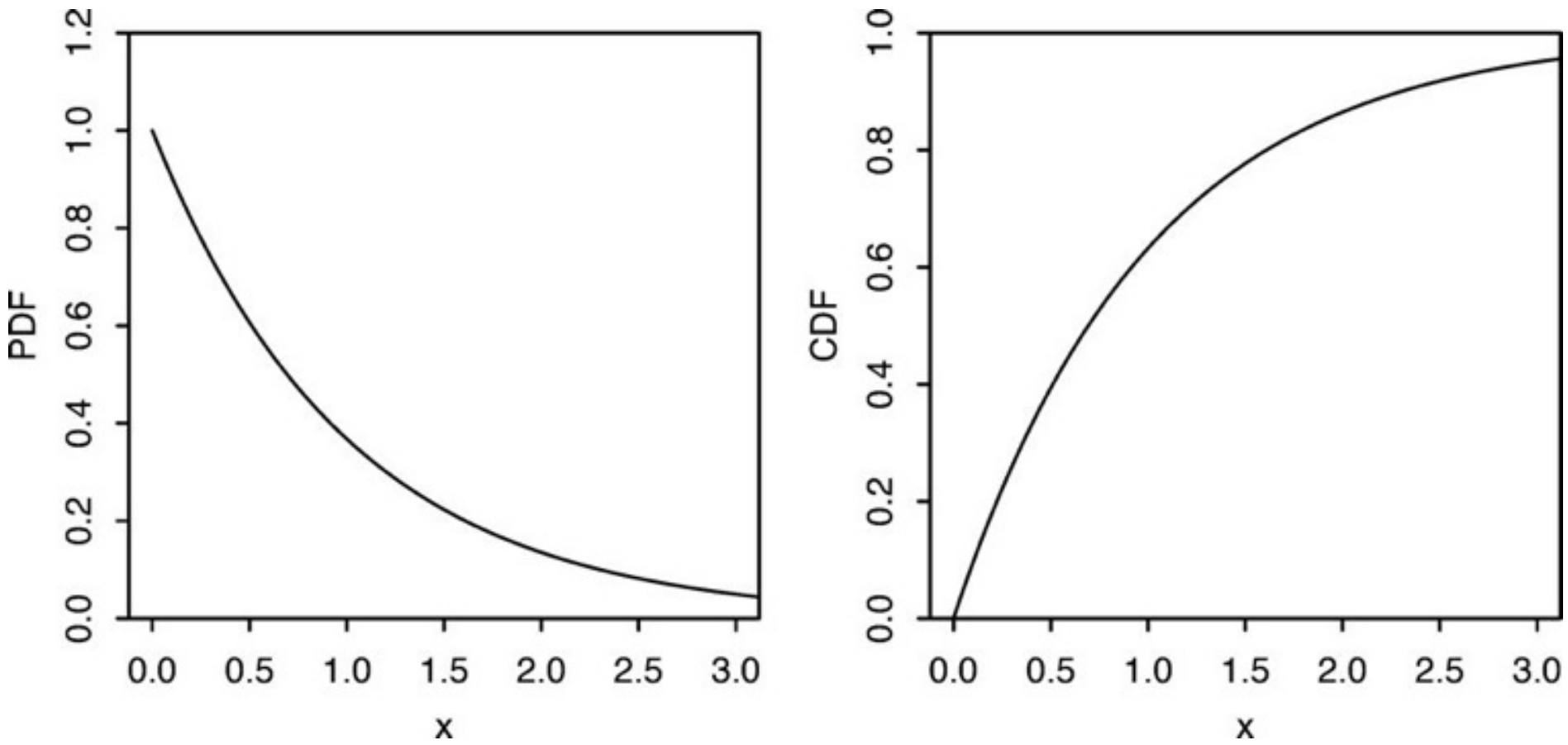
A continuous r.v. X is said to have the *Exponential distribution* with parameter λ if its PDF is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0, \\ 0 & \text{otherwise} \end{cases}$$

We denote this by $X \sim \mathrm{Expo}(\lambda)$.
The corresponding CDF is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Exponential distribution (ctd.)



Expo(1) PDF and CDF

Exponential distribution (ctd.)

Proposition 34

If $X \sim \mathrm{Expo}(1)$ *then* $Y = \frac{X}{\lambda} \sim \mathrm{Expo}(\lambda)$.

Example 4: Nerve Pulses

Exponential data: the likelihood

We view the observations $y_1, y_2, \ldots, y_{139}$ as realisations of random variables $Y_1, Y_2, \ldots, Y_{139}$

A typical starting model for waiting times between events is the exponential distribution, i.e. we model $Y_i \sim \mathrm{Expo}(\lambda)$

We assume that λ is fixed for all observations, and that given λ the random variables are independent of each other

The $\mathrm{Expo}(\lambda)$ distribution has PDF

$$f(y|\lambda) = \begin{cases} \lambda e^{-\lambda y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Example 4: Nerve Pulses

Exponential data: the likelihood (ctd.)

Hence, the joint distribution of $\mathbf{Y}$, $\mathbf{Y} = (Y_1, \ldots, Y_{139})$, is

$$\begin{aligned} f(\mathbf{y}|\lambda) &= \prod_{i=1}^{139} f(y_i|\lambda) = \prod_{i=1}^{139} \lambda \exp(-\lambda y_i) \\ &= \lambda^{139} \exp\left(-\lambda \sum_{i=1}^{139} y_i\right) \\ &= \lambda^n \exp\left(-s\lambda\right) \end{aligned}$$

where $n = 139$ is the *sample size* and $s = \sum_{i=1}^{n} y_i$

Whence the likelihood is

$$L(\lambda|\mathbf{y}) = \lambda^n \exp\left(-s\lambda\right)$$

Example 4: Nerve Pulses

Exponential data: classical analysis

```
library(stats4)
y <- scan("nerve.txt")
nll <- function(lambda) -sum(dexp(y, rate = lambda, log = TRUE))
(res <- mle(nll, start = list(lambda = 2)))
```

```
##
## Call:
## mle(minuslogl = nll, start = list(lambda = 2))
##
## Coefficients:
##   lambda
## 5.089711
```

```
summary(res)
```

```
## Maximum likelihood estimation
##
## Call:
## mle(minuslogl = nll, start = list(lambda = 2))
##
## Coefficients:
##        Estimate Std. Error
## lambda 5.089711  0.4317037
##
## -2 log L: -174.3674
```

Example 4: Nerve Pulses

Exponential data: classical analysis (ctd.)

Using the log likelihood:

$$l(\lambda|\mathbf{y}) = \log(L(\lambda|\mathbf{y})) = n\,\log(\lambda) - s\lambda$$

it is (easy) to show that the maximum-likelihood estimate is

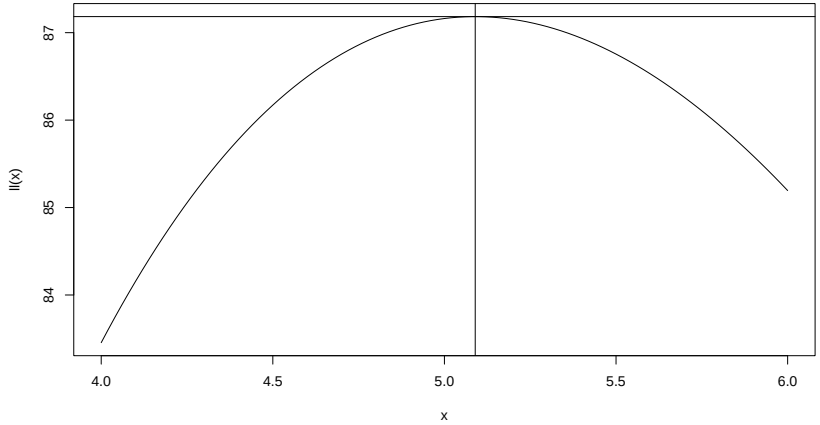
$$\hat{\lambda} = \frac{n}{s} = \frac{n}{\sum_{i=1}^{n} y_i} = \frac{1}{\bar{y}}$$

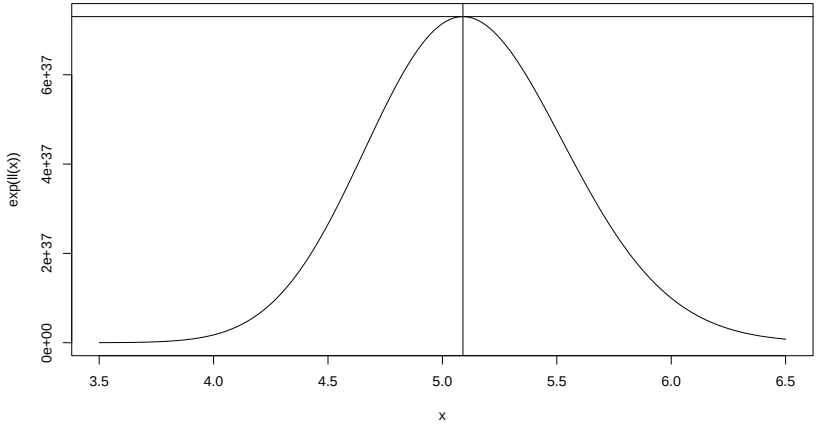
```
1/mean(y)
## [1] 5.089711
```

Example 4: Nerve Pulses

Exponential data: classical analysis (ctd.)

```
ll <- function(lambda) sapply(lambda, function(x) -nll(x))
lambdahat <- 1/mean(y)
curve(ll, from = 4, to = 6)
abline(v = lambdahat)
abline(h = ll(lambdahat))
curve(exp(ll(x)), from = 3.5, to = 6.5, n = 401)
abline(v = lambdahat)
abline(h = exp(ll(lambdahat)))
```





Example 4: Nerve Pulses

Exponential data: how to choose the prior?

With an exponential sampling model for $\mathbf{Y}|\lambda$, $s = \sum_{i=1}^{n} y_i$ and a $\text{Gamma}(\alpha, \beta)$ prior for λ, we have:

$$f(\lambda|\mathbf{y}) \propto \underbrace{\prod_{i=1}^{n} \lambda e^{-\lambda y_i}}_{\text{Likelihood: } L(\lambda|\mathbf{y}) = f(\mathbf{y}|\lambda)} \underbrace{\frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}}_{\text{prior: } f(\lambda)}$$

$$\propto \underbrace{\lambda^n e^{-\lambda s}}_{\text{“essential” part of Likelihood}} \underbrace{\lambda^{\alpha-1} e^{-\beta\lambda}}_{\text{“essential” part of prior}}$$

$$\propto \lambda^{\alpha+n-1} e^{-(\beta+s)\lambda}$$

Example 4: Nerve Pulses

Exponential data: how to choose the prior? (ctd.)

From which we can conclude that

$$f(\lambda|\mathbf{y}) = \frac{(\beta+s)^{\alpha+n}}{\Gamma(\alpha+n)}\lambda^{\alpha+n-1}e^{-(\beta+s)\lambda}$$

That is $\lambda|\mathbf{Y} = \mathbf{y} \sim \mathrm{Gamma}(\alpha+n, \beta+s)$

This shows that the family of gamma distributions is the *conjugate* prior for the exponential distribution

Example 4: Nerve Pulses

Exponential data: Bayesian point estimation?

Traditionally, the posterior expectation of the parameter is used as a point estimate in Bayesian analysis

$$\hat{\lambda} = \mathbb{E}[\lambda | \mathbf{Y} = \mathbf{y}]$$

$$= \int_0^\infty \lambda \, f(\lambda | \mathbf{Y} = \mathbf{y}) \, d\lambda$$

$$= \int_0^\infty \lambda \, \frac{(\beta + s)^{\alpha+n}}{\Gamma(\alpha + n)} \lambda^{\alpha+n-1} e^{-(\beta+s)\lambda} \, d\lambda$$

Example 4: Nerve Pulses

Exponential data: Bayesian point estimation? (ctd.)

Here

$$\begin{aligned}\hat{\lambda} = \mathbb{E}[\lambda|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + n}{\beta + s} = \frac{\alpha}{\beta + s} + \frac{n}{\beta + s} \\ &= \frac{\beta}{\beta + s}\frac{\alpha}{\beta} + \frac{s}{\beta + s}\frac{n}{s} \\ &= (1 - \kappa) \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}} + \kappa \underbrace{\frac{1}{\bar{y}}}_{\text{MLE}}\end{aligned}$$

with $0 \leq \kappa = \frac{s}{\beta+s} \leq 1$

That is, the Bayesian estimate $\hat{\lambda}$ is a weighted average between the prior mean and the maximum likelihood estimate.

Example 4: Nerve Pulses

Exponential data: summary

Sampling model for data (likelihood):

$$Y_i|\lambda \sim \mathrm{Expo}(\lambda) \quad i = 1, \ldots, n, \text{ independent}$$

Prior for the parameter:

$$\lambda \sim \mathrm{Gamma}(\alpha, \beta)$$

Posterior of the parameter ($s = \sum_{i=1}^{n} y_i$):

$$\lambda|\mathbf{Y} = \mathbf{y} \sim \mathrm{Gamma}(\alpha + n, \beta + s)$$

Point estimate $\hat{\lambda}$ for λ:

$$\begin{aligned}\hat{\lambda} = \mathbb{E}[\lambda|\mathbf{Y} = \mathbf{y}] &= \frac{\alpha + n}{\beta + s} \\ &= \frac{\beta}{\beta + s} \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}} + \frac{s}{\beta + s} \underbrace{\frac{1}{\bar{y}}}_{\text{MLE}}\end{aligned}$$

Example 4: Nerve Pulses

Exponential data: how to choose the prior? (ctd.)

One question remains:

How to choose the *hyper-parameters* α and β?

Various proposal exists (all considered *non-informative*):

- *Uniform/flat prior* : $\alpha = 1, \beta = 0$
$$f(\lambda) \propto 1, \quad \text{if } \lambda > 0$$
This is an *improper* prior, i.e. $\int_0^\infty f(\lambda)\, d\lambda = \infty$.
- *Jeffreys' prior* : $\alpha = 0, \beta = 0$
$$f(\lambda) \propto \lambda^{-1}, \quad \text{if } \lambda > 0$$
This is an *improper* prior, i.e. $\int_0^\infty f(\lambda)\, d\lambda = \infty$.

Example 4: Nerve Pulses

Bayesian analysis: Using a flat prior

Prior: $\lambda \sim$ "$\mathrm{Gamma}(1, 0)$"

Data: $n = 139$ observations from $\mathrm{Expo}(\lambda)$

$s = \sum_{i=1}^{n} y_i = 27.31$

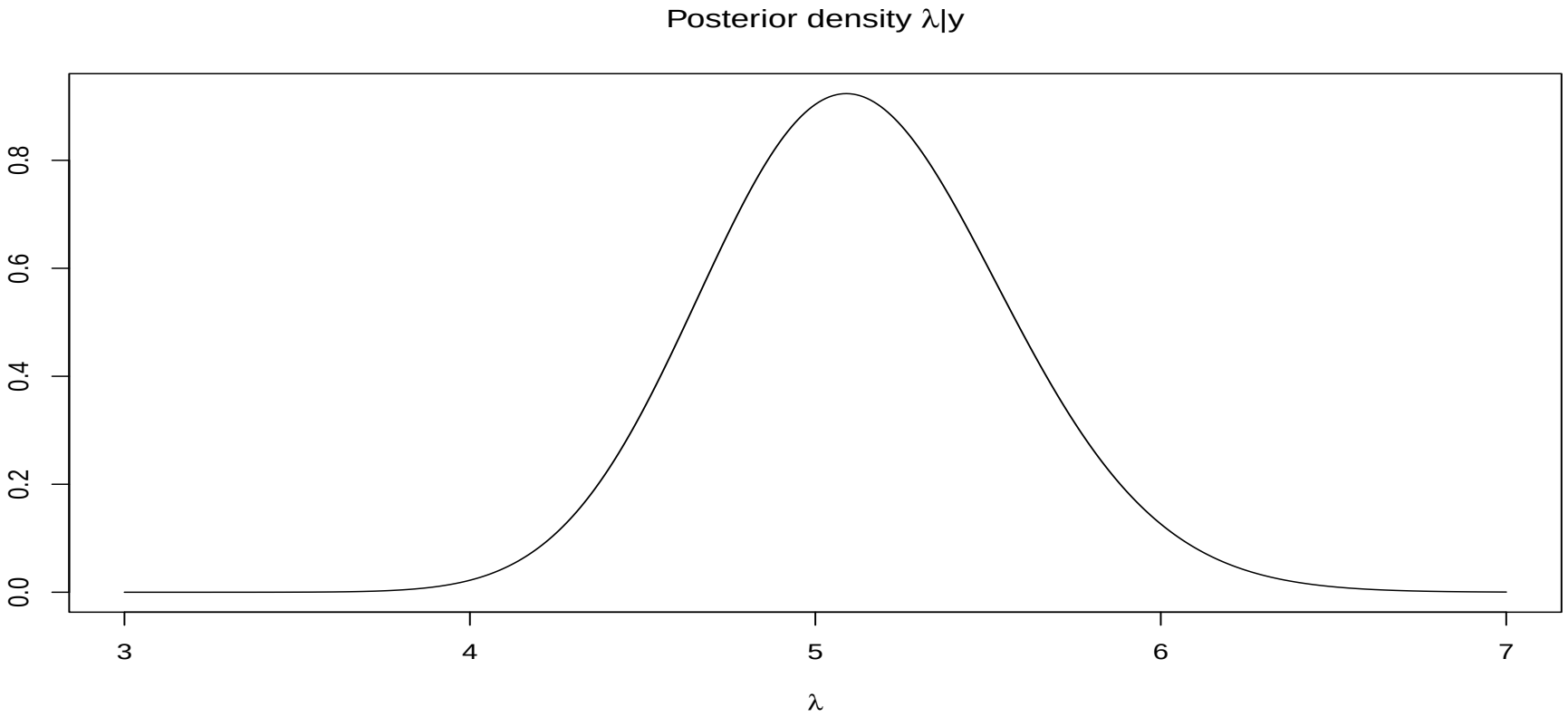
Posterior: $\lambda | \mathbf{Y} = \mathbf{y} \sim \mathrm{Gamma}(140, 27.31)$

```
alpha <- 1
beta <- 0
n <- length(y)
s <- sum(y)
alpha.post <- alpha + n
beta.post <- beta + s
(post.mean <- alpha.post/beta.post)  ## posterior mean

## [1] 5.126327

(post.var <- alpha.post/beta.post^2)  ## posterior variance

## [1] 0.1877088

(post.std <- sqrt(post.var))  ## posterior StdDev

## [1] 0.4332537
```

Example 4: Nerve Pulses

Bayesian analysis: Using a flat prior (ctd.)

```
curve(dgamma(x, shape = alpha.post, rate = beta.post), from = 3, to = 7,
    n = 401, xlab = expression(lambda), ylab = "", main = expression("Posterior density " *
        lambda * "|y"))
```



Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

Writing the BUGS code for this model is fairly simple:

```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dexp(lambda)
  }

  ## prior
  u ~ dflat()
  lambda <- abs(u)
}
```

Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

```
library(BRugs)
modelCheck("Exponential1.bug")

## model is syntactically correct

bugsData(list(y = y, N = length(y)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(u = 1), list(u = 2), list(u = 0.5), list(u = 3))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 4: Nerve Pulses

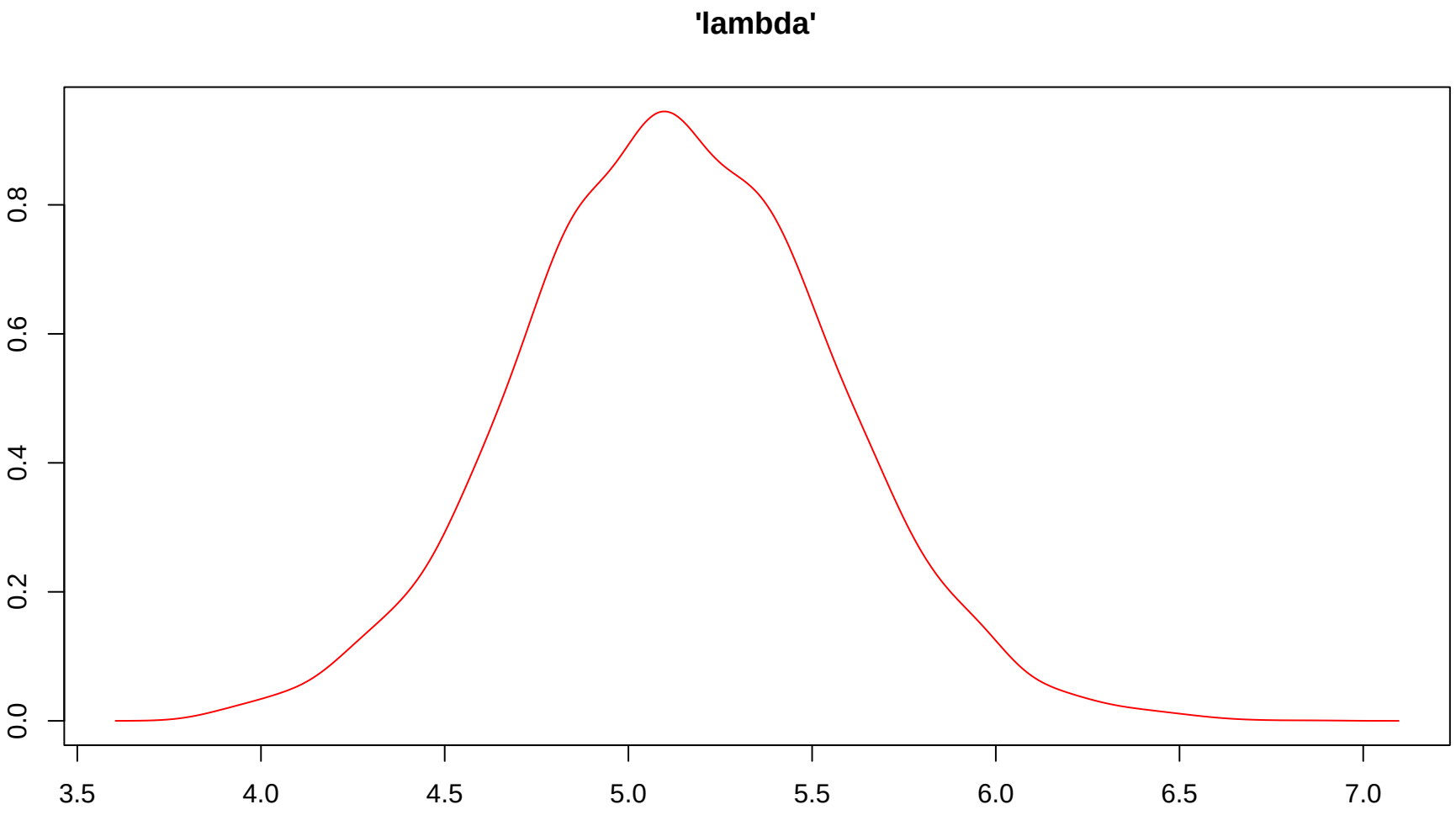
Bayesian analysis: using a flat prior (ctd.)

```
samplesSet("lambda")

## monitor set for variable 'lambda'

modelUpdate(3000)

## 3000 updates took 0 s

samplesStats("*", beg = 1001)

##         mean     sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.133 0.4229  0.00938    4.297  5.126     5.968  1001   8000
```

Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

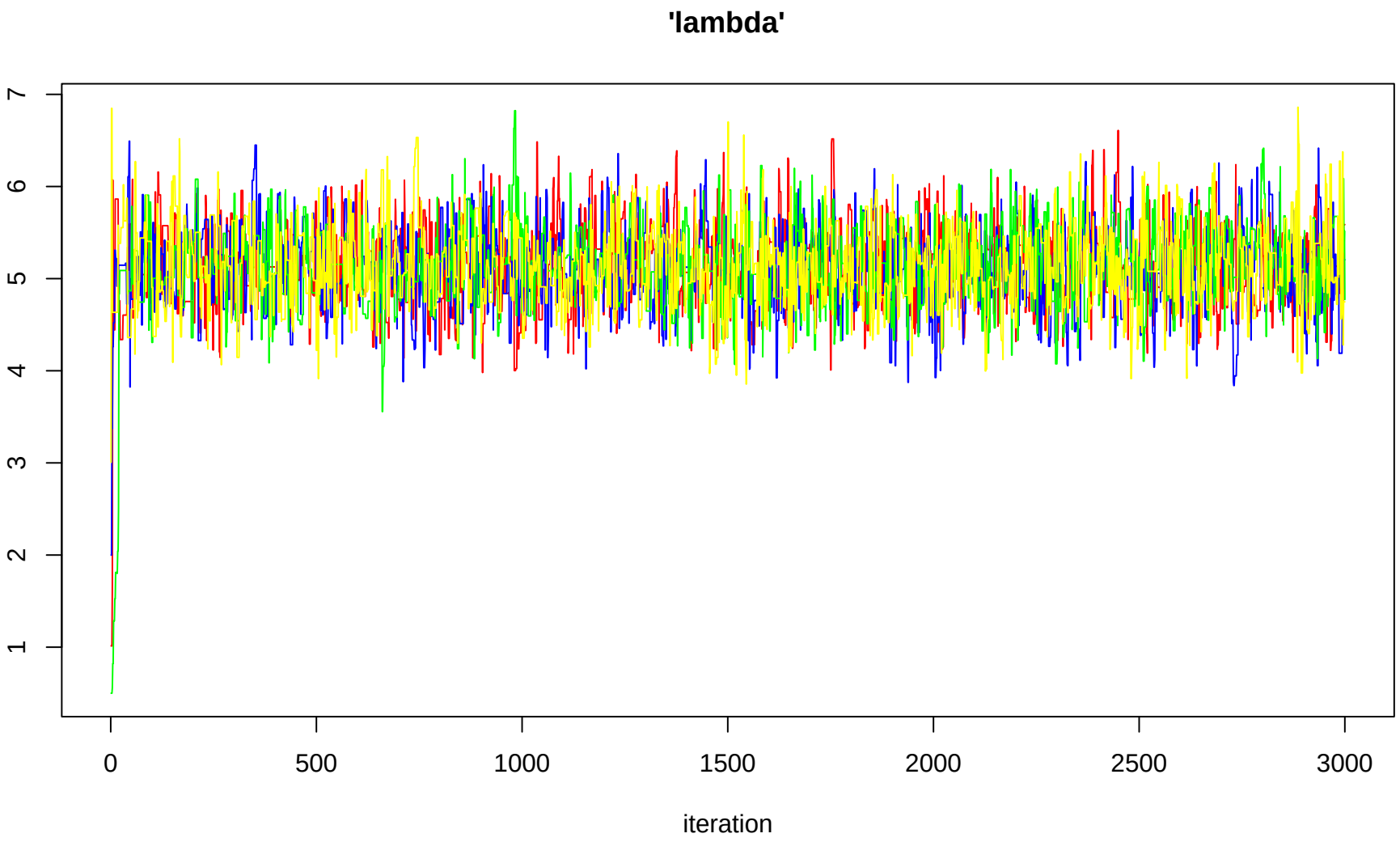
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

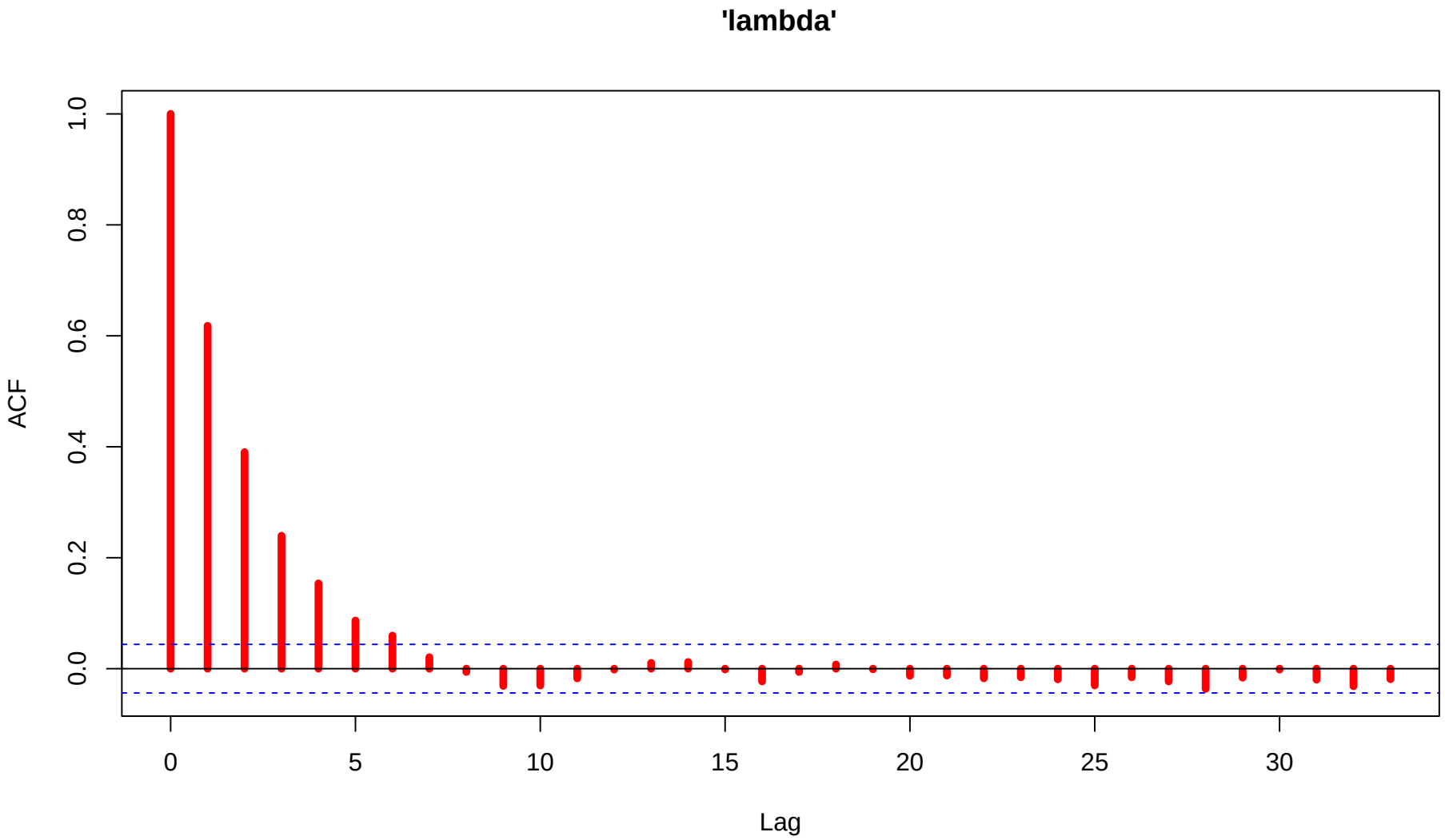
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

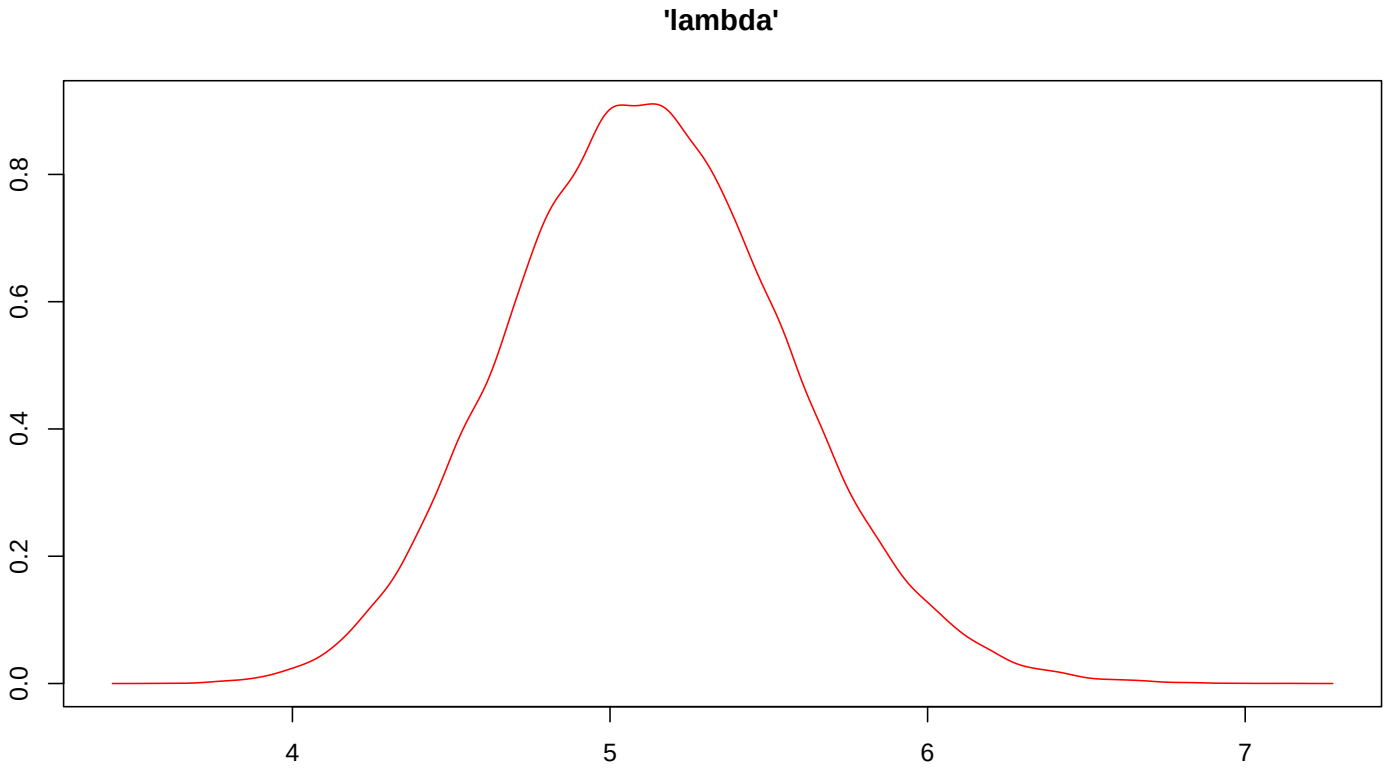
```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

```
modelUpdate(28000)

## 28000 updates took 2 s

samplesStats("*", beg = 1001)

##         mean     sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.122 0.4337 0.002603    4.304  5.112     6.005  1001 120000

samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using a flat prior (ctd.)

Analysing the data on nerve pulses using a flat prior for λ leads to an estimate of $\hat{\lambda} = 5.1$ pulses per second.

This estimate is based on running four chains, each for 31 000 iterations and discarding the first 1 000 iterations as warm-up. Trace plots indicated that the chains had converged and were mixing well.

Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior

Writing the BUGS code for this model is fairly simple:

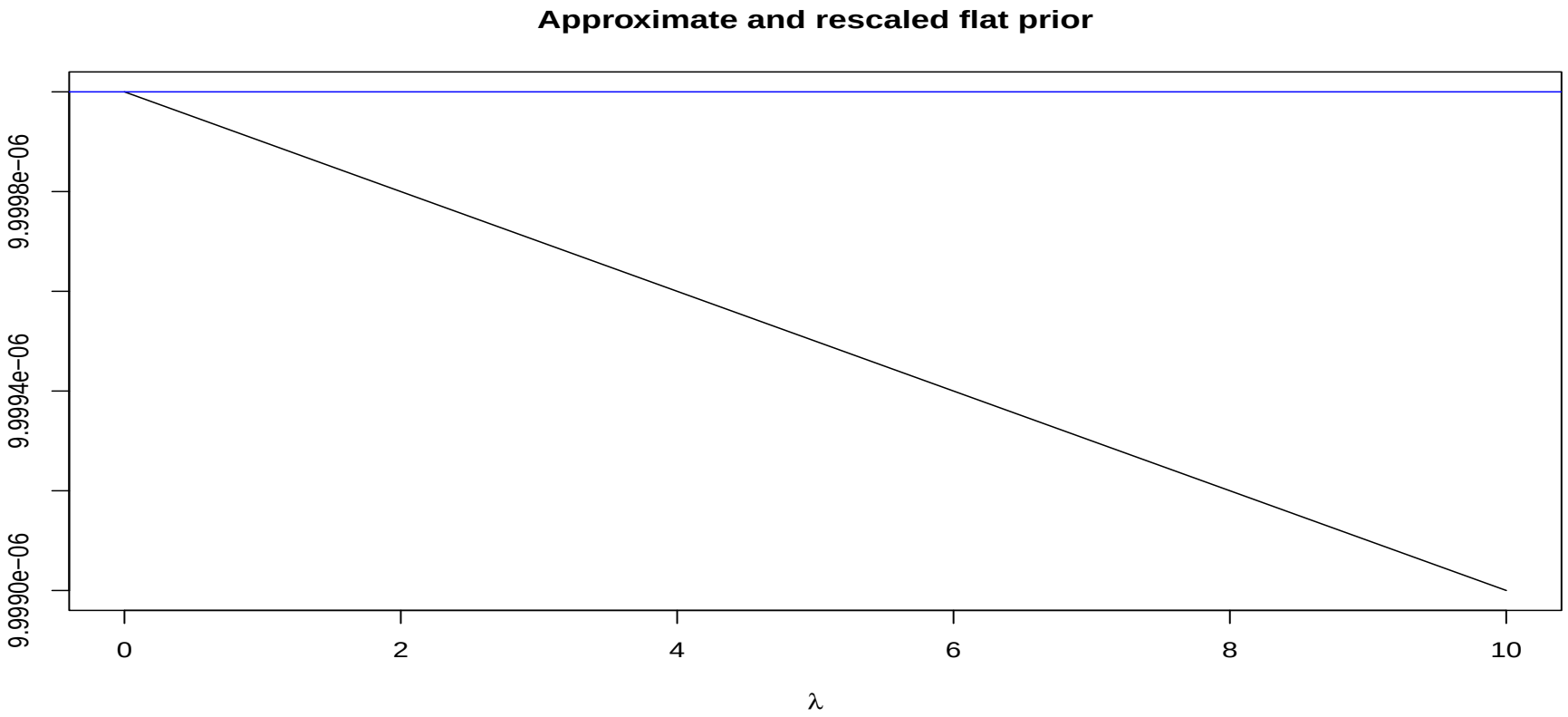
```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dexp(lambda)
  }

  ## prior
  lambda ~ dgamma(1, 0.00001)
}
```

Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

Visually this approximate flat prior (black line) is indistinguishable from the (rescaled) flat prior (blue line):

```
curve(dgamma(x, 1, 1e-05), from = 0, to = 10, n = 401, xlab = expression(lambda),
    ylab = "", main = "Approximate and rescaled flat prior")  ## approximate prior
abline(h = 1e-05, col = "blue")
```

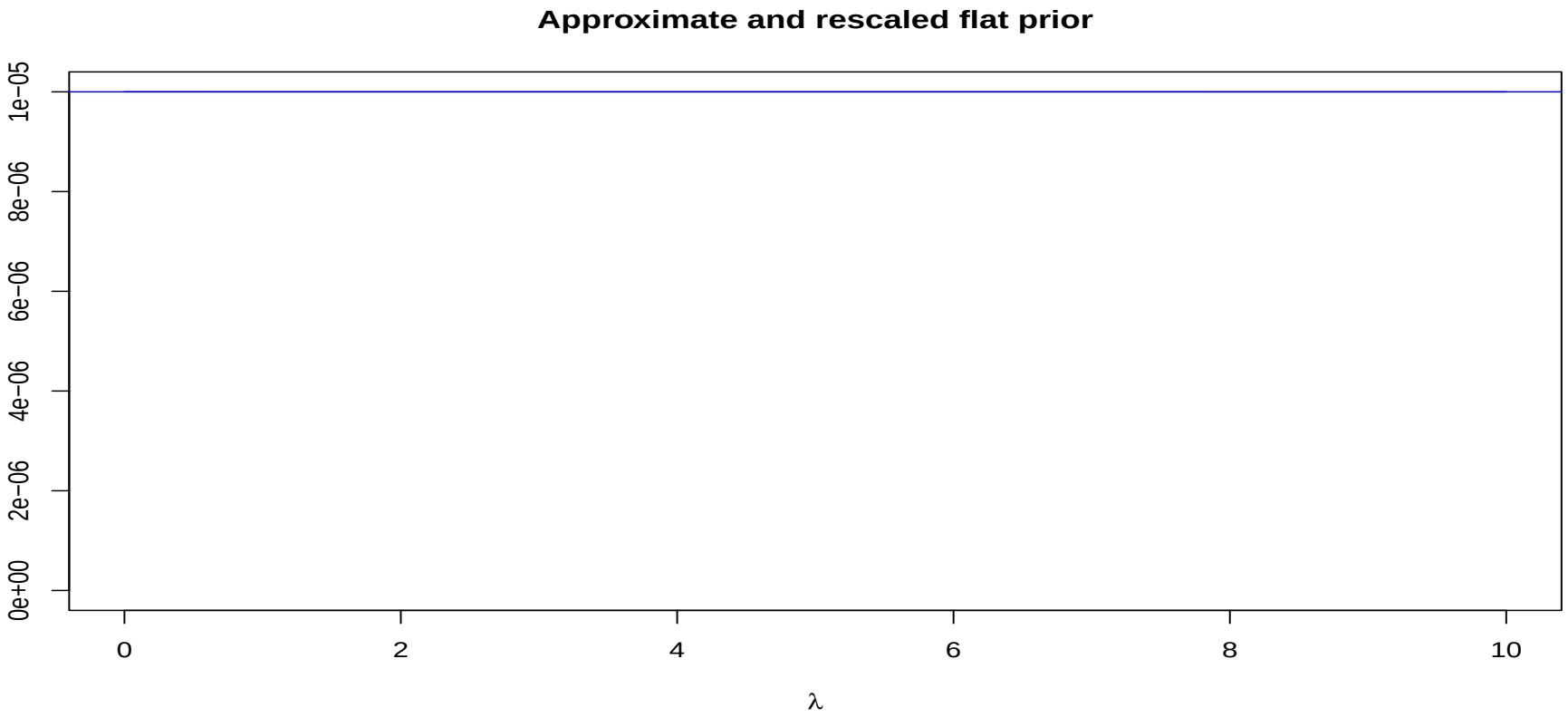


Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

Visually this approximate flat prior (black line) is indistinguishable from the (rescaled) flat prior (blue line):

```
curve(dgamma(x, 1, 1e-05), ylim = c(0, 1e-05), from = 0, to = 10, n = 401,
    xlab = expression(lambda), ylab = "", main = "Approximate and rescaled flat prior")  ## approximate prior
abline(h = 1e-05, col = "blue")
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

```
library(BRugs)
modelCheck("Exponential2.bug")

## model is syntactically correct

bugsData(list(y = y, N = length(y)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(lambda = 1), list(lambda = 2), list(lambda = 0.5), list(lambda = 1.5))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 4: Nerve Pulses

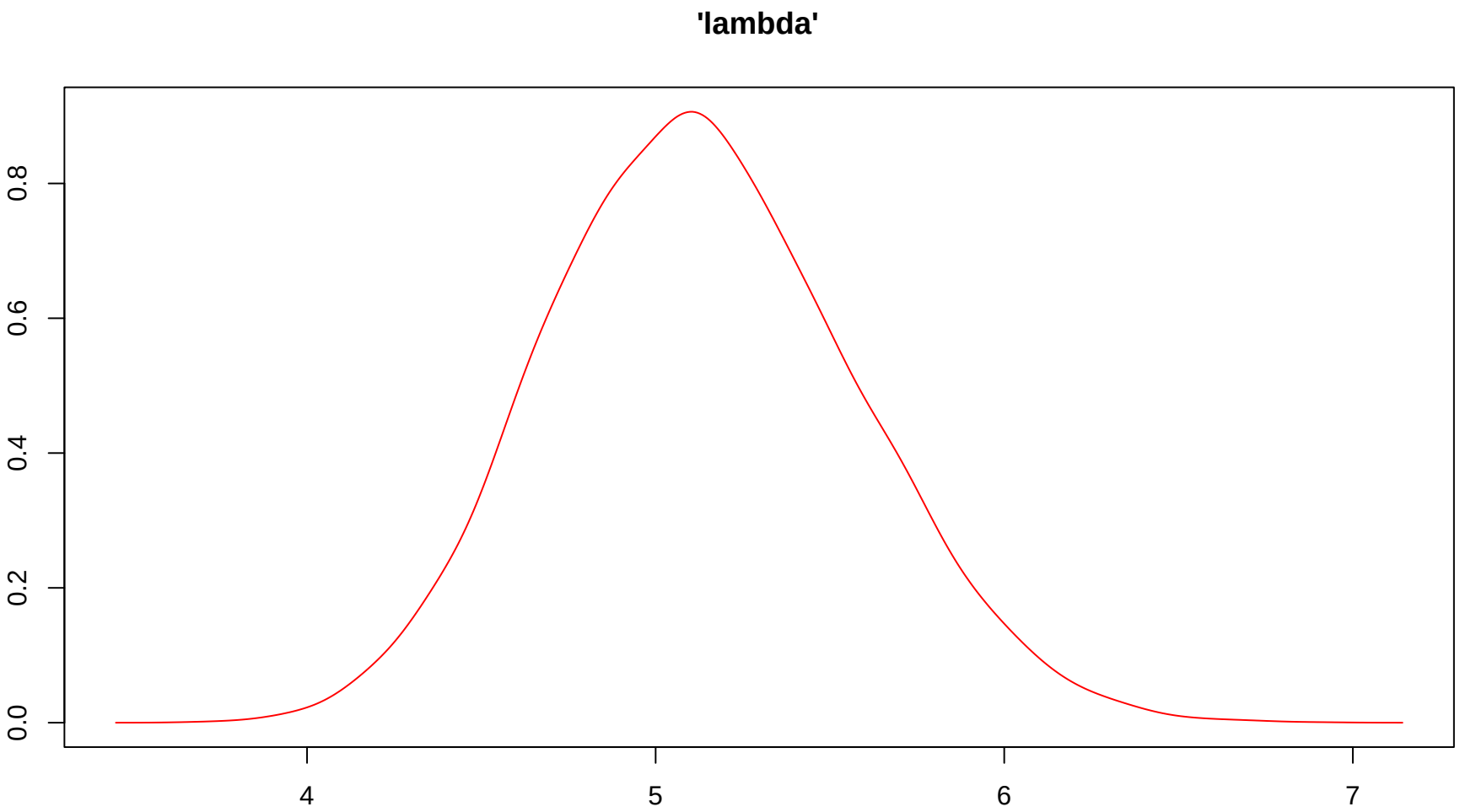
Bayesian analysis: using an approximate flat prior (ctd.)

```
samplesSet("lambda")

## monitor set for variable 'lambda'

modelUpdate(3000)

## 3000 updates took 0 s

samplesStats("*", beg = 1001)

##         mean     sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.131 0.4385 0.004177    4.323  5.114     6.026  1001   8000
```

Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

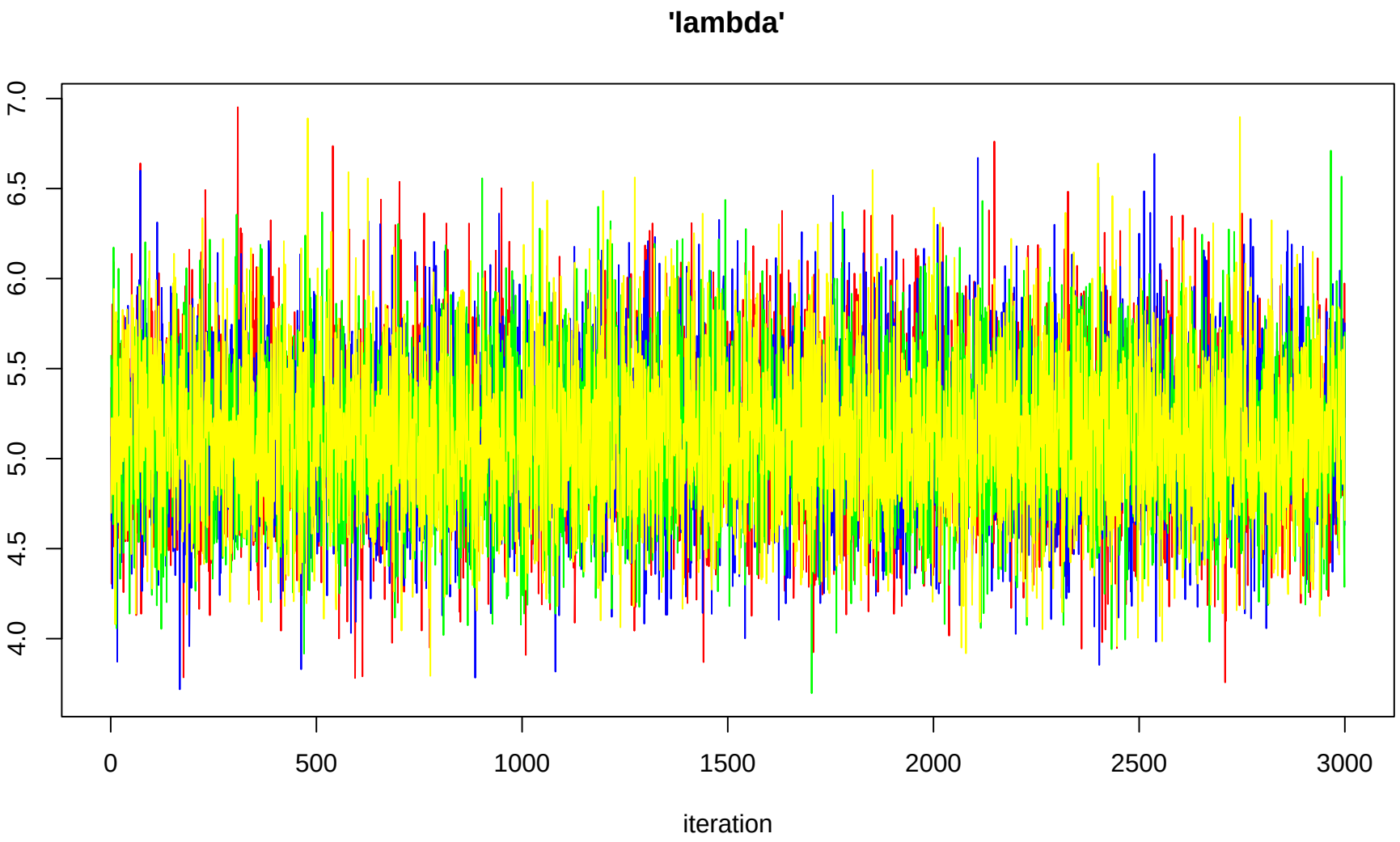
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

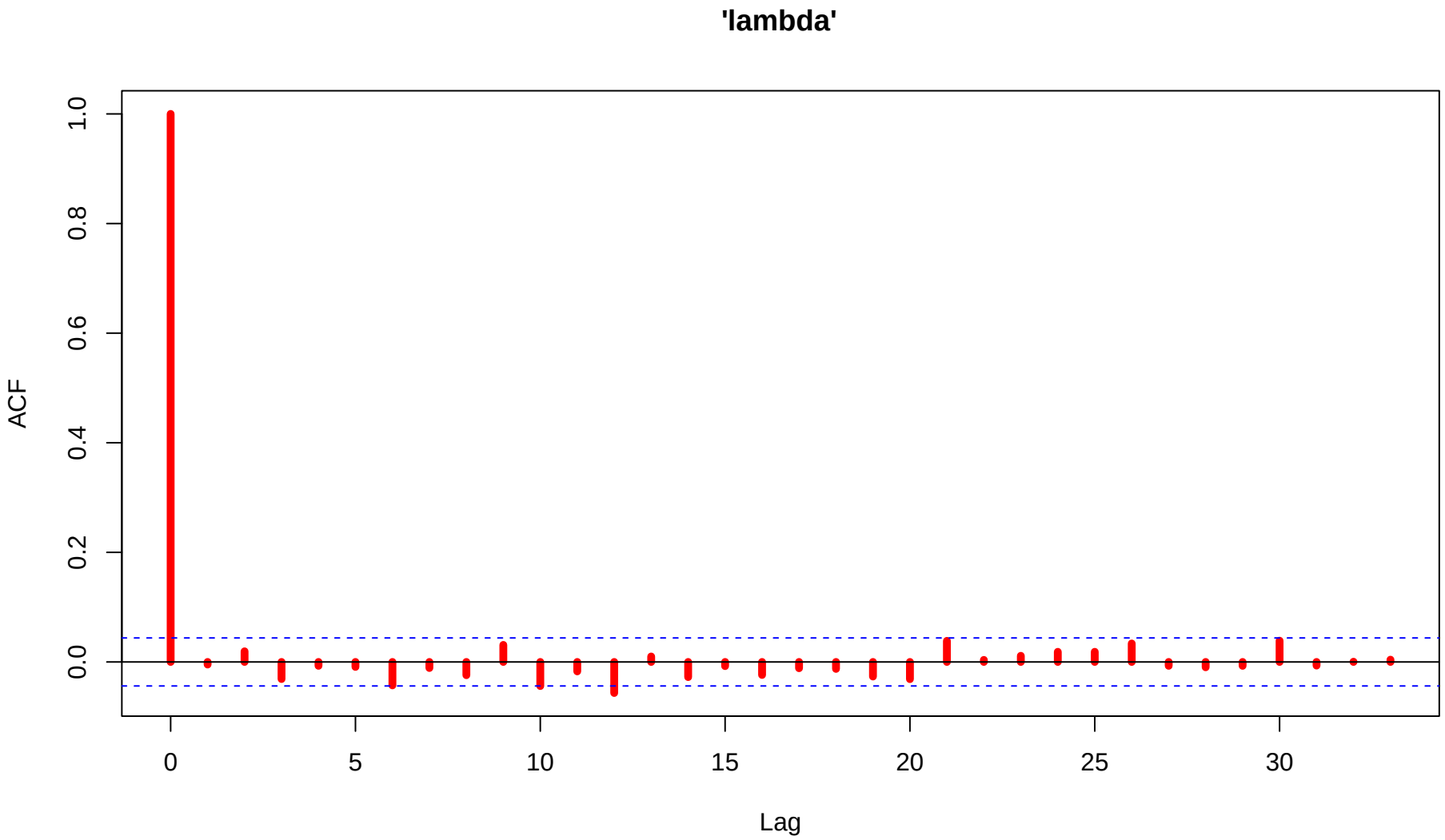
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate flat prior (ctd.)

Analysing the data on nerve pulses using a a gamma prior with parameters $\alpha = 1$ and $\beta = 10^{-5}$, which can be viewed as an approximation to a flat prior (over a wide range of values of the parameter), for λ leads to an estimate of $\hat{\lambda} = 5.1$ pulses per second.

This estimate is based on running four chains, each for 3 000 iterations and discarding the first 1 000 iterations as warm-up. Trace plots indicated that the chains had converged and were mixing well.

Example 4: Nerve Pulses

Bayesian analysis: Using a Jeffreys' prior

Prior: $\lambda \sim$ "$\text{Gamma}(0, 0)$"

Data: $n = 139$ observations from $\text{Expo}(\lambda)$

$s = \sum_{i=1}^{n} y_i = 27.31$

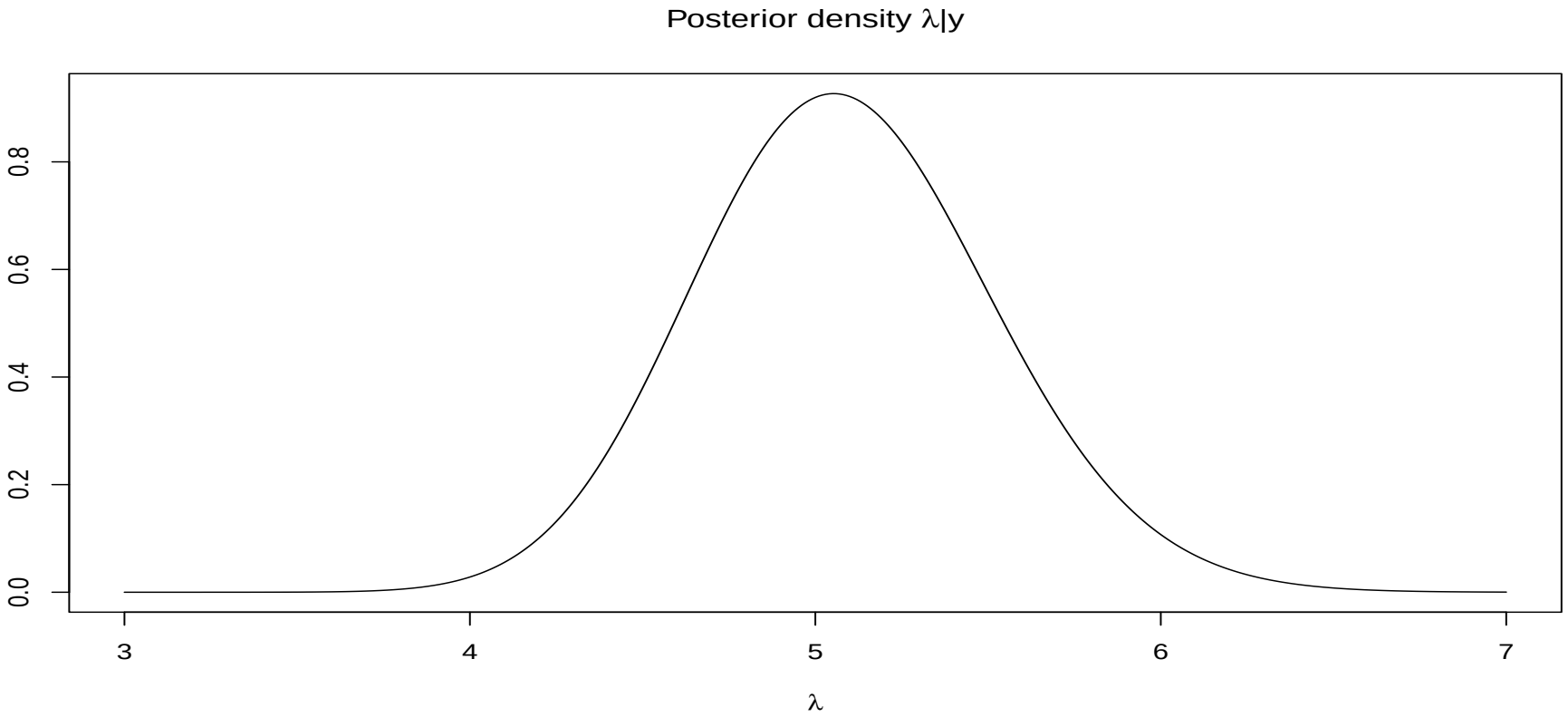
Posterior: $\lambda | \mathbf{Y} = \mathbf{y} \sim \text{Gamma}(139, 27.31)$

```
alpha <- 0
beta <- 0
n <- length(y)
s <- sum(y)
alpha.post <- alpha + n
beta.post <- beta + s
(post.mean <- alpha.post/beta.post)  ## posterior mean

## [1] 5.089711

(post.var <- alpha.post/beta.post^2)  ## posterior variance

## [1] 0.186368

(post.std <- sqrt(post.var))  ## posterior StdDev

## [1] 0.4317036
```

Example 4: Nerve Pulses

Bayesian analysis: Using a Jeffreys' prior (ctd)

```
curve(dgamma(x, shape = alpha.post, rate = beta.post), from = 3, to = 7,
    n = 401, xlab = expression(lambda), ylab = "", main = expression("Posterior density " *
        lambda * "|y"))
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

To use Jeffreys' prior, we have to use the following trick.

Let U be a non-negative continuous random variable with "uniform" distribution on $(-\infty, \infty)$, i.e.

$$f_U(u) \propto 1, \quad \text{for all } u \in \mathbb{R}$$

Then the distribution of

$$\lambda = \exp(U)$$

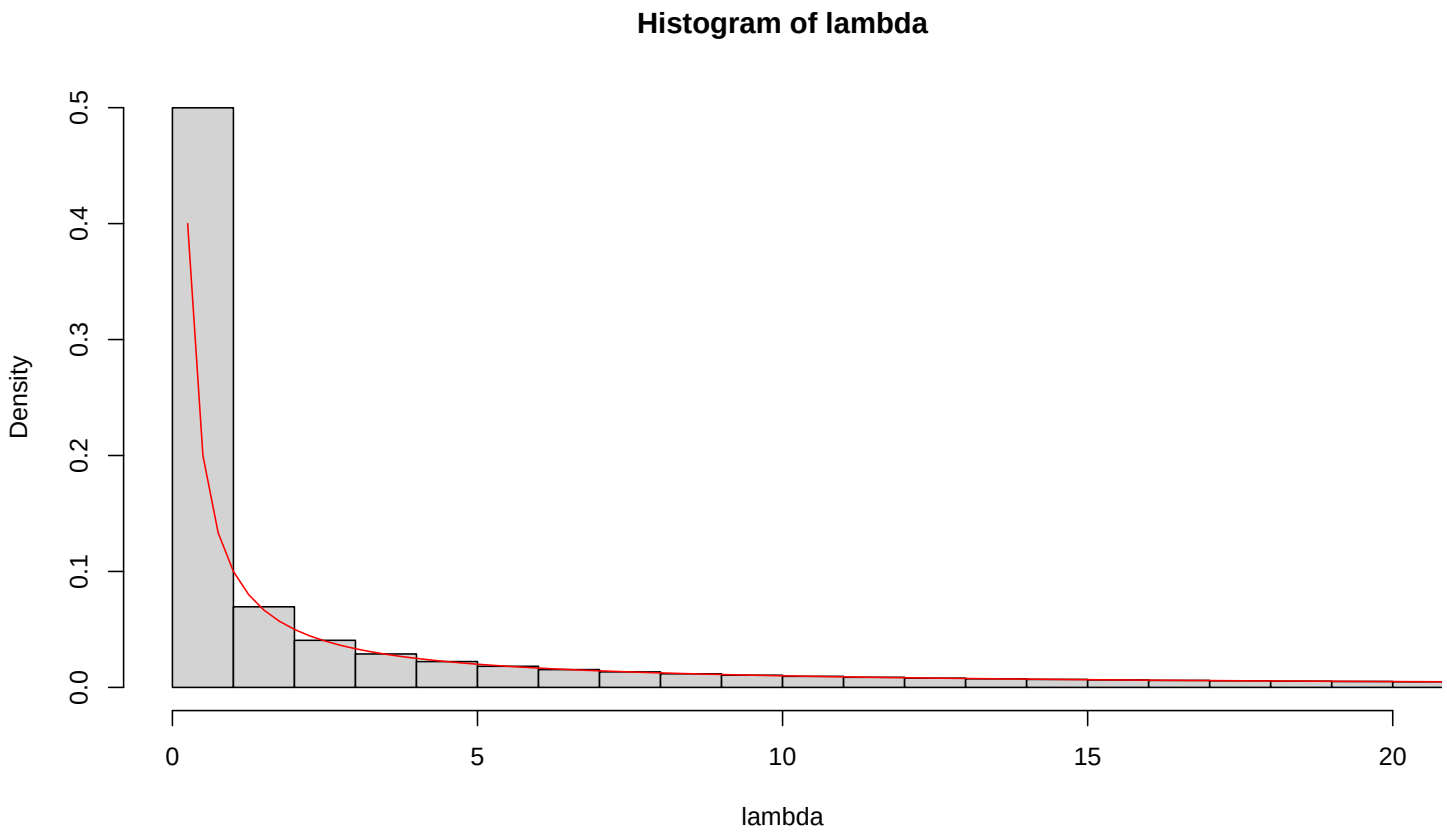
is

$$f(\lambda) \propto \lambda^{-1}$$

Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

```
u <- runif(1e+07, min = -5, max = 5)
lambda <- exp(u)
hist(lambda, probability = TRUE, breaks = 200, xlim = c(0, 20))
curve(0.1 * 1/x, from = 0, to = 100, n = 401, add = TRUE, col = "red")
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

Writing the BUGS code for this model is fairly simple:

```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dexp(lambda)
  }

  ## prior
  u ~ dflat()
  lambda <- exp(u)
}
```

Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

```
library(BRugs)
modelCheck("Exponential3.bug")

## model is syntactically correct

bugsData(list(y = y, N = length(y)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(u = 1), list(u = 2), list(u = 0.5), list(u = 1.5))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 4: Nerve Pulses

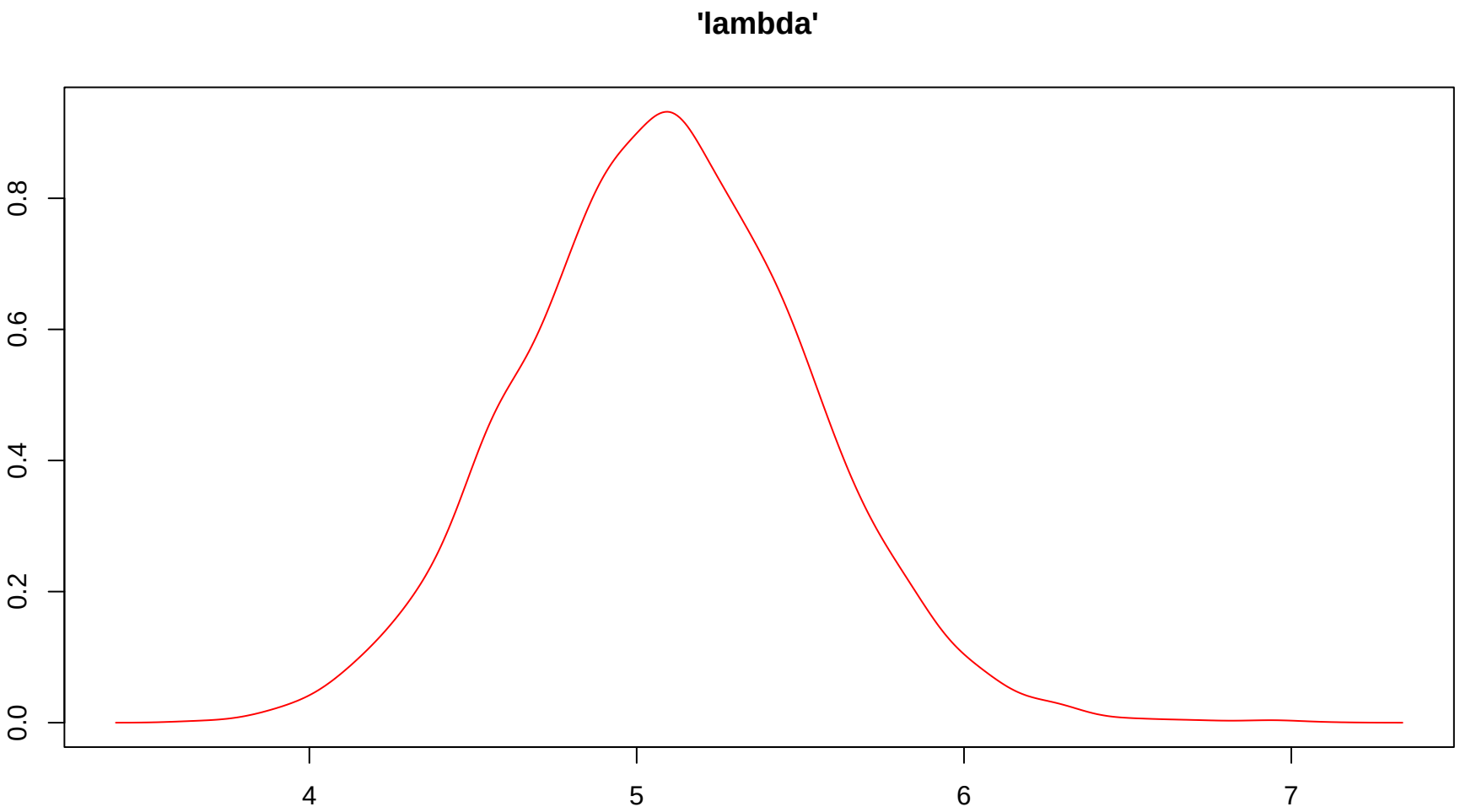
Bayesian analysis: using Jeffreys' prior (ctd.)

```
samplesSet("lambda")

## monitor set for variable 'lambda'

modelUpdate(3000)

## 3000 updates took 0 s

samplesStats("*", beg = 1001)

##         mean     sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.089 0.4384   0.0104    4.249  5.088     5.968  1001   8000
```

Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

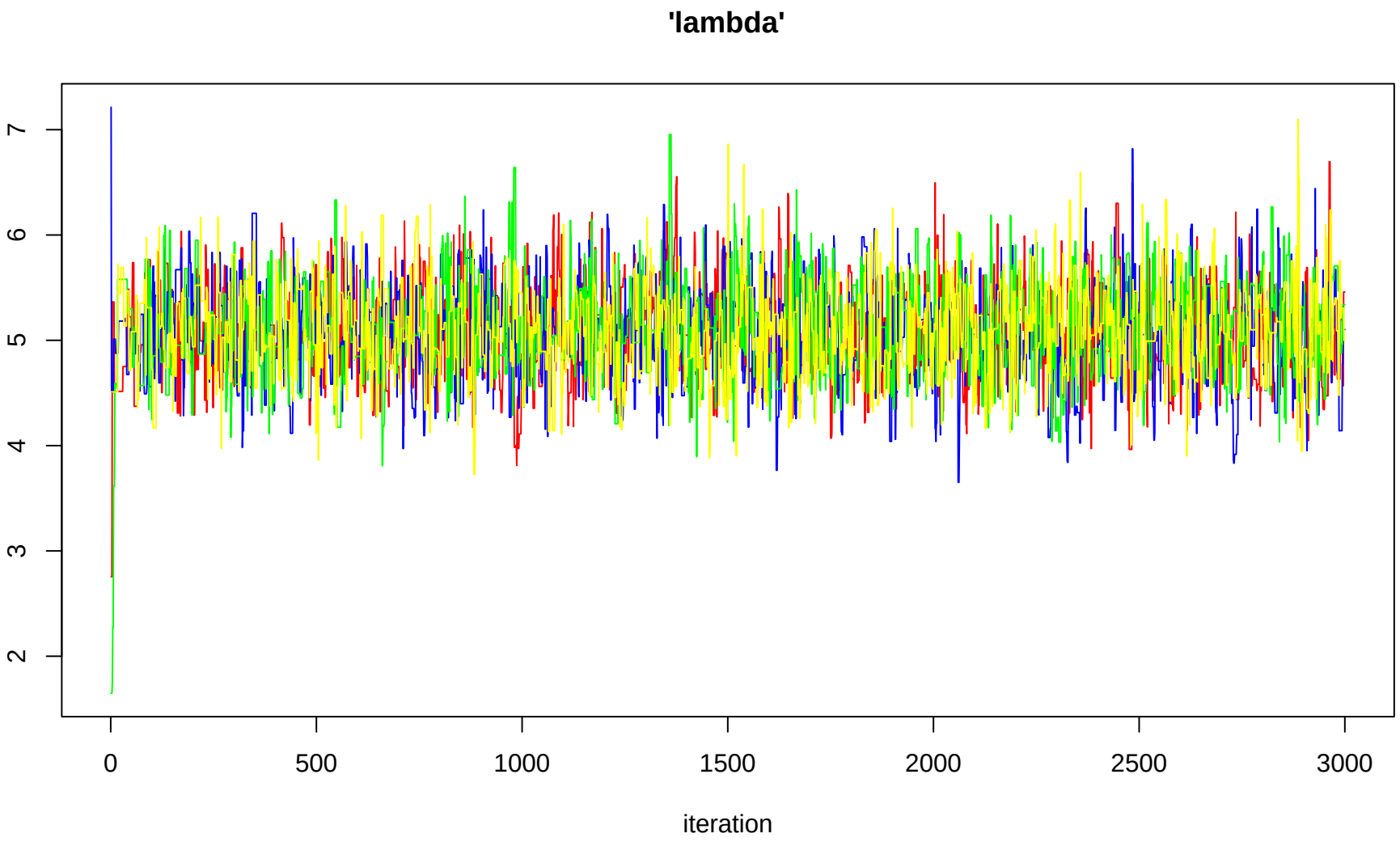
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

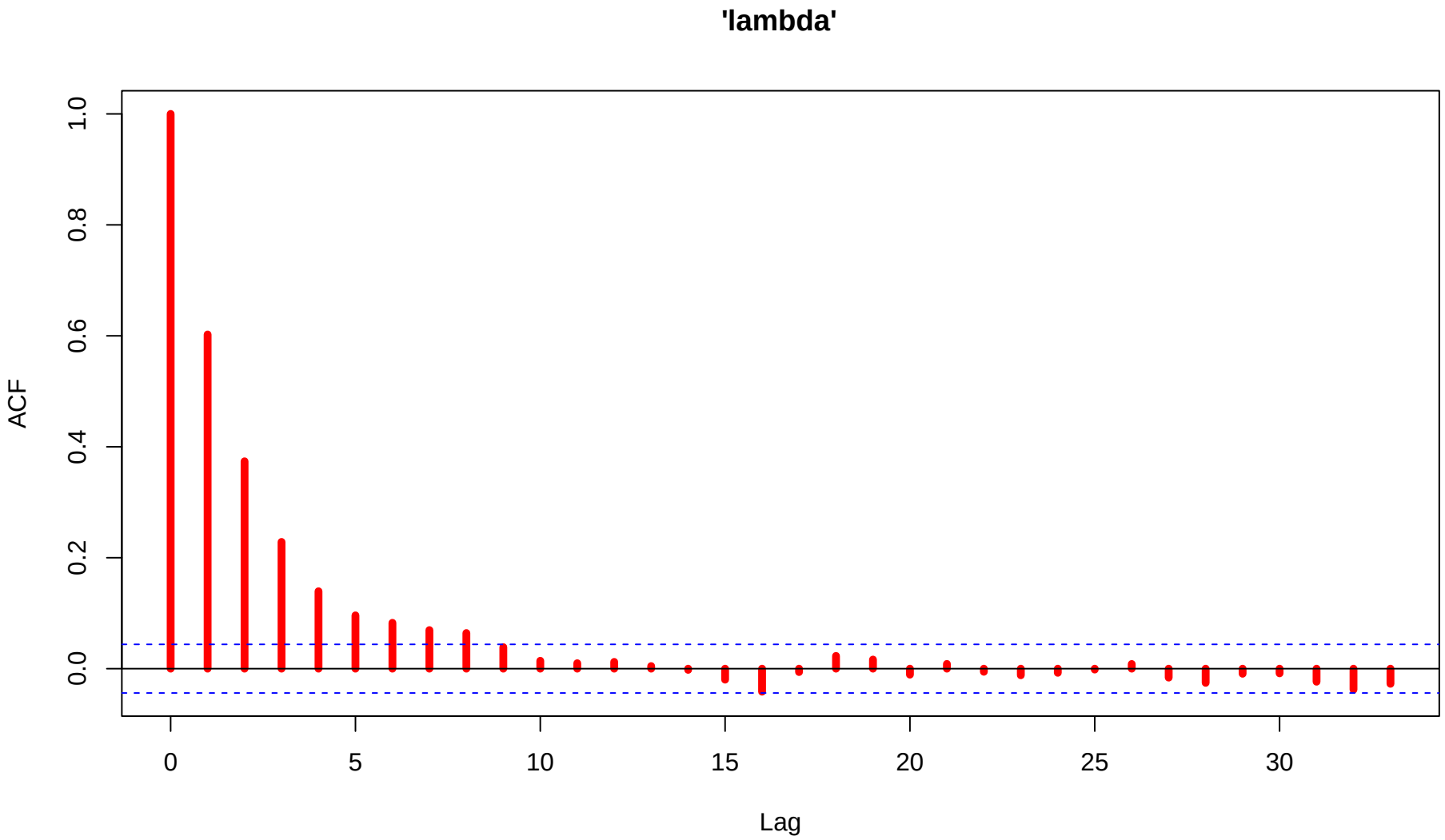
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

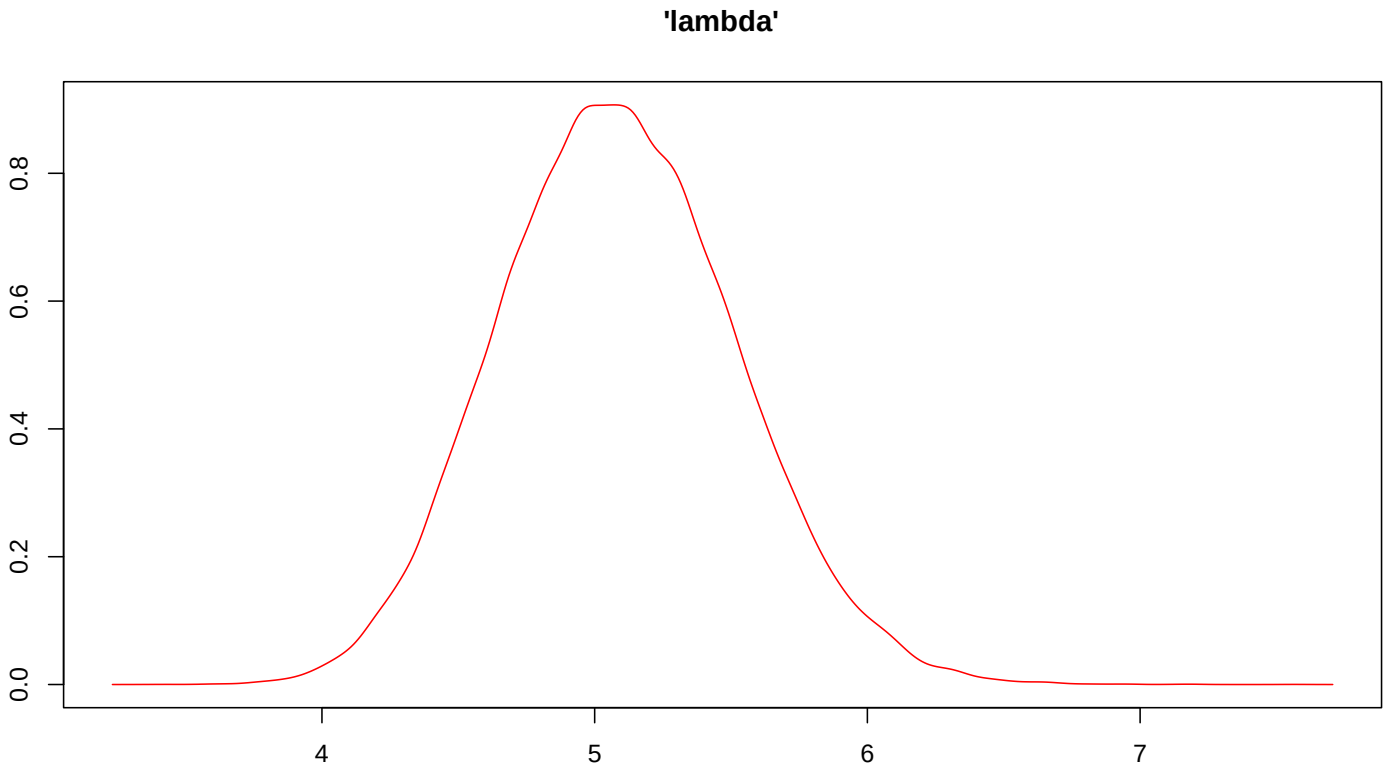
```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

```
modelUpdate(28000)

## 28000 updates took 3 s

samplesStats("*", beg = 1001)

##         mean     sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.086 0.4309 0.002622    4.277  5.076     5.963  1001 120000

samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using Jeffreys' prior (ctd.)

Analysing the data on nerve pulses using Jeffreys' prior for λ leads to an estimate of $\hat{\lambda} = 5.1$ pulses per second.

This estimate is based on running four chains, each for 31 000 iterations and discarding the first 1 000 iterations as warm-up. Trace plots indicated that the chains had converged and were mixing well.

Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior

Writing the BUGS code for this model is fairly simple:

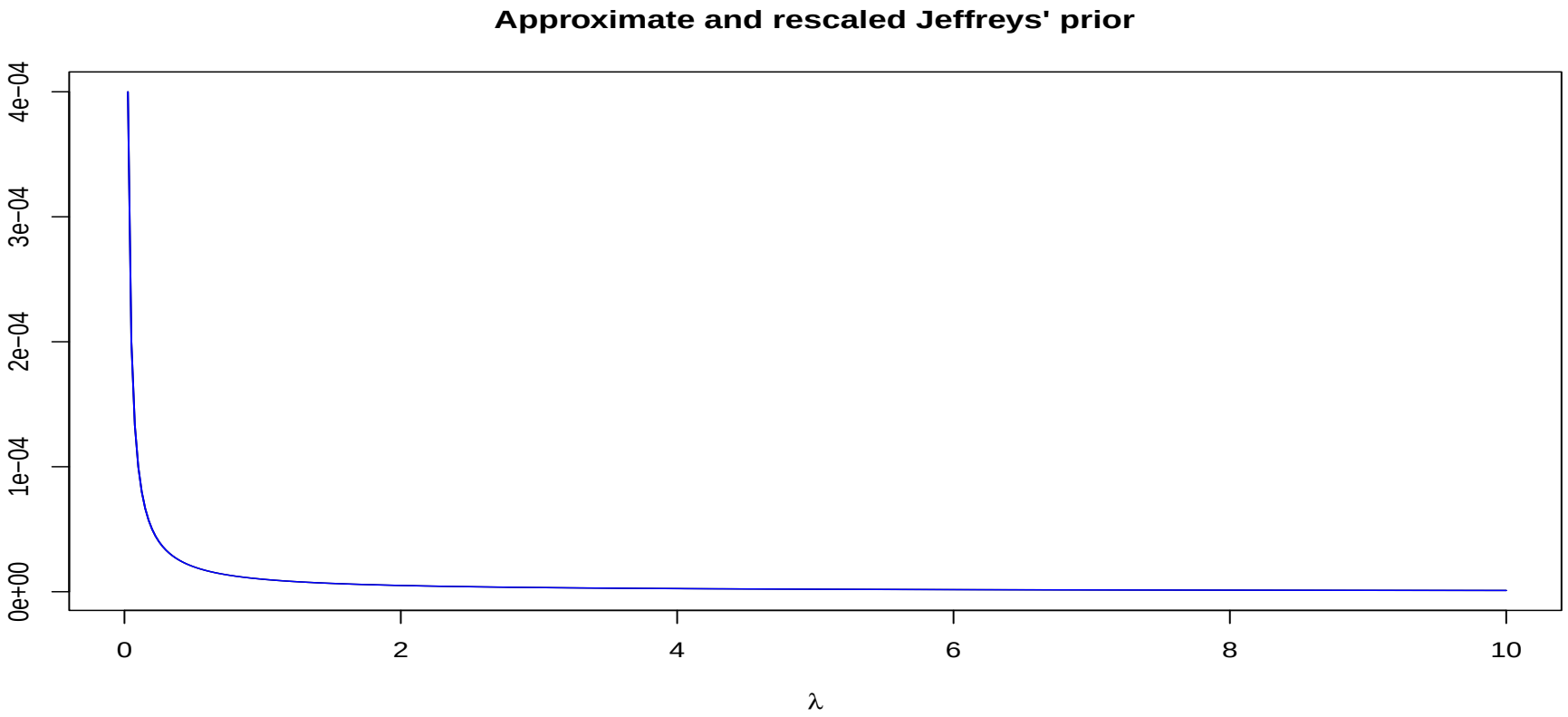
```
model{
  ## likelihood
  for(i in 1:N){
    y[i] ~ dexp(lambda)
  }

  ## prior
  lambda ~ dgamma(0.00001, 0.00001)
}
```

Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

Visually this approximate Jeffreys' prior (black line) is indistinguishable from the (rescaled) Jeffreys' prior (blue line):

```
curve(dgamma(x, 1e-05, 1e-05), from = 0, to = 10, n = 401, xlab = expression(lambda),
    ylab = "", main = "Approximate and rescaled Jeffreys' prior")  ## approximate prior
curve(1e-05/x, from = 0, to = 10, n = 401, col = "blue", add = TRUE)
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```r
library(BRugs)
modelCheck("Exponential4.bug")

## model is syntactically correct

bugsData(list(y = y, N = length(y)), file = "data.txt")
modelData("data.txt")

## data loaded

nch <- 4
modelCompile(numChains = nch)

## model compiled

inits <- list(list(lambda = 1), list(lambda = 2), list(lambda = 0.5), list(lambda = 1.5))
fnames <- paste0("init", 1:nch, ".txt")
bugsInits(inits, numChains = nch, fileName = fnames)
modelInits(fnames)

## Initializing chain 1:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 2:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 3:
## initial values loaded and chain initialized but another chain contain uninitialized variables
## Initializing chain 4:
## model is initialized
```

Example 4: Nerve Pulses

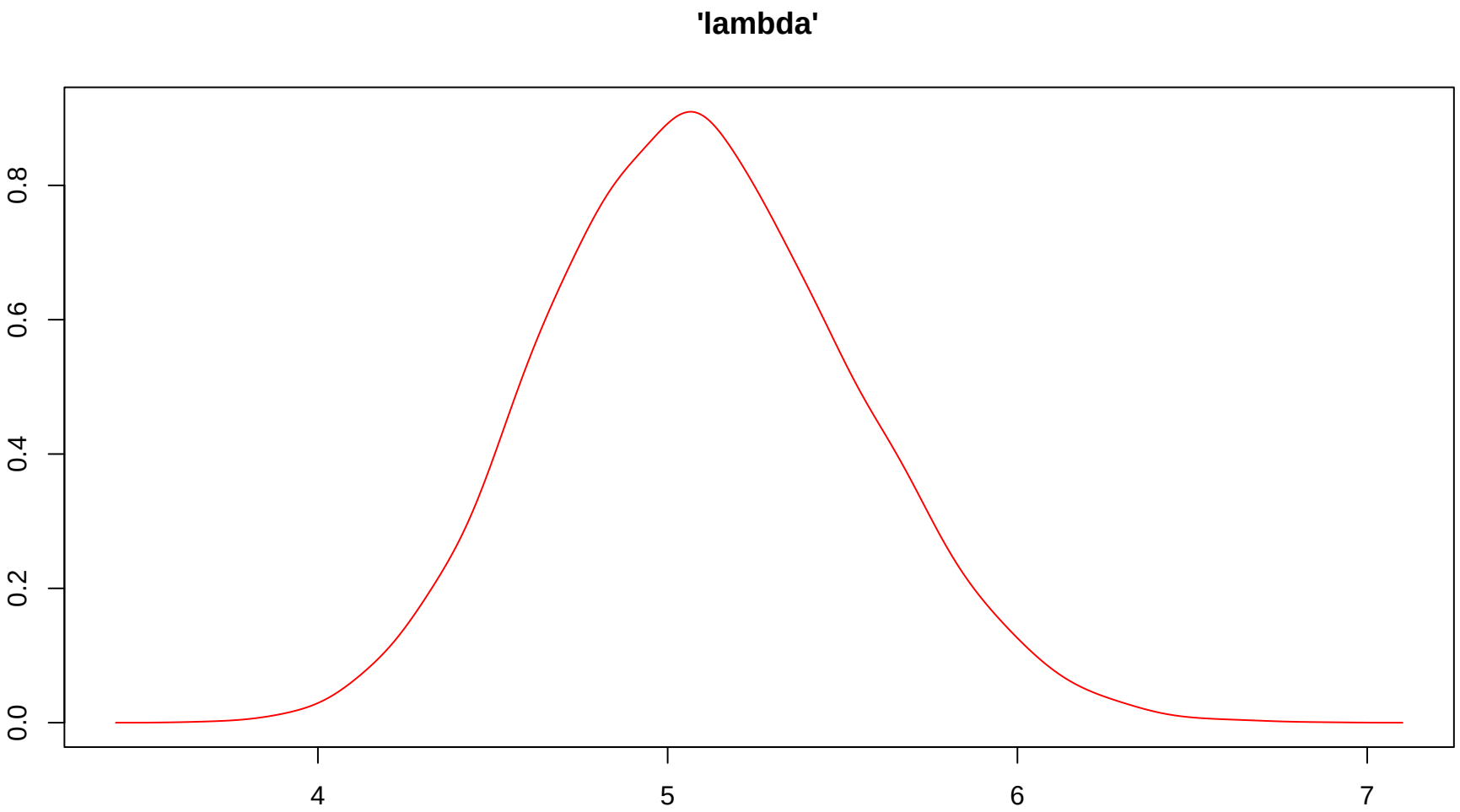
Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
samplesSet("lambda")

## monitor set for variable 'lambda'

modelUpdate(3000)

## 3000 updates took 0 s

samplesStats("*", beg = 1001)

##         mean    sd MC_error val2.5pc median val97.5pc start sample
## lambda 5.094 0.437 0.004213     4.29  5.078     5.987  1001   8000
```

Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

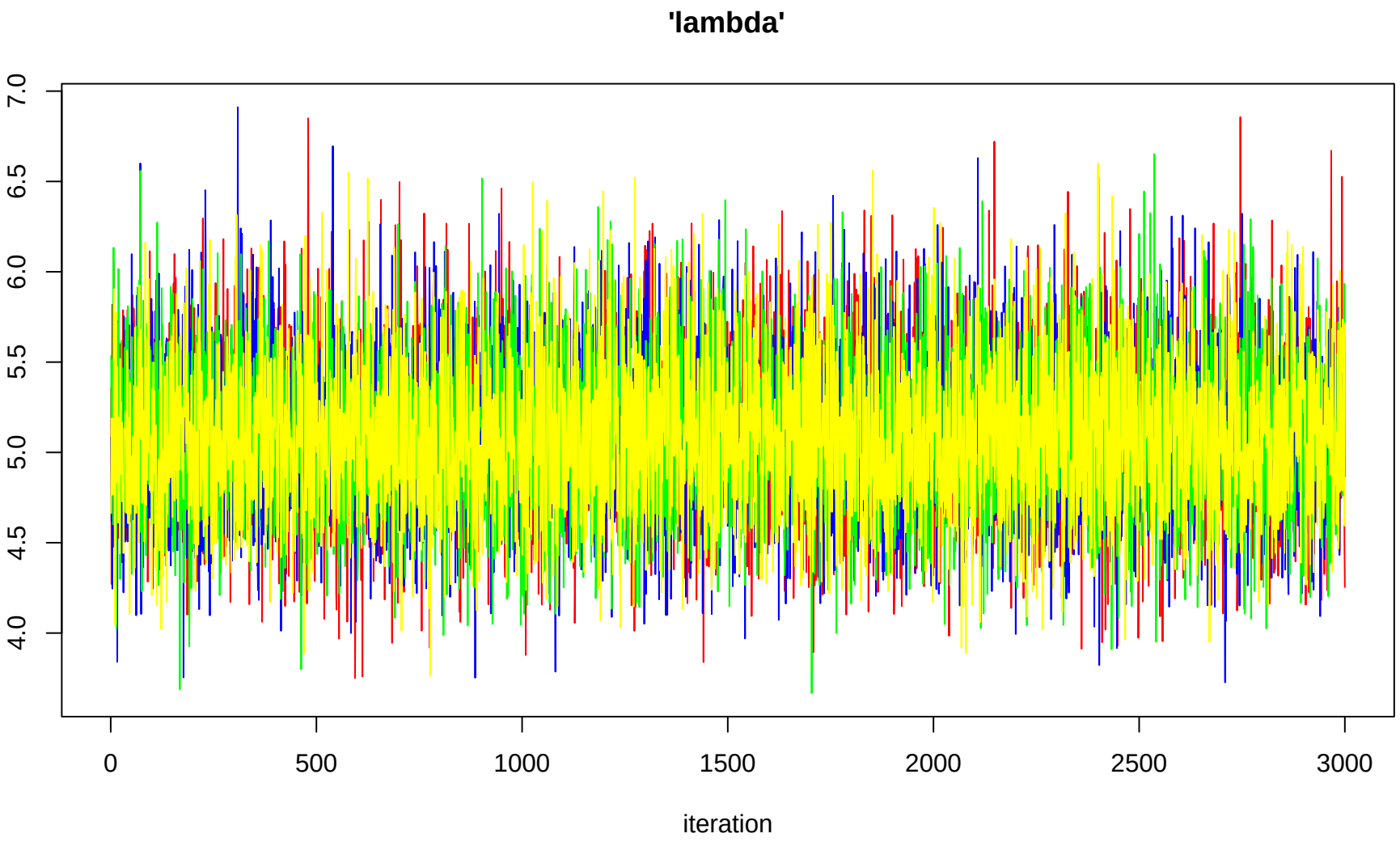
```
samplesDensity("*", beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

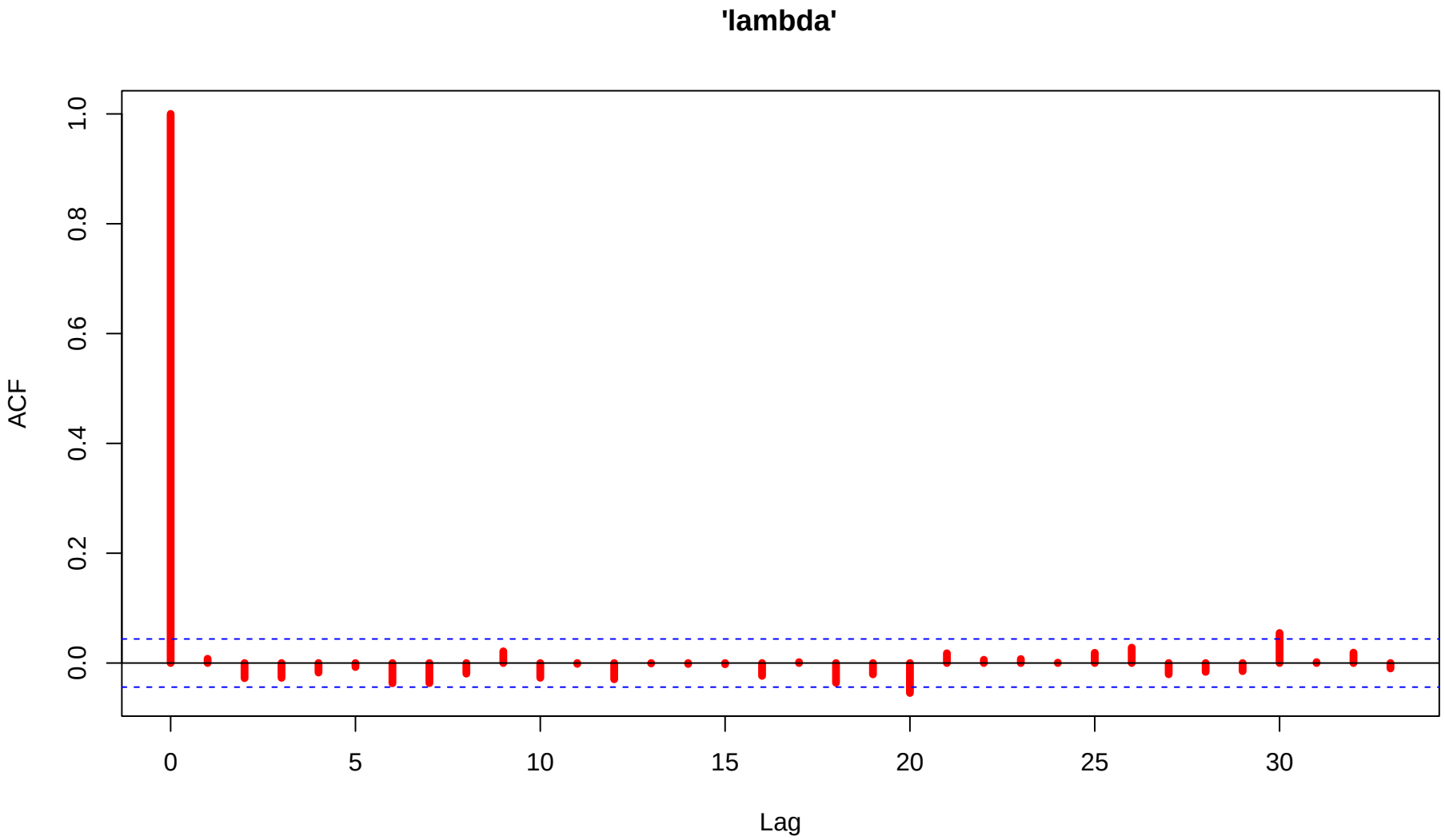
```
samplesHistory("*", ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

```
samplesAutoC("*", chain = 1, beg = 1001, ask = FALSE, mfrow = c(1, 1))
```



Example 4: Nerve Pulses

Bayesian analysis: using an approximate Jeffreys' prior (ctd.)

Analysing the data on nerve pulses using a a gamma prior with parameters $\alpha = \beta = 10^{-5}$, which can be viewed as an approximation to Jeffreys' prior (over a wide range of values of the parameter), for λ leads to an estimate of $\hat{\lambda} = 5.1$ pulses per second.

This estimate is based on running four chains, each for 3 000 iterations and discarding the first 1 000 iterations as warm-up. Trace plots indicated that the chains had converged and were mixing well.